Union Membership and Native-Immigrant Labour Market Gaps

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Abstract

We estimate the causal effects of union membership on labour market outcomes for natives and immigrants, exploiting quasi-experimental variation from large increases in Norway's tax deduction for union dues. Union membership raises earnings and improves employment protection for native workers, while the private returns for non-Western immigrants are substantially smaller and often close to zero. Complementary survey evidence suggests that this gap is not explained by differences in information, preferences, or engagement with unions: immigrants receive similar information when joining, report similar preferences over job amenities, and are more likely than natives to seek union assistance. Instead, the combined administrative and survey evidence is consistent with differences in union-side behaviour. Immigrants report lower satisfaction after contacting unions for support and are more likely to believe that native workers and those with strong local networks receive greater benefits from membership. The impact is most pronounced in concentrated labour markets, indicating that relying on unions to address monopsony-based market failures may unintentionally amplify group inequalities.

JEL Codes: J1, J5, J6
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1 Introduction

Labour unions were established to balance the bargaining power between employers and employees. By organising workers and controlling the supply of labour, unions can raise wages and improve working conditions beyond what workers may obtain individually. Unions might also facilitate communication between workers and firms, channeling worker voice into workplace rules and practices.

By shifting the balance of power between workers and firms, unions can directly influence how immigrants fare in the labour market and whether disparities with natives widen or narrow. The direction of this effect, however, is ambiguous. On the one hand, immigrants possess weaker individual bargaining power than natives and may therefore benefit disproportionately from the application of group-level bargaining, potentially narrowing the wage gap and facilitating immigrant integration.[1] On the other hand, unions might prioritise native workers or struggle to address immigrant-specific challenges, leading to unequal treatment and potentially widening disparities. Moreover, differences in how natives and immigrants engage with unions could influence whether unions mitigate or intensify these inequalities.

Studying these dynamics is empirically challenging because workers do not join unions randomly, and natives and immigrants may self-select into membership for different reasons. As a result, simple comparisons of members across groups can reflect differences in selection rather than causal effects.

In this paper, we make progress on this issue by exploiting policy-induced exogenous shifts in the cost of joining unions, allowing us to estimate the causal effect of union membership on labour market disparities between natives and immigrants. We first examine the effect of union membership among natives and immigrants across a wide range of career outcomes, including earnings, work hours, job security, work environment, and advancement opportunities. We then study how labour market power shapes these effects, noting that unions may have greater scope to influence worker outcomes in less competitive labour markets where firms earn supernormal profits. Finally, we combine survey and administrative evidence to examine the mechanisms behind our findings and to assess the extent to which the observed gaps reflect differences in worker behaviour, firm environments, or unions themselves.

Our identification strategy exploits exogenous changes in the financial cost of joining a union. Assuming union membership is a normal good, a reduction in the cost of membership should increase take-up. The relevant cost changes come from a series of national reforms that introduced tax deductions for union dues without affecting any other part of the tax and transfer system. Because these deductions were subject to maximum caps, workers at firms

[1] Less familiarity with the regulatory framework of labour markets, greater language barriers, greater exposure to discriminatory and predatory hiring practices of firms, workplace segregation, and fewer outside options, are all factors that may affect immigrants more than natives in terms of their individual bargaining power (e.g., Algan et al. (2010); Chiswick (1978); Lehmer and Ludsteck (2011); Åslund and Skans (2010); Cutler et al. (2008); Hirsch and Jahn (2015)).

with relatively high baseline union dues were more likely to be constrained before the reform. As the caps were gradually relaxed, their net cost of joining a union fell. This generates quasi-random variation in the incentive to join a union based on the firm a worker was attached to prior to the reform.

We use an instrumented difference-in-differences design that exploits variation in subsidy exposure across firms. In the design, the intensity of instrument exposure is determined by baseline union dues. Conditioning on current firm fixed effects restricts comparisons to workers within the same workplace, allowing us to isolate how individual union enrolment affects native–immigrant labour market gaps. This allows us to estimate the private returns to union membership across groups, beyond the collective effects of unionization. We also examine the effects of spillovers from changes in firm-level union density across immigration status.

To operationalise our approach, we combine rich administrative and survey data from Norway. Our core data come from matched employer–employee registers containing detailed information on union membership, dues, wages, and occupations. These are linked to population-wide administrative records on education, income, taxes, and social benefits, allowing us to track career outcomes for each individual worker. To complement these data, we draw on two large worker surveys that measure preferences, beliefs, and attitudes toward unions, including sensitivity to membership dues. These surveys are not designed to identify the same complier population as the administrative IV analysis, but they provide complementary evidence on workers' experiences, perceptions, and interactions with unions. We use them to help interpret the patterns observed in the administrative data.

Our main finding is that union membership disproportionately benefits native workers, widening labour market gaps and increasing inequality between natives and immigrants. Evidence from the surveys suggests that these disparities likely stem from union behaviour—rather than differences in how the two groups engage with unions.

The main takeaway from our paper is based on four key results. First, and consistent with prior research, we identify a significant wage premium associated with union membership (e.g., Farber et al. (2021); Sojourner et al. (2015); Card et al. (2004)). However, unlike prior work, we show that this wage premium varies greatly by migration status. Union membership raises earnings by approximately 0.10 log points for native workers. The corresponding effect is roughly half as large for Western immigrants, while non-Western immigrants experience no short-run earnings premium.

Second, we show that the heterogeneity in the union wage premium between natives and immigrants extends to another core function of unions: employment protection. While union membership reduces layoff risk for both groups, the protection afforded to natives is markedly stronger—especially compared to non-Western immigrants.

Third, the only career dimension in which unions appear to narrow the native–immigrant labour market gap is sick leave utilisation. Specifically, union membership has a large neg-

ative effect on sick leave benefits, and this effect is considerably larger among non-Western immigrants. One interpretation consistent with this pattern is that unions disproportionately improve the work environment of non-Western immigrants, reducing their reliance on the sick leave insurance system. This may reflect that non-Western immigrants are more exposed to poorer working conditions and possess weaker baseline bargaining power, making workplace environment protections particularly valuable. Because improvements in working conditions are also less excludable than other union benefits, they may be harder to allocate selectively across workers.

Fourth, by comparing the effects of union membership across labour markets with varying degrees of employer concentration, we find that the career returns to unionization are considerably larger in more concentrated markets. As a result, the differential effects of union membership across groups are amplified in less competitive labour markets. More broadly, these findings suggest that relying on unions to address monopsony-driven labour market distortions may unintentionally increase between-group inequality.

A key question that emerges from our analysis is what drives the differential impact of union membership on natives and immigrants. Three explanations are possible: (i) the two groups may differ in how they use or engage with unions, (ii) unions may treat immigrant and native members differently, and (iii) the groups affected by our instrument may differ in composition. The evidence consistently points away from the first explanation: immigrants receive similar information from unions upon joining, hold comparable preferences over work amenities, report similar beliefs about unions' roles and effectiveness, and are even more likely than natives to seek union assistance. The evidence also points away from the third explanation. In particular, the results remain stable when immigrant compliers are reweighted to match native compliers along the worker characteristics most strongly associated with responsiveness to the union subsidy. Similarly, the estimated union premia remain broadly similar across firms with different levels of baseline union density and non-Western immigrant concentration, suggesting that the results are unlikely to be driven primarily by immigrants sorting into firms where unions are systematically less able or willing to generate wage gains. Furthermore, when we examine union density spillovers onto workers who were not directly induced to join unions by the policy, natives tend to benefit marginally more than immigrants from the same increase in firm-level union density.

Taken together, the evidence is most consistent with differences in how unions allocate benefits and representation across workers, rather than differences in member engagement, observable complier composition, or sorting into different types of firms. While the survey evidence is based on observed union members rather than the complier population identified in the administrative IV analysis, it is broadly consistent with this interpretation: many workers believe that unions treat members differently, non-Western immigrants are more likely to believe that unions favour natives, and they report lower satisfaction despite being more likely

to seek help from unions than natives. Taken together, these patterns suggest that internal allocation choices and institutional incentives within unions may contribute to unequal returns to membership across groups.

To shed light on why unions may generate larger gains for natives, we conduct three auxiliary analyses. First, we rule out the idea that unions are simply more effective for the majority group within a firm: immigrants do not experience larger gains when they constitute a greater share of the workforce. Second, we document that more than 90 percent of within-firm union dues are paid by native workers, reflecting both higher unionization rates and higher average earnings. This financial structure may help explain why representation and support tilt toward natives, consistent with survey responses. Third, we show that non-Western immigrants are substantially underrepresented within union leadership and membership organisations, suggesting weaker representation within unions themselves. However, other related mechanisms may also contribute. For example, survey respondents frequently identify workers with stronger personal ties or longer tenure as those receiving the most attention from unions, suggesting that informal network effects could also play a role.

This paper studies how the gains from union membership are distributed among members, focusing on whether the wage, career, and employment protections associated with membership extend equally to natives and immigrants. It is related to a growing body of work using Norwegian register data to study unions, but prior studies examine different margins. Dodini et al. (2025) analyse the incidence of unionization from the firm side, tracing how a higher wage bill is absorbed in firms' decisions, accounts, and prices rather than how benefits are allocated across workers. Dodini et al. (2022) study the balance of power between unions and employers, documenting in which market structures union density is most effective at counteracting employer power at an aggregated wage-setting level. Dodini et al. (2023) use the same reform to estimate long-run impacts of membership on individual careers, but focus on average effects across the workforce. A related paper by Silliman and Willén (2024) study immigrant sorting into firms through an insider–outsider framework, focusing on the access margin after the 2004 EU enlargement. The present paper complements these studies by studying how union rents are distributed across members: using quasi-experimental variation in membership combined with detailed survey evidence, we examine how bargaining support and job protection are distributed across natives and immigrants inside unions. In doing so, we provide new evidence on whether unions reproduce or mitigate native–immigrant inequality within their own membership, speaking directly to debates on immigrant integration, institutional constraints, and the extent to which collective institutions operate as inclusive organisations.

Our paper contributes to several established literatures. First, there is a long-standing literature on the role of unions, the unions' ability to extract rent from employers, and how they affect aggregate measures of inequality and efficiency (e.g., DiNardo and Lee (2004); Lee and Mas (2012); Frandsen (2021); Sojourner et al. (2015); Card and De La Rica (2006); Bryson

(2002); Fortin et al. (2023); Barth et al. (2020); Dodini et al. (2022)). Of particular relevance is Barth et al. (2020), who first used the union dues instrument in the Norwegian setting.

Our contribution to this literature lies in demonstrating that union membership can have distinctly different effects on workers based on their migration status, in terms of both the size of the effect and the specific career dimensions it impacts. Given the differences in individual bargaining power between natives and immigrants, this finding speaks to the nuanced role of unions in shaping distributional outcomes in the labour market. It highlights how the benefits of worker representation are allocated across worker groups and informs broader discussions on inequality and representation.

Second, there is a large and growing literature on the integration of immigrants into domestic labour markets (e.g., Rica et al. (2015); Chin and Cortes (2015); Martín et al. (2016); Becker and Ferrara (2019); Dorn and Zweimüller (2021); Brell et al. (2020)),[2] and the effectiveness of specific integration programmes such as language training, social network facilitation, job search aid and mentoring, and internship and transitional job programmes (e.g., Arendt and Bolvig (2020); Arendt et al. (2020); Lochmann et al. (2019); Battisti et al. (2022); Bratu et al. (2020); Butschek and Walter (2014); Ottosson (2022)).[3] We contribute to this field by offering causal evidence on the role of labour unions in shaping immigrant-native gaps. Although unions have the potential to counteract employer market power and enhance worker outcomes, theory offers no clear prediction about their relative impact across nativity groups, and empirical evidence is extremely limited. Our findings deepen the understanding of how market dynamics and institutions interact to produce inequality across social groups, particularly in labour markets characterised by insider–outsider divisions. Related evidence comes from Turner et al. (2014), who provide descriptive survey evidence on how unions impact natives and immigrants in Ireland. We contribute by using quasi-experimental variation to estimate causal effects and by providing evidence on the mechanisms behind the gaps.

Third, there is a rapidly growing literature that has directly measured labour market concentration and how concentration affects wages and employment (e.g., Schubert et al. (2020); Azar et al. (2020b); Qiu and Sojourner (2019); Rinz (2018); Prager and Schmitt (2021); Azar et al. (2020a); Benmelech et al. (2022); Marinescu et al. (2021); Hershbein et al. (2018); Bassanini et al. (2022); Dodini et al. (2024); Barth et al. (2020)). On average, these studies show that labour market concentration reduces worker wages and has negative effects on workers' careers. A smaller set of studies has shown that unions may counteract the labour market power of employers and reduce the negative effects of imperfect competition in labour markets

[2]For the Nordic region, see Schultz-Nielsen (2017) for Denmark; Sarvimäki (2017) for Finland; Bratsberg et al. (2017) for Norway; and Åslund et al. (2017) for Sweden.

[3]Recent studies have also explored the role of firms in immigrant integration (e.g., Damas de Matos and Parent (2019); Dostie et al. (2023); Arellano-Bover and San (2023)). Most of this literature finds substantial native-immigrant wage gaps across firms, but smaller gaps within firms. Our paper contributes to this discussion by showing that within-firm wage gaps can be sizable and account for a meaningful share of the overall native–immigrant wage disparity, thereby adding to our understanding of firms' roles in immigrant integration.

by equipping workers with additional bargaining power (e.g., Dodini et al. (2022); Azkarate-Askasua and Zerecero (2023). This provides an important policy solution to the increasing market power of firms over the past decades. Our core contribution to this literature is to demonstrate that the countervailing force of unions differs substantially across demographic groups. Specifically, we show that union membership can mitigate the negative labour market effects of concentration for native workers, while providing substantially smaller benefits for immigrants. As a result, relying on unions to address labour market power may unintentionally increase inequality across demographic groups.

Lastly, a nascent literature has focused on monopsony and immigration (e.g., Amior and Manning (2020); Amior and Stuhler (2024)), suggesting that unions may act as a counterweight to employer wage-setting power that is exacerbated by immigration. We show that unions do act as a counterweight to firm wage-setting power (which also is consistent with Dodini et al. (2022) and Dodini et al. (2023)), but that this counterweight is primarily benefiting natives and, to some extent, Western immigrants. The finding highlights that unions may only partially offset labour market distortions linked to employer power and migration.

2 Background

2.1 Immigration in Norway

Over the past five decades, Norway has transitioned from a largely homogeneous to a more diverse society with a substantial immigrant population. Between 1970 and 2020, the number of immigrants increased from about 57,000 to 711,000—roughly 15 percent of the total population.[4] Approximately half have a Western background. Immigrants reside in all municipalities, and while residential segregation persists, it has declined over the past 15 years (Kornstad et al. (2018)). Figure A1 shows that in 2002 both Western and non-Western immigrants were concentrated in a few municipalities, whereas by 2014 most areas had experienced substantial growth in immigrant shares—especially of non-Western immigrants.

Like other OECD countries, Norway has seen a marked shift in immigration patterns over recent decades—from predominantly European inflows to migration from other regions of the world. As a result, immigrants have become increasingly distinct from natives in terms of ethnicity and cultural background, a change that may facilitate discriminatory practices by employers and institutions (e.g., Chiswick and Miller (2005)). While ethnic and racial segregation takes many forms, in Scandinavia it is most clearly reflected in persistent gaps between non-Western immigrants and other groups (e.g., Böhlmark and Willén (2020); Aldén et al. (2015); Grand and Szulkin (2002)).

Figure 1 illustrates the evolution of Norway's immigrant population over time. The share of

[4]The ten largest immigrant groups are from Poland (97,197), Lithuania (37,638), Sweden (36,315), Somalia (28,696), Germany (24,601), Iraq (22,493), Syria (20,823), the Philippines (20,537), Pakistan (19,973), and Eritrea (19,957).

immigrants has more than tripled in the past three decades, rising from below 5 percent to over 15 percent of the total population. This increase almost exclusively comes from non-Western immigrants. However, although immigration increased markedly during this period—especially after the 2004 EU enlargement—the distribution of native employment across industries and occupations remained remarkably stable, and the average immigrant share within firms rose only modestly, from about 8 percent in 2002 to roughly 13 percent in 2010 (Appendix Figure A2). Our empirical design mitigates concerns that these developments affect identification: it exploits variation in exposure to the reform conditional on each worker's baseline workplace in the year before implementation, so newly arrived immigrants entering after the reform do not influence the estimates. The inclusion of fixed effects for both baseline and current firm and occupation–industry cells further ensures that any sorting across firms or occupations is absorbed by the specification.

2.2 Unions in Norway

All workers in Norway have the legal right to join a union if they so wish, but this has to be on a voluntary basis (i.e., closed-shop union agreements are not allowed). Similar to other countries, the primary goal of unions is to improve members' rights and work conditions through collective bargaining. Not only do unions play a central role in wage negotiations, but they also are involved in decisions related to job protection, work environment, work time, and other non-monetary amenities. In addition, they offer mediation and legal help in the event of work disputes.

Norwegian research has documented non-negligible tensions between unions and immigrants (Silliman and Willén, 2024), echoing patterns observed across other OECD countries and in the United States (Tabellini, 2020; Medici, 2024). For instance, in Ireland, survey evidence indicates that Irish nationals benefit more from union membership than immigrants (Turner et al., 2014).[5] These findings suggest that the relationship between union membership and immigrant integration reflects a broader institutional challenge rather than a country-specific feature. Such tensions may arise because immigrants engage with unions differently from natives or because unions allocate representation unevenly—mechanisms we examine directly in our empirical and survey analyses.

Although union density has declined across most OECD countries over the past two decades, the fall in Norway has been comparatively modest.[6] Existing research attributes this largely to the government's union dues subsidy: Barth et al. (2025) estimate that union density would be at least 5 percentage points lower in the absence of this policy. Currently, roughly half

[5]In Canada, unionization appears to do little to close wage gaps for racial minority immigrants (Reitz and Verma, 2004). In the United States, craft unions historically excluded minority workers, widening racial wage gaps (Ashenfelter, 1972; Leigh, 1978). These dynamics mirror "insider–outsider" theories of the labour market, where union members act as insiders and immigrants as outsiders (Lindbeck and Snower, 1989).

[6]Between 2000 and 2020, unionization dropped from 81 to 67 percent in Sweden, from 75 to 67 percent in Denmark, and from 74 to 59 percent in Finland. See https://stats.oecd.org/Index.aspx?DataSetCode=TUD. In Norway, the decline was only about 4 percentage points (from 54 to 50 percent).

of Norwegian workers are union members, with substantial variation across sectors—about 79 percent in the public sector versus 40 percent in the private sector—and industries, ranging from 70 percent in mining to 20 percent in hospitality.

Union involvement in wage bargaining in Norway operates on two levels. At the national level, industry-wide collective agreements set wage floors for each occupation. Unions play a central role in these negotiations, and if talks break down, both sides may resort to industrial action such as strikes or lockouts. These agreements are typically renegotiated every 2-4 years. At the local level, unions bargain directly with employers over establishment- and individual-specific wage adjustments. These annual negotiations determine how total wage increases are distributed across union members, and unions often rank members or convey specific requests on behalf of individuals when bargaining with employers. More than 80 percent of Norwegian firms participate in local bargaining arrangements.[7] Recent work in Norway shows that industry-level wage floors appear to bind for only a small share of workers (about 5-8%), and that wage drift is, on average, about 15% for the typical worker (Bhuller et al., 2022), reflecting the importance of local negotiations above the industry mandates. Indeed, since the late 1990s, local negotiations have accounted for over 70 percent of total negotiated wage growth (Mogstad et al., 2021). Non-union workers have no statutory right to participate in these processes; their pay adjustments are determined unilaterally by employers.

Employers in Norway are legally required to recognise and negotiate with local unions if they are present in the workplace, regardless of their membership size. This stands in contrast to the United States, where union representation in the private sector requires majority support through a workplace election, and to Germany, where entire firms are either covered by a union agreement or not. In Norway, unions can operate within any workplace with at least one member, functioning on top of the national collective agreements already in place. As a result, individual union membership is essential for workers to fully benefit from union representation and services. In this respect, the Norwegian model resembles that of Sweden, Denmark, and the United Kingdom, where both central and local bargaining play central roles and the private, member-specific component of union membership is relatively large (e.g., Dodini et al. (2023)).

2.3 Union Tax Deductions

To join a labour union in Norway, workers must pay monthly membership dues. These dues, common across most unionized labour markets, finance a wide range of union activities, including leadership salaries and benefits, legal assistance, lobbying, strike funds, and member-oriented programmes.

Union dues are set at the annual meeting of the union, and the amount as well as the calculation of dues can vary significantly across different unions. For example, some unions collect a percentage of each worker's pay, while others allow this percentage to fluctuate on a sliding scale

[7] See Table 4.2 in Dale-Olsen et al. (2018).

based on worker earnings, and some set dues at a fixed level. On average, dues typically range from 1-3.5 percent of a worker's monthly pre-tax income. It is not necessarily the case that higher-income individuals pay higher dues. In fact, there is only a weak correlation between the gross amount of dues charged by the union and workers' hourly wages (approximately 0.19 for individual dues paid and 0.03 for the imputed dues measure we use to construct our instrument).[8] Due to these varying approaches, it is clear that workers in higher-paying occupations or industries do not always pay more dues, which explains the widespread identifying variation across occupations and the lack of correlation between base dues levels and income. At the median, native Norwegians and non-Western immigrants pay the same amount of their total pre-tax annual income in gross union dues (1.34% versus 1.35%).

To promote union membership, the Norwegian government provides a tax deduction for union dues. The deduction functions as a direct and highly salient subsidy, automatically applied to workers' annual tax returns. Since the early 2000s, the government has raised the maximum allowable deduction several times, increasing its value by more than 300 percent during the first decade of the 21st century (Panel A of Figure 3). These reforms altered the cost of membership without affecting other aspects of the policy environment, generating sharp and predictable changes in the after-tax price of union membership. In our empirical analysis, we exploit the variation created by these reforms between 2002 and 2010 to instrument for union membership. Because membership is a normal good, a lower price increases take-up by prompting previously non-unionized workers to join once the subsidy expands and leading union members who might otherwise leave a union due to costs to remain in the union.

Importantly, the changes in tax subsidies for union members in Norway led to significant changes in the net price of union membership for some workers but not for others (Barth et al., 2020; Dodini et al., 2022). In particular, these changes only reduced the monetary cost of joining a union for workers whose union dues deduction was previously bounded by the deduction cap. As a result, workers at firms with higher union dues in 2001 experienced a more substantial increase in union dues subsidies compared to those at firms with lower union dues. This creates exogenous variation in predicted union membership status, allowing us to utilize an instrumented difference-in-differences design to examine its causal effect.

The expansions in the tax deductibility of union dues are best understood as part of a broader policy approach to sustaining union participation in Norway, rather than a response to contemporaneous labour market conditions. In Norway's collective bargaining system, maintaining high union density is central to coordinated wage-setting, and the tax deduction has

[8]To provide some examples, in 2024, TEKNA, a professional association for scientists and technologists, charged NOK 4,980 in annual dues to the central union, along with NOK 144-300 in local branch dues. Similarly, NITO, a union for engineers and technologists, set annual dues at NOK 5,340. Other unions, such as Fellesforbundet, charge dues ranging from 1.5% to 2.2% of annual salary, with the exact percentage varying by local branch. Industri Energi, representing workers in petroleum, metals, pharmaceuticals, and chemicals, charges 1.47% of annual salary, with a cap of NOK 692 per month (NOK 8,602 per year), and local unions may impose additional dues.

historically been used as a policy tool to support participation in organized labour. While similar adjustments have been implemented at different points in time, these changes occur infrequently and take the form of discrete national policy shifts rather than continuous or automatic updates (see Barth et al. (2025) for more background).

3 Conceptual Framework

In this section, we outline a simple analytical framework linking union representation to the native–immigrant wage gap. The goal is to fix ideas and provide context for the empirical analysis, rather than to offer a full formal model of bargaining. As described earlier, the Norwegian bargaining system operates in two stages. Here, we abstract from the first stage—treating the national wage floors as fixed—and focus on the local bargaining process, where unions directly influence within-firm wage setting.

We begin by writing down a simple earnings expression for the within-firm market wage of individual i as a function of individual characteristics and union status (abstracting away from any match-specific component):

$$\log(w_i) = \mathbf{X}_i\beta + U_i\gamma + \varepsilon_i \tag{1}$$

where $\mathbf{X}_i$ is a vector of individual characteristics, U_i is a union membership indicator, and ϵ_i is an idiosyncratic error term.

The $X_i\beta$ vector represents the individual-specific wage component and directly links the skills, qualities, and experiences of worker i to the wage compensation received at the firm. For simplicity, we assume these characteristics perfectly predict the worker's outside option value. Workers are assumed to understand and leverage this information, making it a proxy measure of their individual bargaining power at the firm.[9]

The indicator $U_i\gamma$ denotes whether individual i is a union member and should be viewed as the additional bargaining power granted by leveraging a group-level bargaining component of the wage equation. This variable measures how much individual i benefits from the bargaining power that the union membership bestows upon the worker, above and beyond the level of compensation that the individual worker can secure through his/her individual bargaining power. For simplicity, we do not differentiate between individual union membership and firm-level union density, whose relative importance depends on the extent to which unions can be viewed as a public or a private good. Importantly, our stylised results would not change if we adjusted the expression above on this dimension. (We examine this empirically in the results section).

This simple expression provides a helpful foundation for considering how unions affect the

[9]Silliman and Willén (2025) show that workers generally form accurate beliefs about their outside options and adjust their behaviour accordingly. Importantly, there is no reason to expect immigrants to have more accurate perceptions than natives, so the assumption that natives possess stronger individual bargaining power remains plausible.

native–immigrant wage gap and highlights the theoretical ambiguity of the question. Immigrants tend to hold weaker individual bargaining positions than natives, due to factors such as less familiarity with workplace regulations, language barriers, discriminatory hiring practices, and weaker outside options.[10] If so, $\beta_{Immigrant} < \beta_{Native}$, and in the absence of union representation, this gap in individual bargaining power would lead to a native–immigrant wage differential.

Since the marginal benefit of collective bargaining is decreasing in individual bargaining power, group-level bargaining should benefit workers with weaker outside options more. In particular, union membership is unlikely to raise wages for workers with a skill monopoly, who already command high wages on their own, but it can substantially increase wages for workers who face significant markdowns. Thus, if immigrants have weaker individual bargaining power, we would expect $\gamma_{Immigrant} > \gamma_{Native}$, and unions should help narrow the native–immigrant wage gap.

The above reasoning holds only if (1) immigrants and natives use unions similarly, and (2) unions treat both groups equally in bargaining. In practice, neither need hold. Immigrants may interact with unions differently, and unions may prioritise natives. In such cases, $\gamma_{Immigrant} < \gamma_{Native}$, and union membership could widen rather than reduce inequality.[11]

The discussion above highlights the theoretical ambiguity surrounding unions' effects on the native–immigrant wage gap. If both groups use unions similarly and unions are both willing and able to represent them equally, then $\gamma_{Immigrant} > \gamma_{Native}$ and union membership should narrow the gap. Conversely, if unions are unable or unwilling to extend equal benefits, then $\gamma_{Immigrant} < \gamma_{Native}$ and union membership will instead widen it.[12]

4 Data

We use population-wide administrative data from multiple registers managed by Statistics Norway, covering all individuals aged 16–74 between 2001 and 2015. The population register provides information on demographics (gender, age, education, marital status, country of origin, migration year, and location). These data are linked to matched employer–employee files, which include information on employers, industries, occupations (three-digit level), establishments, and contractual hours. For years prior to 2015, when exact work hours are unavailable, we convert categorical hours to continuous measures using midpoints, assigning 37 hours to the top category (30+), based on the post-2015 distribution.

We use the matched data to construct two promotion measures: (i) a within-firm indicator

[10] E.g., Algan et al. (2010); Chiswick (1978); Lehmer and Ludsteck (2011); Åslund and Skans (2010); Cutler et al. (2008); Hirsch and Jahn (2015).

[11] This relates to classic debates on union behaviour—whether unions follow a median-voter model (Booth, 1995) or a hierarchical structure where leaders' interests diverge from members' (Pemberton, 1988). In some contexts, unions may also reinforce discrimination (Ashenfelter, 1972).

[12] For simplicity, we abstract from differences between Western and non-Western immigrants. Incorporating this distinction would imply that Western immigrants, who share more institutional familiarity with natives, fall between natives and non-Western immigrants in terms of baseline bargaining power.

for upward occupational moves[13], and (ii) a cross-firm indicator for moving to higher-paying firms. We then merge these data with tax and transfer registers containing annual earnings, unemployment insurance (UI), and sick leave (SL) benefits, as well as information on union membership and dues payments.

We measure employment protection using unemployment insurance (UI) receipt. In Norway, UI is only available to workers who lose at least 50 percent of their hours, are registered as job seekers, and meet minimum earnings requirements. Voluntary separations are ineligible. As a result, UI receipt serves as a helpful proxy for involuntary job loss and employment protection.

We measure workplace environment using medically certified sick-leave absences. Sick leave compensates for income loss due to temporary illness or injury at a replacement rate of 100 percent from day one (up to a ceiling of roughly $62,000 in 2019). Eligibility requires at least four weeks of prior employment, and absences longer than three days must be certified by a physician (or chiropractor for musculoskeletal injuries). Because coverage is universal and standardised, certified sick-leave days provide a consistent and comparable measure of medically certified absences across workers and firms. Union membership could reduce such absences if it improves workplace conditions, or increase them if workers feel more secure taking certified leave when needed. We estimate the reduced-form effect on certified sick-leave days, capturing the net outcome across these channels.

We restrict the sample to workers averaging at least 20 weekly hours and earning above one basic amount (*1G*, approximately NOK 90,000 in 2015).[14] These restrictions remove marginally attached workers and ensure meaningful labour market engagement.

We divide workers into three groups: natives, Western immigrants, and non-Western immigrants. Following prior research on this topic in Scandinavia, we define Western immigrants as immigrants with backgrounds in Sweden, Denmark, Finland, Iceland, Belgium, France, Ireland, Luxembourg, the Netherlands, Great Britain and Northern Ireland, Germany, Austria, Switzerland, Israel, the United States, Canada or Oceania (Böhlmark and Willén (2020); Aldén et al. (2015); Korpi et al. (2023)). Non-Western immigrants, therefore, constitute all migrants coming from countries other than those listed above. We will also show results using other groupings of immigrants.

Descriptive statistics are shown in Table 1. Non-Western immigrants are younger, more likely to be male, and have fewer children than natives and Western immigrants. They are also less educated: 37 percent have a college degree (versus 39 percent for natives and 55 percent for Western immigrants), and nearly 30 percent have less than high school (versus 17 and 13 percent). These differences translate into sizable labour market gaps. Non-Western immigrants

[13]Specifically, we generate an indicator variable that takes the value of one if a worker shifts to an occupation located higher in the earnings distribution. Since we include firm fixed effects in our main empirical specification, this outcome examines the impact of union membership on within-firm vertical occupation moves.

[14]The "1G" designation (also called *Grunnbeløpet*), is used to calculate whether individuals qualify for certain government welfare payments and transfers, and how large those payments should be.

earn about 80 percent of native earnings, while Western immigrants earn roughly 9 percent more. They are more likely to collect unemployment benefits, whereas certified sick-leave receipt is similar across groups. Natives work in firms with higher union density and higher native shares, consistent with sorting across industries and occupations (Figure 2). Finally, union membership rates differ substantially: more than half of natives are members, compared with 34 percent of non-Western immigrants and 37 percent of Western immigrants.[15]

We complement the administrative data with two surveys. The first, fielded by Norstat on 5,000 workers, is screened to ensure national representativeness on baseline demographics and collects information on immigrant background, union membership, preferences over career amenities, beliefs about union influence, and price sensitivity to membership (vignette experiments randomised across NOK 500–2,500 intervals). The second survey, fielded by Ipsos, targeting about 1,000 union members (screened on migration status to ensure sufficient sample size), focuses on how workers interact with unions—what information they receive, how satisfied they are, and whether they perceive unions to prioritise certain groups. This survey was implemented after the completion of the register-based data analysis. Taken together, the surveys provide evidence on what workers know about unions, how they engage with them, and how they perceive the support they receive.[16] Note that the large worker survey was designed in collaboration with Norstat to support two related research projects (this one as well as Dodini et al. (2023), while the smaller survey—fielded with IPSOS—was conducted specifically for the present study).[17] Importantly, the survey evidence is based on observed union members rather than the complier population identified in the administrative IV analysis. As such, the survey results should not be interpreted causally or extrapolated directly to the marginal workers induced to join unions by the policy. Instead, they provide complementary evidence that helps interpret the patterns observed in the register-based results.

5 Method

5.1 Overview and Intuition

Identifying the causal effect of union membership is challenging because joining is not random: workers self-select based on characteristics that also influence labour-market outcomes, generating standard omitted-variable concerns. We address this issue using exogenous changes in the effective price of union membership created by government reforms to the union-dues tax deduction. When dues fall, membership becomes more attractive to non-members. Because the size of the deduction is set nationally rather than at the individual or firm level, the resulting price changes are unrelated to workers' characteristics or labour-market trajectories.

[15] When estimating causal models we draw a 50% random subsample to reduce computational burden from tens of thousands of fixed effects on 24 million observations.

[16] NHH IRB 42/22.

[17] Simple descriptive statistics for survey one are in Appendix Table A1.

We therefore use these policy-induced changes as an instrument to identify the causal effect of union membership on wages, job stability, and related outcomes.

The identifying variation comes from national reforms to the government-mandated tax deduction for union dues. Between 2002 and 2010, the maximum deduction increased by more than 300 percent, substantially reducing the net cost of union membership. Crucially, these reforms only affected workers whose union dues were high enough to be constrained by the deduction cap, while leaving unaffected those whose dues were below it. Consequently, individuals in firms with higher pre-reform dues (2003) experienced a much larger effective price reduction than those in low-dues firms. This structure allows us to implement an instrumented difference-in-differences design that compares changes in unionization over time between high- and low-subsidy firms, leveraging the differential "subsidy bite" to identify causal effects on union membership and downstream labour market outcomes.

Figure 3 illustrates the policy variation underlying our identification strategy. Panel A shows the evolution of the maximum tax deduction for union dues, which rose from 900 NOK to over 3,800 NOK between 2002 and 2010. Panel B plots the relationship between base (pre-deduction) dues and net dues after the subsidy.

Increases in the maximum deduction cap may affect workers differently depending on their prior dues in three ways (Panel B). First, workers with dues below the old cap ($D < c_0$) experienced no change, as illustrated by those with base dues under 900 NOK in 2002. Second, those with dues between the old and new caps ($c_0 < D < c_1$) saw their net dues fall by $\tau(D - c_0)$—the sloped gap opening between base and net dues. Third, workers with dues above the new cap ($D > c_1$) received a fixed reduction of $\tau(c_1 - c_0)$, reflected by the constant vertical gap. As the deduction cap increased over time, this vertical gap widened, reducing the net cost of membership most strongly for workers facing higher base dues. Panels C and D confirm that these reforms substantially lowered the effective price of union membership. Panel C shows the decline in net dues relative to income (the mean within each group), while Panel D illustrates the corresponding rise in subsidy generosity relative to the total membership cost.

Our research design can be illustrated with two otherwise identical non-unionized workers (A and B) of the same age, in the same occupation–industry cell and year, but employed by different firms. Worker A's firm has union dues high enough that the subsidy is capped, whereas Worker B's firm faces dues below the cap. When the government raises the maximum tax deduction, the reform reduces the effective cost of membership for workers at Firm A but not Firm B in that year. If union membership is a normal good, workers at Firm A should therefore be more likely to join. We exploit this exogenous, firm-level variation in the cost of joining to identify the causal effect of union membership. Our main comparison of interest—the immigrant–native gap—is estimated within firms, holding constant firm characteristics and market conditions. By comparing the change in unionization for natives at Firm A versus Firm B to the corresponding change for immigrants, we isolate the within-firm change in the native–immigrant gap induced

by the policy.

5.2 Empirical Implementation

To implement our empirical approach, we begin by calculating the hypothetical cost of joining a union for workers who are currently not members.

We calculate this hypothetical cost by taking the mean union dues paid by workers in each occupation-industry cell each year and applying this to all non-members. This approach is designed to capture the average cost of union membership within local unions, as unions are often defined by occupation and industry. Using the mean membership cost within these cells provides an accurate representation of what a non-member would have to pay, on average, in order to join the union, and thereby captures the variation in union dues costs across local unions. We rely exclusively on mean dues within these cells when calculating the counterfactual cost to ensure the exogeneity of the instrument.[18]

To ensure that all workers are treated equally and to abstract from any information on individual union dues or wages that may be endogenously determined by individual or firm characteristics, we apply this imputation approach to union members as well.[19] We then define the union dues of the firm as the average imputed union dues across all workers at the firm.

We fix each worker's imputed union dues, $\overline{D^0_{f_b}}$, at the first firm in which they appear in their first year in the data (2001 for the majority of workers in our sample). The reason for doing this is that firms and unions may endogenously respond to the subsidy legislation by altering the occupations they decide to employ or by changing the union dues directly. Individual union membership may also change the likelihood that workers switch firms. By fixing each worker's imputed union dues at their first year in the data, we remove the risk of these potential biases breaking the exogeneity of our instrument. We adjust the imputed union dues forward using the Consumer Price Index in Norway to nominal NOK, yielding $\overline{D^0_{f_b t}}$, which represents the nominal cost of joining a union for a worker in base firm f_b in year t if that firm's imputed dues grew at the same rate as overall price levels. This adjustment ensures that changes in the instrument reflect changes in the statutory deduction cap relative to workers' nominal union dues, rather than general price inflation.

The subsidy is equal to the lesser of the legislated maximum deduction ($MaxDeduction_t$) and the worker's imputed union dues ($\overline{D^0_{f_b t}}$) multiplied by the applicable tax rate:[20]

$$Subsidy_{f_b t} = T_t \times (min\{\overline{D^0_{f_b t}}, MaxDeduction_t\}) \ , \tag{2}$$

[18] Incorporating individual characteristics would risk endogeneity in the counterfactual dues calculation, as they may correlate with unobserved factors affecting union membership and labour market outcomes.

[19] The correlation between the imputed and non-imputed union dues for workers who are members is close to 1.

[20] To isolate changes in the guaranteed *statutory* subsidy from changes in the *realized* subsidy that may depend on marginal tax rates, we multiply the subsidy value by the country's base tax rate (28 percent from 2001 to 2013 and 27 percent from 2014 onward)

where T_t is the base tax rate in year t.

The identifying variation in our subsidy measure comes from differences in the occupation-industry mix of the firm in each worker's base year (which generates differential exposure to the raising of the cap due to differences in the level of the pre-increase union dues) combined with changes in the legislated maximum deduction over time (which more than quadrupled over 12 years and generated several sequential increases).

High union dues may not be randomly assigned across firms, and baseline differences could in principle correlate with our outcomes of interest. Empirically, however, these differences are very small: Appendix Table A3 shows only weak correlations between baseline firm characteristics and the instrument. Moreover, identification does not require high- and low-dues firms to be identical in levels. Our empirical strategy is an instrumented difference-in-differences that compares changes over time as a function of the subsidy bite.[21] We, therefore, do not need these firms to be identical in the base year – we only need them to trend similarly to each other absent the policy shift. We show strong supportive evidence of this assumption in the next subsection.[22]

Having obtained our subsidy measure, we calculate the net-of-subsidy union dues by subtracting the value of the subsidy from the gross imputed baseline union dues ($ND_{f_bt} = \overline{D^0_{f_bt}} - Subsidy_{f_bt}$). This measure changes within a worker's base firm over time only through the legislated subsidy channel and represents our instrument. Using this instrument, we estimate the following equations:

$$y_{iocf,t+1} = \alpha + \beta_1 \hat{U}_{iocft} + \beta_2[\hat{U}_{iocft} \times WI_i] + \beta_3[\hat{U}_{iocft} \times NWI_i] + \beta_4 WI_i + \beta_5 NWI_i + \gamma_t + \zeta_{a_b} + \eta_a + \iota_{oc_b} + \kappa_{oc} + \lambda_{f_b} + \phi_f + \delta_{i\bar{U}} + \epsilon_{iocft}, \tag{3}$$

$$U_{iocft} = \tau + \pi_1 ND_{f_bt} + \pi_2[ND_{f_bt} \times WI_i] + \pi_3[ND_{f_bt} \times NWI_i] + \pi_4 WI_i + \pi_5 NWI_i + \gamma_t + \zeta_{a_b} + \eta_a + \iota_{oc_b} + \kappa_{oc} + \lambda_{f_b} + \phi_f + \delta_{i\bar{U}} + \mu_{iocft}, \tag{4}$$

where Equation 4 represents the first-stage and Equation 3 represents the second-stage.

In Equation 4, U_{iocft} is the union membership status of individual worker i in occupation-industry cell oc and firm f in year t. The instrument, ND_{f_bt} is assigned to individuals as

[21] While one could classify firms as capped or uncapped in each year, this binary measure shifts mechanically as the deduction cap rises, groups together firms experiencing very different price changes, and discards continuous variation within the uncapped range (the $\tau(c_1 - c_0)$ segments above). Our approach, which compares high- and low-dues firms based on pre-reform dues levels, captures the full intensity of exposure and can be interpreted as a dose–response difference-in-differences design.

[22] In Panel B of Table A3, we show the union membership rates by immigration status and industry at baseline (2001-2002). Lower levels of baseline union membership for non-Western immigrants persist across all major industry groups. While this is not directly relevant for identification, it does help provide context. In Panel C, we estimate placebo regression in which we use the instrument value lagged by six years as the instrument for union membership and outcomes before 2003. In short, this determines if future exposure to the instrument predicts union membership or outcomes. We find clear evidence that this is not the case.

described above. We interact the instrument with dummies for whether the individual is a Western immigrant (WI_i) or non-Western immigrant (NWI_i).[23] The immigrant coefficients reflect how the impact of union membership differs from the native baseline, so they capture relative rather than absolute effects. Statistical significance in these coefficients indicates that the effect for the immigrant group differs significantly from the corresponding effect for natives.

We include fixed effects for both current as well as baseline characteristics. Specifically, t represents year fixed effects, a (a_b) denotes age (baseline age) fixed effects, oc (oc_b) signifies occupation-industry (at baseline) fixed effects, and f (f_b) indicates firm (at baseline) fixed effects.[24]

The $\delta_{i\bar{U}}$ coefficient is an indicator for whether the worker was an "always-taker" (i.e., a union member throughout our sample period). We account for always-taker status for two reasons. First, always-takers are employed in the same firms, occupations-industry cells, and years as marginal union members and contribute to variation in the fixed effects for all of these cells. Second, while always-takers contribute to variation in the fixed effects, they contribute nothing to identification because there is no variation in union membership among this group.[25]

In Equation 3, $y_{iocf,t+1}$ represents an outcome of interest for individual i at time $t+1$ and β_1 measures the average effect of union membership for all workers on that outcome using the net union dues ND_{f_bt} as an instrument. β_2 and β_3 are the estimates of interest and measure the differential effect of union membership depending on worker i's immigration status within the same firm, industry-occupation, and age cells. We measure outcomes in $t+1$ to capture the effect of the union with a full year of membership, as individuals could choose to join a union partway through the year. All fixed effects included in Equation 3 are also included in Equation 4. We cluster the standard errors at the individual worker level, the level of treatment assignment.[26]

5.3 Assumptions and Threats to Identification

Our estimation strategy relies on four key assumptions. First, workers in low-exposure firms must provide a credible counterfactual for those in high-exposure firms absent the subsidy changes (common trends). Second, workers must respond to changes in the price of union membership (relevance). Third, the subsidies must affect outcomes only through their impact on union membership (exclusion). Finally, there can be no defiers (monotonicity).

The relevance assumption is strongly supported by the data: workers respond sharply to

[23] Following Wooldridge (2010), the interactions between each nativity group (i.e., Western and Non-Western immigrants) and the net dues (ND_{f_bt}) serve as instruments for the interaction between nativity group and union membership, resulting in three combined instruments for three endogenous treatments.

[24] The results are also robust to educational controls (Appendix Table A6).

[25] Not accounting for always-takers means that the estimated first-stage coefficient of the instrument will be smaller because there is no variation in the union membership choice of always-takers, leading to larger and less precise second-stage estimates.

[26] Because treatment is individual and instrument variation arises at the base-firm level, the choice of clustering level is not obvious. Results are robust to clustering at the firm level (Table A2).

price changes, as shown in the next section and in the survey's hypothetical price-change experiment. The exclusion restriction cannot be tested directly, but since the subsidies were implemented nationally and the identifying variation stems from pre-existing firm differences, we can think of no alternative channel linking the policy to worker outcomes. Monotonicity is also untestable, but would only fail if union membership were a Giffen good at certain prices - a scenario we find unlikely.

Regarding the common-trends assumption, Figure 4 plots trends in union membership and earnings for native and immigrant workers in firms with high versus low subsidy exposure. Panels A and B, respectively, show raw trends when controlling for base firm fixed effects, which determines exposure. Panels C and D are event study analogs of Panels A and B and more formally measure the relative gaps between the top quartile of exposure in comparison to the bottom within each nativity group while controlling for nativity group-specific time dummies. These event studies show that trends in union membership and earnings were parallel within groups before the policy change, that union membership and earnings gradually evolved as policy exposure grew, and that immigrants (including Western and non-Western immigrants) were more responsive, on average, to the cumulative change in net dues in their union membership choices. Panel D also shows that despite a greater union membership response, the cumulative earnings changes were typically smaller and noisier for immigrants than for natives, implying lower returns to union membership. The earnings gap in Panel B between immigrants and natives widened over the period, most notably for high-exposure firms. This divergence—strong wage gains for natives despite smaller membership increases—suggests that union membership may have amplified earnings inequality between the two groups. Both gaps stabilized around 2010, coinciding with the end of large subsidy increases. This provides strong suggestive support for the common trends assumption. We have also formally tested for differential pre-trends by examining the joint significance of the pre-reform coefficients (1998–2001) in the event study specifications. The corresponding p-values for immigrants in Panels C and D are 0.46 and 0.42, respectively, providing no evidence of systematic differences in pre-trends across exposure groups.

When examining this figure, two things are worth noting. First, not all variables in our combined registers are available prior to 2001 (e.g., occupation), so we cannot estimate a full event-study model with contemporaneous controls. The fact that parallel trends hold even without the addition of a large set of controls supports our identification strategy and rules out strong differential sorting of immigrant groups into high-exposure status based on firm. A key reason we see clear parallel trends—even without the rich fixed effects in the main specification—is that high- and low-exposure firms are very similar at baseline (Appendix Table A3). Second, our main interest lies in the differential effect of union enrolment between natives and immigrants. This means the common-trends assumption is in fact stronger than required, since any violation would also have to differ systematically by nativity to bias our results.

To further assess this empirically, we perform a placebo test that shifts the instrument six years backward in time to examine whether later exposure predicts outcomes prior to 2002. As shown in Panel C of Table A3, the placebo first-stage F-statistic is only 3.79—well below conventional thresholds—indicating no relationship between future exposure and pre-reform union membership or earnings.

Before turning to the main results, it is useful to clarify that our estimates capture the group-specific local average treatment effect (LATE) for compliers—those who joined a union in response to the subsidy-induced reduction in membership costs, conditional on their baseline firm. Differences in estimated effects across nativity groups could therefore reflect either (1) heterogeneity in treatment effects or (2) differences in the composition of compliers across groups. To interpret our findings, we first examine compliance rates by nativity and across relevant subgroups within each classification.

Our instrument is continuous, which requires a slight adaptation from the standard binary case. Following Dahl et al. (2014), we first residualize the one-year change in the instrument for each worker on all fixed effects. We then estimate a first-stage regression of the one-year change in union membership on the one-year change in this residualized instrument (Z in (Dahl et al., 2014)). To quantify the first-stage compliance rate, we compare predicted treatment take-up between the 1st and 99th percentiles of the residualized change in net dues—effectively identifying the workers most and least exposed to the subsidy. We repeat this same exercise for different subgroups within each immigrant group (immigrant status and immigrant status intersected with other demographic identifiers), including across income quartiles.

We report the results of this exercise in Table A4. The goal of the income-based compliance rates in Panel B is to show that immigrants and natives are not drawn from systematically different parts of the income distribution. Panel A presents one-year compliance rates across demographic and job characteristics. One-year compliance is slightly higher for natives than immigrants, for women than men, for public- than private-sector workers, and for higher- than lower-educated workers. However, these patterns do not consistently differ by nativity. For instance, compliance rates among non-Western immigrants with low or high education closely match those of natives, while middle-educated non-Western immigrants comply only marginally less. Compliance rates are also similar across the income distribution with a small difference being that, while compliance is fairly constant over income for NWIs, native Norwegians have higher compliance rates in the lowest earnings quartile.

Overall, the substantial overlap in compliance rates across demographic, tenure, sectoral, and income subgroups indicates that compliers are broadly comparable across nativity groups. To assess this more formally, we implement a propensity-score reweighting exercise that reweighs immigrant compliers to match the characteristics of native compliers along education, sector, and income (see Section 7.3 and Appendix Table A5). These reweighting variables differ from the high-dimensional fixed effects used in the OLS specifications, as they explicitly tar-

get the worker characteristics associated with native responsiveness to the union subsidy. The reweighted estimates are nearly identical to our baseline results, confirming that differences in complier composition do not drive the heterogeneous effects of union membership by immigration status.

6 Results

6.1 First-stage

Table 2 shows the effect of our instrument on the probability that workers enroll in unions. The first row shows that a 1,000 NOK decrease in net union dues via the tax deduction raises the probability that a native worker joins a union by 11 percentage points. This corresponds to a sizable price elasticity of union membership among marginal workers in Norway, and closely matches the responsiveness reported in our survey of Norwegian workers.[27]

Rows 2 and 3 show that the price elasticity of union membership for marginal Western immigrants is the same as that of natives, while that for marginal non-Western immigrants is slightly higher. That the price elasticity of union membership is relatively similar across nativity status is interesting in light of the large differences in baseline union enrolment shown above, but helps reinforce the idea that the subsidy change had a large effect on workers' propensity to unionize, irrespective of migration background. The last row of Table 2 reports the Kleibergen–Paap F-statistic, which exceeds 150, providing strong support for the instrument's relevance for causal inference.

To validate the interpretation of our first-stage results, we use survey evidence that directly measures whether cost deters union membership and how workers would respond to changes in dues. Figure 5 shows that cost is among the most common barriers to joining a union: 45 percent of non-Western immigrants and 32 percent of natives list dues as their primary reason for non-membership. Figure 6 further shows substantial price sensitivity. More than half of current members report that a 500 NOK increase in dues would lead them to reconsider membership, and roughly 40 percent of non-members would join under an equivalent reduction. Sensitivity is strongest among non-Western immigrants. Taken together, these patterns are consistent with the tax-deduction reforms increasing union membership by lowering the effective price of joining, especially for immigrants.[28] Price sensitivity is highest among non-Western immigrants.[29]

[27]Specifically, the survey shows that 35% of the non-unionized native workforce would consider joining a union if the dues fell by 500-2,500 NOK per month.

[28]The hypothetical change was randomised between 500 and 2,500 NOK in 500 NOK increments.

[29]There are some interesting differences in price sensitivity across demographic subgroups in Figure A4. Across all nativity groups, younger respondents, respondents with a college degree, and those in the private sector are more cost sensitive than their counterparts outside these groups. The gaps across these subgroups are larger for NWIs, particularly when it comes to age. However, when we control for each of these categories (age group, public sector, and education), the average NWI is still 9.9 percentage points (s.e. = 3.3) more likely than Norwegians to reconsider their union membership if the price of union membership increased/decreased by at least 500 NOK. The average Western immigrant is 3.65 percentage points (s.e. = 3.3) more sensitive than Norwegians. This means that demographic composition differences across respondents do not explain the differences in cost sensitivity between immigrants and natives.

6.2 OLS Benchmarks and Selection into Union Membership

To help illustrate the importance of accounting for selection into union membership, Table A7 reports the full set of results from conventional OLS models. These estimates show that union membership is positively associated with wages, hours, and promotions, with especially large coefficients for non-Western immigrants. For unemployment benefits, the associations are smaller, while for sick leave, union members appear more likely to be on leave, again particularly among non-Western immigrants. These patterns suggest that immigrants appear to benefit disproportionately from union membership in simple correlations. As we show below, once we exploit the quasi-experimental variation from the union dues reforms, these apparent advantages disappear, highlighting non-random selection into membership. This non-random, positive selection appears more significant for non-Western immigrants, which underscores the importance of our IV strategy for identifying causal effects.[30]

6.3 Compensation

Columns (1) and (2) of Table 3 report the effects of union membership on total labour earnings and hours worked for natives, Western immigrants, and non-Western immigrants, based on Equations 4 and 3. Several patterns emerge. First, consistent with prior evidence on the union wage premium, we find a significant earnings effect of roughly 0.10 log points for native workers (row 1, Column (1)).

Second, the union wage premium varies markedly by migration background. While native workers gain roughly 0.115 log points, Western immigrants about 0.05 log points, and non-Western immigrants experience no short-term wage benefit from joining a union. These differences imply that unions widen the native–immigrant wage gap and, in doing so, exacerbate inequality between the groups.

Third, the differential wage effects across natives and immigrants do not appear to stem from differences in hours worked. Row 1 of Column (2) shows that union membership increases weekly hours by about four among natives, while rows 2 and 3 indicate similar gains for Western and non-Western immigrants. Hence, the wage premium in Column (1) is not driven by a redistribution of work hours across groups.[31]

While these results indicate that differences in hours cannot explain the wage premium, they also suggest a modest overall rise in contractual hours with no accompanying wage increase for non-Western immigrants. One possible interpretation is that this reflects the formaliza-

[30] In Table A8, we show evidence from our survey sample that non-Western immigrants who are union members may be positively selected on certain characteristics. NWI union members tend to be younger than their native or Western immigrant counterparts with slightly less work experience. However, they are also located in higher-earning regions of Norway (particularly Oslo), are just as likely to be working full-time, are less likely to be working in lower-earning industries such as construction and trade, and are more likely to work in health, other services, education, information/communication, and public administration.

[31] Using hourly wages yields a premium of about 0.06 for natives and 0.00 for non-Western immigrants. As noted in the data section, however, the hours variable is imputed from categories and reflects contractual rather than actual hours, so these estimates should be interpreted with caution.

tion of work arrangements—previously unpaid or irregular overtime becoming part of regular contracts—rather than a genuine increase in total labour supplied, which would explain why earnings do not rise proportionally.

To interpret the magnitude of these effects, it is useful to relate the earnings gains to the policy-induced change in the cost of union membership. The first-stage estimates imply that a NOK 1,000 reduction in net union dues increases union enrolment by approximately 11 percentage points for native workers. Combining this with the IV estimates implies that such a cost reduction raises average earnings by about 0.011 log points (1.1 percent) per 1,000 NOK reduction in net dues. Given average annual earnings of roughly NOK 400,000–420,000, this corresponds to an increase of approximately NOK 4,400–4,600 per year—several times larger than the underlying change in membership costs.

6.4 Worker Protection

Column (3) of Table 3 reports the effect of union membership on unemployment benefits. We use information on UI as a way to identify if individuals are laid off from work or not (workers who voluntarily exit a job do not qualify for UI), enabling us to better understand the job protection benefits of unions and to what extent they differ by nativity status.

Row 1 shows that union membership reduces UI benefits among natives by about 15,000 NOK (USD 1,500), indicating stronger employment protection. Western immigrants experience a smaller reduction of roughly 12,900 NOK, suggesting weaker protection. Thus, similar to our findings for the compensation effects of unions, Western immigrants appear to benefit less from union take-up than natives.

For non-Western immigrants, the effect is much smaller—about 8,000 NOK. This result either suggests that unions prioritise the compensation and protection of natives over immigrants in the negotiation process with firms or that the unions are unable to offer the same benefits to immigrants due to other reasons (e.g., less leverage over firms with respect to these workers, for example due to shorter tenure).

6.5 Work Environment

Column (4) of Table 3 provides estimates of the union membership effect on the work environment of employees, which we proxy with sick leave usage of individuals. For natives, the coefficient is negative but statistically not significant, indicating no measurable change in sick leave following union enrolment. Interpreting sick leave as a proxy for work environment quality, this suggests that unions have little effect on natives' working conditions on an individual basis.[32]

The point estimate in row 2 of Column (4) suggests that unions affect the sick leave usage of Western immigrants to an even lesser extent than natives, with a point estimate of only

[32]Dodini et al. (2023) document substantial heterogeneity in the sick-leave effects of unions across worker age, which is beyond the scope of this paper.

about 134 NOK (combining the $Union$ and $Union * WesterImmigrant$ coefficients). However, neither the effect on natives nor Western immigrants is statistically significantly different from zero.

For non-Western immigrants, row 3 of Column (4) shows a substantial decline in sick leave benefits following union enrolment. We interpret this pattern as suggesting that unions may help improve the work environment of non-Western immigrants, reducing their reliance on the sick leave system. That this effect arises only for non-Western immigrants may reflect their lower baseline bargaining power and greater exposure to exploitative or hazardous work conditions. The application of group-level bargaining leverage and union protection may therefore have a larger positive marginal effect on their work environment. However, we emphasize that this explanation is speculative and not one that we can fully disentangle in the data.

6.6 Career Progression

Regarding career progression, rows 1 in Columns (5) and (6) show that union membership increases the probability that a native worker is promoted to a higher-paying position but reduces the likelihood of switching to a higher-paying firm. The magnitude of these promotion and lock-in effects is similar for Western and non-Western immigrants. This indicates that the differential wage gains associated with union membership are not driven by unions helping natives advance faster in their careers, but rather by unions securing different wage benefits for natives and immigrants who experience comparable upward mobility within firms.[33]

7 Extensions

7.1 Interaction with labour Market Power

A union's ability to extract rents from firms and redistribute them to members depends on two main factors: (1) the presence of firm rents and (2) the union's bargaining power. Both are closely tied to employer monopsony power. Because rents and bargaining leverage vary with labour market concentration, the effects of union membership—and its distribution across natives and immigrants—may differ substantially across markets (Dodini et al., 2022). Given the steady rise in market concentration across the OECD, understanding how union effects vary by concentration is of independent interest.

Table 4 presents results from our baseline model interacting union membership with an indicator for high labour market concentration (above-median HHI). To aid interpretation, we report the marginal effects of union membership by concentration level (the underlying coefficients appear in Table A9).[34]

[33] In our data, the average promoted worker receives a pay raise that is 5.6 log points more in annual earnings compared to their non-promoted counterparts when they are promoted. A ten percentage point increase in the probability of promotion, at most, predicts a 0.56% average increase in earnings through this channel.

[34] HHI is calculated as the sum of squared firm employment shares within each occupation–local labour market cell and ranges from 0 (perfect competition) to 1 (monopsony). Each firm's HHI is fixed at its first observation year. The median HHI across Norwegian local labour markets is approximately 0.05.

Two patterns stand out. First, union benefits are considerably larger in concentrated markets, while there are no statistically significant private returns to individual union membership in competitive markets, consistent with unions extracting rents where employer power is high (Dodini et al., 2022). Second, the native–immigrant gap widens with concentration: union membership disproportionately benefits natives (and, to a lesser degree, Western immigrants) in concentrated markets. Thus, while union membership counteracts employer power, it does so unevenly—reducing monopsony distortions for some groups while amplifying wage inequality across others.[35]

7.2 Expanded Immigrant Categorizations

To further explore whether proximity and similarity to the host country shape these effects, we subdivide Western immigrants into those from Scandinavia and those from outside the region. Scandinavian migrants are culturally, institutionally, and linguistically closest to Norwegians. Across all outcomes in Table A12, the benefits of union membership decline monotonically with distance from the host country: Scandinavian immigrants gain almost as much as natives, non-Scandinavian Western immigrants gain somewhat less, and non-Western immigrants gain the least. This gradient is consistent with the idea that workers who are closer to natives in language, networks, or institutional familiarity are better able to capitalize on union representation.

7.3 Why Do Union Returns Differ for Immigrants and Natives?

What explains the gap in union returns between natives and immigrants? In this section, we use survey and administrative data to shed light on the underlying mechanisms. We organize the discussion around three questions: (1) Do immigrants and natives interact differently with unions? (2) Do unions treat immigrant members differently or prioritise native members more? and (3) Could differences in worker selection and complier composition explain the observed gap?

It is important to note that the survey evidence is based on observed union members rather than the marginal workers induced to join by the policy, and therefore does not identify the same population as the IV estimates. We use the surveys to complement the administrative analysis and to help interpret the patterns in the causal estimates, rather than as stand-alone evidence on mechanisms.

Taken together, the evidence from this section suggests that although we cannot isolate a single mechanism, the combined evidence points toward union-side explanations. Immigrants

[35] One concern with our specification is that there could plausibly be immigration group-specific time-varying shocks that bias our estimates. To examine this, we perform a supplemental analysis in which we include immigration status by year fixed effects to all analyses. The results for the baseline model with additional fixed effects are shown in Table A10. Overall, the results do not significantly change from the baseline specification quantitatively or qualitatively. Similarly, the results for the marginal effect of union membership by labour market concentration with additional fixed effects are shown in Table A11, and they do not differ meaningfully from the baseline model.

receive similar information, engage at least as actively with unions, and hold comparable beliefs about union influence and the nature of union-provided benefits. Yet they report lower satisfaction after seeking support, even in domains where they are more likely to engage. At the same time, administrative evidence shows that natives contribute the majority of union dues and tend to have longer tenure and stronger institutional ties within firms and unions. Non-Western immigrants are also substantially underrepresented within union leadership and membership organisations, consistent with immigrants having weaker representation within unions themselves. In addition, the immigrant–native gap in returns persists after reweighting workers along observable characteristics associated with responsiveness to the union subsidy and after accounting for sorting across firms with different union density and immigrant composition. These patterns are difficult to reconcile with differences in worker engagement, observable complier composition, or firm sorting alone and are most consistent with unions allocating representation and bargaining effort in ways that disproportionately benefit native members. In this sense, the current institutional structure appears to reproduce, rather than narrow, underlying immigrant–native disparities in labour market outcomes.

Do Immigrants and Natives Use Unions Differently? If immigrants received less information or engaged less with unions, this could explain lower returns. To examine this in detail, we field a second survey asking workers detailed questions related to how they interact with unions. Overall, we find little evidence suggesting that differences in the use or interaction with unions drive the wage gap we observe.

We begin by examining whether immigrants and natives receive different information from unions upon joining. Figure 7 shows that there is little variation across nativity groups in reported information provision on dimensions such as salary, job security, promotions, and work environment. This suggests that differential access to information is unlikely to explain the immigrant-native union gaps observed in the administrative data, as both groups report receiving similar guidance and support from unions upon entry. Furthermore, immigrants and natives do not have meaningfully different perceptions of unions' influence over their careers overall (Appendix Figure A5) or within various age, education, and industry groups (Appendix Figure A6).

We next examine how workers engage with unions in practice, and whether some groups make greater use of union support. Figure 8 shows that non-Western immigrants are significantly more likely to contact their local union for help than natives or Western immigrants—especially regarding work environment issues, where contact rates are more than three times higher. This suggests that lower returns to union membership for immigrants are unlikely to reflect lower engagement. Supporting this, Appendix Figure A7 shows that non-Western immigrants are more likely to contact unions than natives even when conditioning on similar beliefs about union effectiveness. The same pattern holds, though less strongly, among those who are sceptical of union influence. In addition, when immigrants expect unions to prioritise all work-

ers equally, they are far more likely than natives to engage with their unions (Appendix Figure A8). The same pattern holds even when immigrants believe unions prioritise some groups over others, though Western immigrants—not non-Western immigrants—report the highest engagement in those cases. Immigrants also engage more frequently than natives regardless of whether their previous experience with the union was positive or negative (Appendix Figure A9), with the largest gaps following negative interactions.

If the gap in returns is not driven by information or engagement, one alternative explanation could be that immigrants and natives simply value different aspects of their jobs, and thus engage with or benefit from unions in different ways. However, the survey provides little support for this idea. Appendix Figure A10 shows that natives and immigrants - and in particular Western immigrants - report very similar preferences when asked about the relative importance of salary, job security, promotion opportunities, and work environment. The same pattern holds across major demographic subgroups (Appendix Figure A11), suggesting that preference heterogeneity is unlikely to explain the differences we observe in union returns.

Another potential explanation is that immigrants view union benefits as public goods—largely accessible regardless of membership—and are therefore less motivated to engage seriously with the union even if they contact them more frequently and more intensely. However, Appendix Figure A12 shows that this is not the case. If anything, non-Western immigrants are actually slightly less likely than natives to perceive union support as non-excludable. This further reinforces the interpretation that the immigrant-native gap is not driven by differences in beliefs about what unions do or who they serve, but by differences in the actual benefits received.

Finally, even when asked more generally whether union membership makes them happier at work (Appendix Figure A13), non-Western immigrants respond positively, but report lower satisfaction than natives—mirroring the broader pattern of lower returns despite equal or greater engagement.

Overall, these results suggest that immigrants are not disengaged from unions—they receive similar information, are more likely to seek help, hold similar preferences over work amenities and are not more likely to consider union benefits as public goods. Yet they experience lower satisfaction and smaller labour market returns, raising the question of whether unions treat immigrant members differently or prioritise certain workers over others.

Do Unions prioritise Natives? To explore whether unions treat some groups more favourably than others, we use both survey responses and administrative data to examine perceptions of prioritization, patterns of satisfaction, and the financial incentives unions face. The evidence suggests that unions may focus their efforts disproportionately on native members, who tend to be longer-tenured, higher-earning, and contribute more in dues.

We first ask union members whether they believe that unions prioritise certain groups of workers, and if so, which groups they think receive more attention. Figure 9 presents the results. In Panel A, nearly 30% of non-Western immigrants report believing that unions treat workers

unequally—substantially higher than the 18% of natives who say the same. Panel B shows that both groups most commonly cite workers with strong personal connections (i.e., "Network") to union leadership (e.g., friends of representatives, local workers, or those with long-standing ties) as the most prioritised—35% of natives and 25% of non-Western immigrants give this response. Among the remaining responses, non-Western immigrants are more likely to identify natives (15%) and high-earning skilled workers (11%) as prioritised, while natives cite skilled workers (25%) and vocal individuals (15%).

These responses point to two main insights: (1) a sizable share of members—especially non-Western immigrants—perceive unequal treatment within unions, and (2) many non-Western immigrants explicitly believe that native workers are prioritised. Notably, this may understate the perceived nativity bias, since the other commonly cited groups that immigrants believe are being favoured by the unions (locals with strong networks and high-skilled workers) are themselves more likely to be natives.[36]

We next examine whether immigrants and natives differ in how they evaluate the help they receive from unions. As shown in Appendix Figure A14, non-Western immigrants are significantly less satisfied than natives after contacting the union for support. This gap persists across most domains—salary, promotion and job security. These patterns are consistent with the idea that unions provide less effective or less tailored support to immigrant members, even when those members actively seek assistance.

One notable exception is the work environment dimension, where non-Western immigrants report slightly higher satisfaction than natives after seeking union help (Appendix Figure A14). This mirrors our main administrative findings. One interpretation is that work-environment improvements (e.g., safer conditions, clearer procedures, or better scheduling) are shared within the workplace and are therefore harder to target toward specific workers, leading to more equal gains across groups.

The evidence above points to unequal treatment within unions. The next question is why these differences might arise. One possibility is that financial and institutional incentives lead unions to focus more on native members. Appendix Figure A15 shows that more than 90 percent of union dues collected within firms come from native workers, reflecting both their higher unionization rates and higher average earnings.[37] If unions allocate effort partly in proportion to dues contributions—or to the seniority, income, or perceived influence of their members—this structure naturally tilts representation toward longer-tenured, higher-paying native workers. These financial patterns align closely with our survey evidence, where respondents identified locally connected members with strong networks and native workers as those most likely to

[36]That both groups commonly cite workers with strong personal connections to union leadership as receiving preferential treatment is relevant because unions are often assumed to focus primarily on supporting workers with weak bargaining power; the survey responses instead suggest that informal ties and seniority may carry more weight in how support is allocated.

[37]This has been surprisingly stable even as immigrants have become a much larger share of the total workforce in Norway.

receive union attention.

Native workers also tend to have deeper institutional ties. In our survey, natives report an average of 16 years of union membership, compared to 13 years for Western immigrants and 10 years for non-Western immigrants, and greater labour market experience overall (21 versus 16 years on average). These differences may contribute to informal prioritization in union support or bargaining processes, particularly if longer-standing members are viewed as more credible, deserving, or influential. This interpretation is reinforced by open-ended survey responses across all groups, which frequently cite local ties, experience, and seniority as key factors shaping who unions prioritise (Figure 9). This survey finding is also consistent with our empirical results in Section 7.3.

While these patterns point to the role of tenure, experience, and informal ties in shaping how unions allocate support, a natural next step is to examine the institutional side more directly—specifically, who occupies decision-making roles within unions.

Using administrative data, we document that non-Western immigrants are substantially underrepresented in the union and membership organisation sector, and especially so in leadership positions. Among workers in management roles within this sector, non-Western immigrants constitute less than 1 percent of employees at the start of the period and remain extremely rare throughout (see Table A19). In addition, union managers have limited prior exposure to non-Western immigrant coworkers in their own work histories. While not conclusive, these patterns are consistent with the idea that decision-making within unions is concentrated among individuals with relatively limited direct experience with immigrant workers, which may contribute to differences in how support and representation are allocated.

In sum, both the survey and administrative evidence point to meaningful asymmetries in how unions serve their members. Immigrants are more likely to perceive unequal treatment, report lower satisfaction after seeking support, and belong to groups that contribute less in dues and have weaker institutional ties. These patterns are more consistent with differences in union behaviour than with differences in worker engagement as an explanation for the lower returns to union membership among immigrants.

Could This Be Driven by Sorting or Complier Differences? The results so far suggest that the immigrant-native gap in union returns is not due to differences in information access, engagement, or willingness to seek help, but rather reflects differences in how unions respond to or prioritise different groups of members. One possibility is that the immigrant-native gap in union returns reflects differences in the types of firms where immigrants work. For example, unions may be more effective at representing the majority group within a firm—either because of strategic prioritization or institutional dynamics—which would lead to lower returns for immigrants in firms with mostly native coworkers.

To test this, we estimate our baseline model separately for firms with high and low immigrant shares—defining "high immigrant share" as above the sample median of 11 percent. The

results (Table A13) show no evidence that non-Western immigrants benefit more when they are in high immigrant worker share firms. In fact, the opposite is true: immigrants experience larger returns to union membership when they work in firms with fewer immigrant coworkers, though the effects are. We obtain similar patterns when focusing specifically on firms with high versus low shares of non-Western immigrant workers (Table A14). This is not explained by observable firm characteristics—such as union density, sector, or average wages—which are similar across firm types. While we cannot fully rule out compositional selection (e.g., more capable immigrants sorting into these firms), the pattern is inconsistent with the "majority-group advantage" hypothesis and suggests that sorting alone is unlikely to drive the gap. Importantly, firms with high- and low native shares are not markedly different from each other on most measures, with the exception of firm size (Appendix Table A15).

That said, it remains important to consider whether broader compositional differences could still explain the results. Natives and immigrants differ in observable characteristics such as education, industry, and public sector employment, and they may also sort into different types of firms. If unions are more effective at bargaining for certain worker profiles or within specific firm environments, then part of the gap may reflect these compositional differences rather than differential treatment. However, even under this interpretation, the findings remain informative: they suggest that the institutional structure of unions—as currently configured—may be better suited to advancing the interests of workers who are more likely to be natives. This would not necessarily reflect explicit discrimination, but it would still point to systematic mismatches between immigrant workers and the union model of representation.

We address this in two ways. First, we present a compliers analysis in Table A4, showing that compliance rates for natives and immigrants are similar overall and across key dimensions such as education, sector, collective bargaining coverage, and white-collar status.

Second, we reweight the immigrant complier sample to match the observable characteristics of the native compliers—specifically along binarized characteristics, including education (less than high school, high school, Bachelors+), white collar status, work in the manufacturing, construction, and public sectors, coverage by collective agreements, firm-specific tenure, and income quartile. We use the compliance rate of Norwegians in each cell of these characteristics to construct inverse probability weights, which we then apply to the immigrants in our two-stage least squares estimation procedure. As shown in Table A5, this reweighting does not meaningfully alter the estimated pattern of earnings returns to union membership for any of the groups.

The logic of this exercise is as follows: If the immigrant–native gap were driven by differences in who complies with the instrument, then reweighting immigrants to match the complier composition of natives would be expected to eliminate, or at the very least substantially attenuate, the gap. It does not—estimates remain essentially unchanged—implying that observable differences in complier composition are unlikely to explain the gap.

A natural follow-up question is whether the observable characteristics used in the reweighting exercise capture relevant dimensions of selection into union membership more broadly. To assess this, we return to the OLS results, which show strong positive selection into union membership—particularly for non-Western immigrants. When we augment the OLS specification to include these same worker-level characteristics, however, the differential selection patterns between natives and immigrants are largely attenuated (see Table A16). This suggests that these variables capture a substantial share of the observable heterogeneity underlying selection into union membership. At the same time, the OLS and IV estimates capture different margins—equilibrium selection versus policy-induced enrolment—and these need not coincide. We therefore interpret the reweighting exercise as a diagnostic for observable differences in complier characteristics rather than as establishing full comparability.

A related question is whether sorting across firms helps explain the differential returns to union membership for natives and immigrants. We examine two related sorting mechanisms. First, Silliman and Willén (2024) show that firms with higher levels of union membership prior to the EU expansion hired incoming EU immigrants at lower rates, generating sorting across firms by baseline union density as unions treated immigrants more as labour market "outsiders." Motivated by this mechanism, we estimate heterogeneous union effects separately for workers in firms with above- versus below-median pre-2004 union density. If immigrants are disproportionately sorted into firms where unions are less able to generate wage gains, then part of the immigrant–native gap in union returns could reflect differences in firm environments rather than differential treatment conditional on membership. Column (1) of Table A14 shows little evidence of this: the estimated returns to union membership are broadly similar across high- and low-union-density firms for both natives and immigrants. This suggests that sorting across firms by baseline union density is unlikely to be the primary driver of the differential returns we document.

The effect of union membership on native–immigrant labour market gaps is economically important regardless of the precise underlying mechanism, as it highlights that unions can shape inequality across groups in meaningful ways. If the effects primarily operate through differences in the workers or firms induced into unionization, this would imply that unions affect inequality through composition and sorting. If instead the gap reflects differences in the returns generated conditional on membership, this points more directly toward differences in how unions represent and allocate benefits across groups. Distinguishing between these channels is therefore important for understanding how unions shape labour market inequality and for evaluating which policy margins are most relevant.

Taken together, the evidence suggests that the immigrant–native gap in union returns is unlikely to be driven primarily by differences in the composition of marginal workers or firms induced into unionization by the policy. The gap remains stable when we reweight immigrants to match native compliers along observable characteristics associated with responsiveness to

the union subsidy and when we account for several forms of firm sorting. This suggests that the heterogeneous effects primarily reflect differences in the returns unions generate conditional on membership, rather than differences in who is induced to join. More broadly, the evidence points toward institutional dynamics within unions—such as representation, incentives, and the allocation of bargaining effort—as important determinants of how collective bargaining shapes labour market inequality. Understanding these dynamics is important for evaluating when unions mitigate inequality, when they may reproduce existing disparities, and which institutional features determine these outcomes.

Public Goods and Firm-Level Union Density Spillovers. Our empirical strategy captures the causal effect of individual union membership among compliers. This captures the private goods component of individual membership in comparison to non-compliers in the same firm. However, because changes in union density might entail public goods and union density spillovers, it is instructive to see if the spillover component of firm-level union density affects those who did not newly join a union from our instrument. To measure this, we calculate the predicted union densities from Equation 4 at the firm level in each year and estimate a variant of Equation 3 in which we replace individual union membership with predicted firm-level union density and limit this sample only to those who did not switch their union status over the prior three years (a simple proxy for likely compliers). This approach isolates the causal effect of changes in firm-level union density (driven by compliers) on those who were not compliers, and, therefore, not reaping new private benefits from their membership. This includes never-union members and always-union members. We also allow the effects of union density to vary based on whether the worker was always a member of a union. This approach also allows us to check if firm-level union density dynamics could possibly explain the differential effects by immigration status in Table 3.

Appendix Table A17 column (2) shows that among non-compliers, increasing predicted union density by one percentage point increases log annual earnings by 0.45% on average for Norwegians, by 0.25% for Western immigrants, and 0.15% for NWIs. Columns (3) and (4) allow these effects to differ based on whether the worker was always a member of a union, and, therefore, had longstanding connections to the union. On average, those with close ties to the union received slightly larger earnings gains from firm-level union density. For NWIs, public goods gains accrue only to those with close ties to the union via longstanding membership. These results show that immigrants benefit from the public goods and union density spillovers that unions provide, albeit less than their Norwegian counterparts, and that those with longstanding connections to the unions benefit more. Importantly, these results also show that firm-level dynamics cannot explain the magnitude of the private earnings effects of individual union membership we find in Table 3. These results also reinforce the conclusions from our surveys, suggesting that those with closer ties to the unions disproportionately benefit from greater union bargaining power.

A related concern is that the estimated individual-level effects may reflect omitted firm-level union exposure rather than a distinct individual channel. To address this directly, we estimate a specification that includes both individual union membership and firm-level union density as endogenous regressors, each instrumented separately.

To do so, we proceed in two steps. First, we construct a leave-out measure of firm-level union density that excludes the individual's own union status. This ensures that the firm-level variable captures the broader union environment within the firm rather than mechanically reflecting the contribution of the individual focal worker's membership. Second, we build a firm-level analog of our instrument that mirrors the structure of the individual-level instrument but varies at the firm level. Specifically, we assign baseline firm-level dues to each firm in its first year in the data (the firm-level counterpart to assigning baseline dues in the worker's first observed year) and apply the same imputation procedure for net dues.

Crucially, the two instruments exploit distinct sources of variation: the individual-level instrument follows the worker over time, while the firm-level instrument follows the firm. As a result, when a worker moves across firms, their firm-level exposure changes while their individual instrument remains specific to the worker. This provides the independent variation needed to separately identify the individual and firm-level channels.

The results, reported in Table A18, show that the estimated effect of individual union membership is virtually unchanged when including the instrumented firm-level union density. The point estimates remain very similar in magnitude, and the confidence intervals overlap closely with those from our baseline specification. Taken together, these results indicate that the individual-level estimates are not being driven by omitted firm-level union exposure. Instead, they reflect a distinct individual channel, while the firm-level "public good" component can be separately identified without materially affecting the main conclusions.

8 Discussion and Conclusion

This paper provides the first comprehensive causal assessment of how union membership shapes labour market disparities between natives and immigrants. We exploit exogenous variation in the cost of joining a union—generated by government-mandated changes in union-dues subsidies—within an instrumented difference-in-differences framework.

Our main finding is that union membership generates substantially larger labour market returns for native workers than for immigrants, thereby widening native–immigrant labour market gaps. Native workers experience sizable earnings gains and stronger employment protection, while the corresponding effects for non-Western immigrants are substantially smaller and, for earnings, close to zero.

We then examine a range of mechanisms that may help explain these differential returns. One possibility is that the results reflect differences in the composition of workers who marginally join unions in response to the subsidy reforms. However, the gap remains after reweighting na-

tives and immigrants on detailed worker and job characteristics, suggesting that observable differences in complier composition are unlikely to fully account for the findings. A second possibility is that immigrants are disproportionately sorted into firms where unions generate weaker returns. While immigrants are more concentrated in certain firms and sectors, the differential returns to union membership persist across firms with different immigrant shares and different levels of union density, suggesting that firm sorting alone is also unlikely to fully explain the results.

The remaining evidence points toward differences in how unions represent and support workers across groups. Descriptive survey evidence indicates that immigrants are at least as likely as natives to seek union assistance and report similar broad preferences regarding union priorities, making differences in engagement less likely to explain the patterns. At the same time, non-Western immigrants report lower satisfaction following contact with unions and are more likely to perceive that unions prioritise native workers and workers with stronger local connections. We also document that non-Western immigrants are substantially underrepresented in union leadership and membership organisations. While the survey evidence is descriptive and does not isolate causal mechanisms, the combined evidence is consistent with differences in representation, responsiveness, or bargaining support contributing to the unequal returns to union membership across groups.

The distributional implications are particularly pronounced in concentrated labour markets. Union membership generates larger returns where employer market power is stronger, consistent with unions extracting and reallocating rents. However, these gains accrue disproportionately to native workers. This suggests that institutions designed to counter monopsony power, such as unions, may simultaneously mitigate one form of labour market inequality while reinforcing another.

At the same time, the results shed light on how unions create value for non-Western immigrants. For this group, the benefits of union membership appear to arise primarily through improvements in employment stability and workplace quality rather than through higher pay. We find lower take-up of unemployment and sick-leave benefits, alongside modest increases in contractual hours and promotion probabilities—patterns consistent with more stable and formalized employment relationships. More broadly, these findings suggest that unions do provide benefits to non-Western immigrants, but that these gains are more limited in scope and concentrated in non-pecuniary dimensions of employment.

The paper's broader contribution is to integrate two defining features of modern labour markets—migration and collective representation—to study whether core labour market institutions mitigate or amplify inequality even among workers employed in the same firms, occupations, and industries. In an era of rising migration, persistent labour market disparities, and growing concern about labour market power, understanding these interactions is central to both economic research and policy design. A key question for future research is to further disentangle

the relative roles of worker composition, firm sorting, employer behaviour, outside options, and internal union dynamics in shaping the returns to collective representation across demographic groups.

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Tables

Table 1: Summary Statistics by Immigration Status

	Natives	Western Immigrants	Non-Western Immigrants
Age	43.55	42.35	38.58
Gender (1=male, 2=female)	1.49	1.45	1.44
Number of children	1.55	1.32	1.24
Years since arrival	-	19.57	12.14
Education (%)			
Less than high school	0.17	0.13	0.29
High school	0.45	0.33	0.33
College	0.39	0.55	0.37
Earnings	467,857	506,225	377,229
Hours	33.36	33.61	32.26
Union	0.56	0.37	0.34
Always union	0.34	0.19	0.12
Firm union density	0.53	0.47	0.41
Share of native workers at firm	0.897	0.798	0.721
Firm labour market power (HHI)	0.09	0.08	0.06
Unemployment benefit	2,694	3,917	7,265
Conditional unemployment benefit	54,305	62,039	65,408
Sick leave	12,040	11,797	11,661
Conditional sick leave	50,241	51,968	50,968
N	24,075,082	1,104,113	2,109,859

Source: Authors' calculations of Norwegian registry data from 2001 to 2015. Education statistics of immigrants are self-reported. Earnings are measured as pre-tax income from labour and self-employment. Firm labour market power (HHI) is the sum of squared employment shares across firms in each occupation and labour market, and ranges from 0 to 1. A value of 1 implies a perfectly monopolistic market, and a value of 0 implies a perfectly competitive market. We fix each firm's HHI at the first year in which the firm appears in the data. Unemployment benefit and sick leave benefit are calculated based on the cumulative amount of benefits received in a given year. Conditional unemployment and sick leave benefits are calculated from those who receive positive amounts.

Table 2: First-Stage Results

	(1) Union Membership
Net Dues (1,000 NOK)	-0.1135***
	(0.0053)
Net Dues×Western immigrant	-0.0016
	(0.0024)
Net Dues×Non-Western immigrant	-0.0136***
	(0.0020)
Observations	11,640,111
Kleibergen-Paap F	156.99

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the specification in Equation 4 with log total earnings as the regression outcome. Standard errors are clustered at the individual level. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status.

Table 3: Effect of Union Membership on Career Outcomes
Baseline Specification

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.115***	4.444***	-15,119***	-6,619	0.122***	-0.125***
	(0.0440)	(0.851)	(1,884)	(5,304)	(0.0249)	(0.0256)
Union × Western Immigrant	-0.0641***	-0.444*	2,188***	6,485***	0.00267	0.0224***
	(0.0172)	(0.229)	(566.0)	(1,470)	(0.00634)	(0.00674)
Union × Non-Western Immigrant	-0.101***	-0.290	7,163***	-14,416***	-0.0159	-0.0120
	(0.0250)	(0.385)	(1,044)	(2,522)	(0.0106)	(0.0115)
Total Effect for Immigrant Groups After Linear Combination of Coefficients						
Western Immigrant	0.0511	4.001***	-12,930***	-134.0	0.124***	-0.102***
	(0.0474)	(0.881)	(1,982)	(5,550)	(0.0260)	(0.0269)
Non-Western Immigrant	0.0137	4.154***	-7,955***	-21,035***	0.106***	-0.137***
	(0.0563)	(1.036)	(2,379)	(6,621)	(0.0306)	(0.0316)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	156.99	120.75	163.68	164.03	163.33	163.33

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table 4: Marginal Effect of Union Membership on Career Outcomes by labour Market Concentration

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Natives in Low HHI Firms	0.0688	3.602***	-11,018***	-6,069	0.169***	-0.114***
	(0.0420)	(0.807)	(1,915)	(4,995)	(0.0235)	(0.0244)
Western Imm in Low HHI Firms	-0.0349	2.942***	-6,818***	4,407	0.171***	-0.0792***
	(0.0499)	(0.868)	(2,125)	(5,511)	(0.0260)	(0.0267)
Non-Western Imm in Low HHI Firms	-0.0484	4.308***	3,106	-18,800***	0.135***	-0.0739**
	(0.0558)	(0.983)	(2,592)	(6,355)	(0.0294)	(0.0308)
Natives in High HHI Firms	0.324***	7.440***	-31,275***	-7,367	0.00850	-0.136***
	(0.0590)	(1.164)	(2,707)	(6,940)	(0.0330)	(0.0343)
Western Imm in High HHI Firms	0.242***	7.211***	-31,463***	3,627	-0.0653	-0.107**
	(0.0871)	(1.478)	(3,919)	(8,998)	(0.0425)	(0.0440)
Non-Western Imm in High HHI Firms	-0.217	3.404*	-30,830***	-45,355***	-0.254***	-0.520***
	(0.154)	(2.064)	(6,888)	(15,681)	(0.0746)	(0.0926)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	61.47	49.27	63.98	64.51	64.54	64.54

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.

Notes: Estimates represent the marginal effects from the two-stage least squares specification in Equations 3 and 4, and are interpreted independently. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. High HHI is an indicator for the worker's firm being above the sample median in local labour market concentration, which we measure at the occupation-local labour market level. In our analysis sample, the median HHI is approximately 0.05.

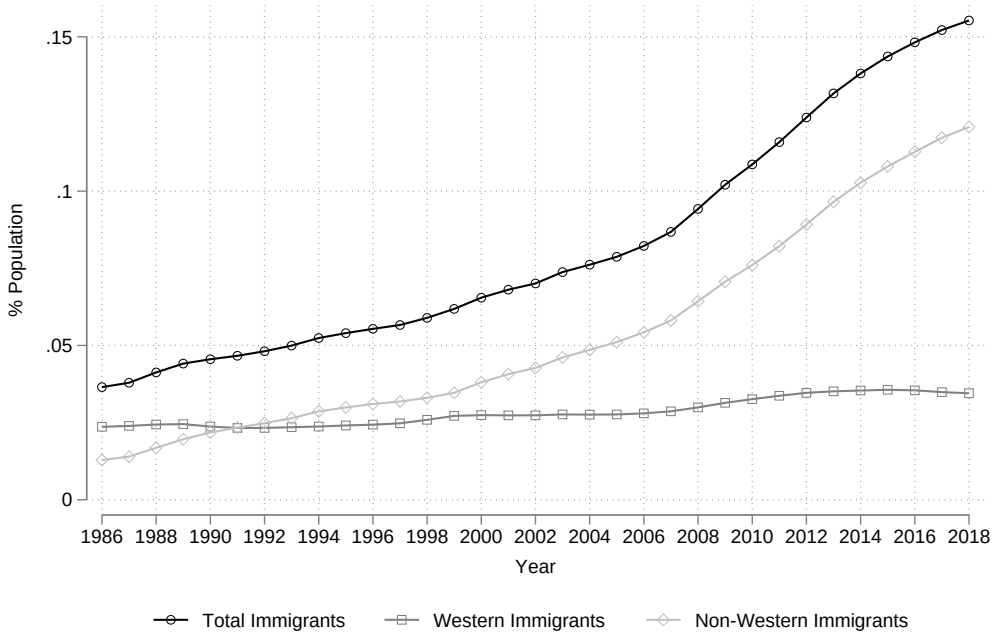
Table 5: Marginal Effect of Union Membership on Career Outcomes by Firms' Immigrant Worker Share

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Natives in Low ImmShare Firms	0.117***	4.864***	-17,104***	-5,674	0.133***	-0.114***
	(0.0444)	(0.869)	(1,923)	(5,338)	(0.0251)	(0.0259)
Western Imm in Low ImmShare Firms	0.0620	4.644***	-16,604***	-903.1	0.117***	-0.116***
	(0.0517)	(0.956)	(2,154)	(5,990)	(0.0282)	(0.0291)
Non-Western Imm in Low ImmShare Firms	0.132**	4.111***	-18,644***	-12,145*	0.119***	-0.195***
	(0.0628)	(1.153)	(2,764)	(7,312)	(0.0342)	(0.0355)
Natives in High ImmShare Firms	0.128***	4.119***	-12,957***	-5,985	0.119***	-0.158***
	(0.0433)	(0.839)	(1,866)	(5,214)	(0.0245)	(0.0253)
Western Imm in High ImmShare Firms	0.0405	3.777***	-9,249***	2,650	0.131***	-0.112***
	(0.0498)	(0.895)	(2,074)	(5,635)	(0.0264)	(0.0275)
Non-Western Imm in High ImmShare Firms	-0.0530	4.777***	745.6	-28,601***	0.0805**	-0.156***
	(0.0608)	(1.092)	(2,634)	(7,020)	(0.0321)	(0.0335)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	78.92	59.77	82.38	82.57	82.3	82.3

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates represent the marginal effects from the two-stage least squares specification in Equations 3 and 4, and are interpreted independently. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues.

Figures

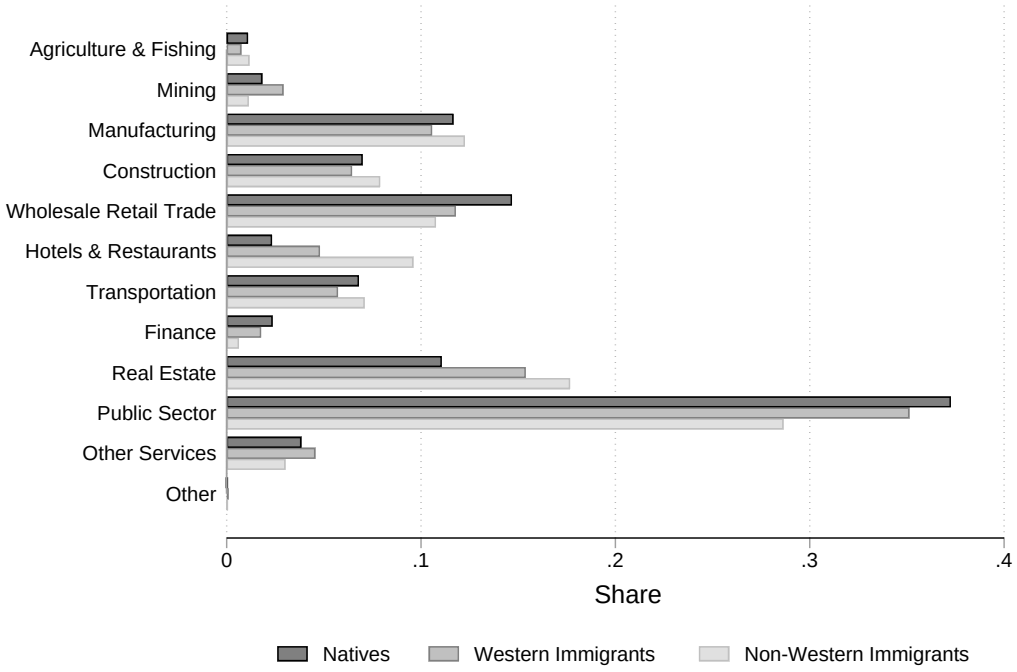
Figure 1: Trends in Immigration in Norway



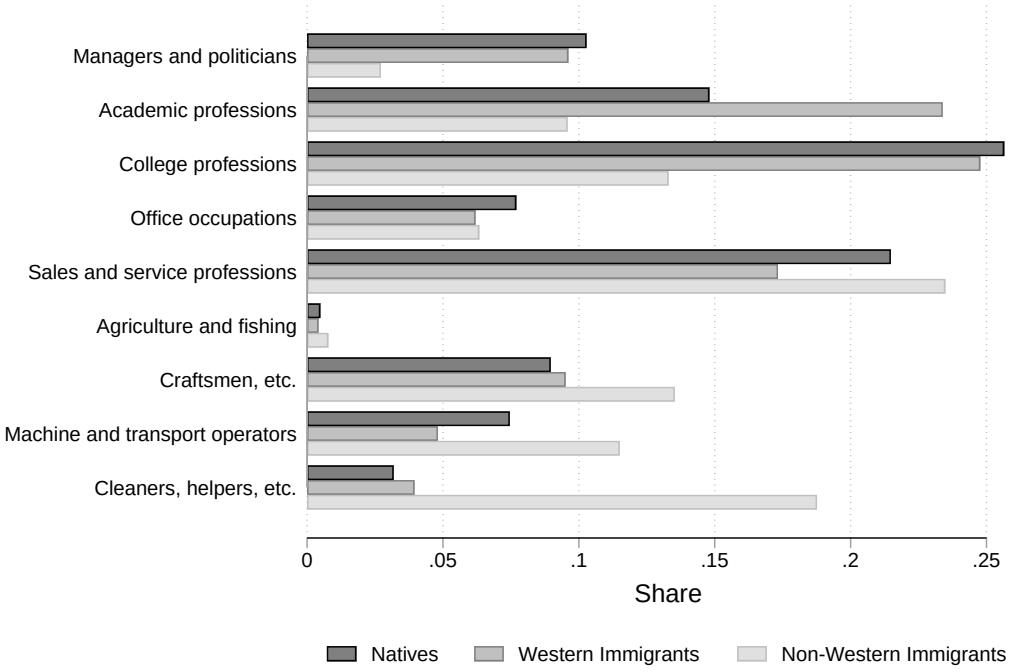
Source: Authors' calculations of Norwegian registry data.

Figure 2: Worker Industry and Occupation Share by Immigration Status

Panel A: Industry



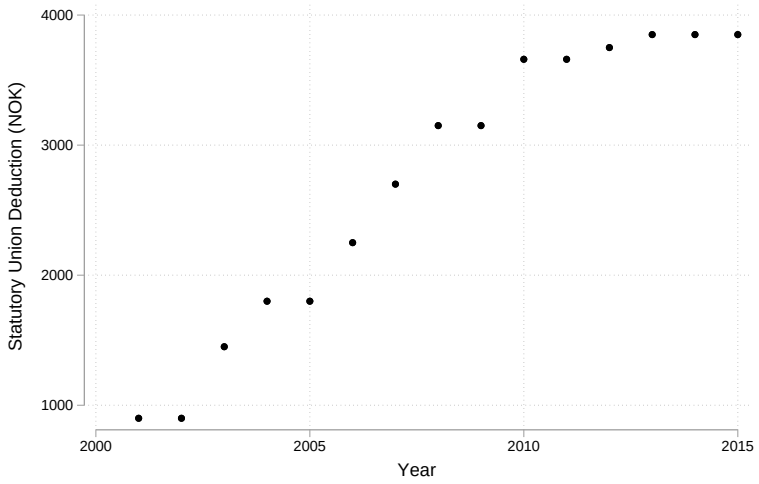
Panel B: Occupation



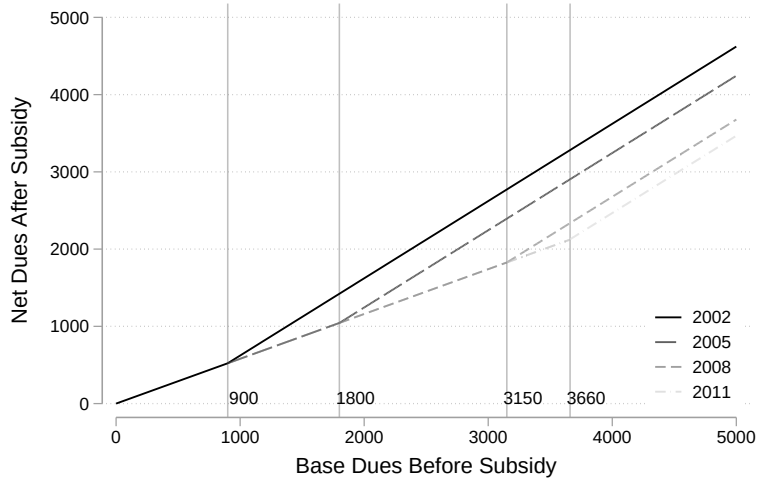
Source: Authors' calculations of Norwegian registry data from 2001 to 2015.

Figure 3: Maximum Deduction, Policy Variation, and Generosity Measures Over Time

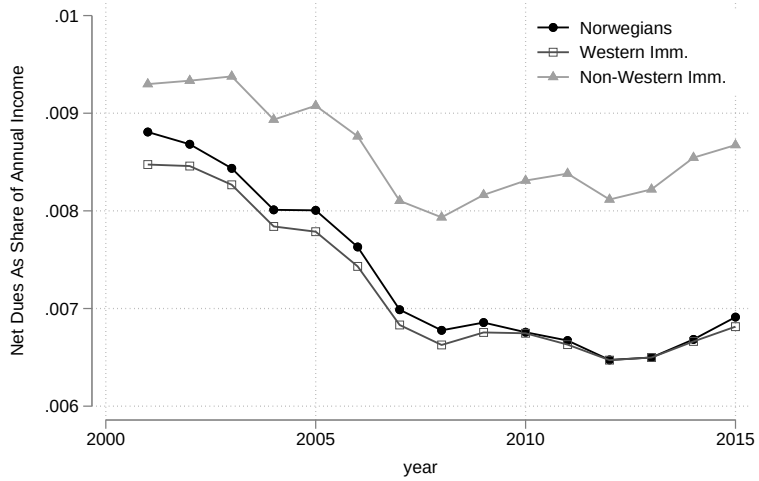
Panel A: Changes in Maximum Deduction, 2001-2015



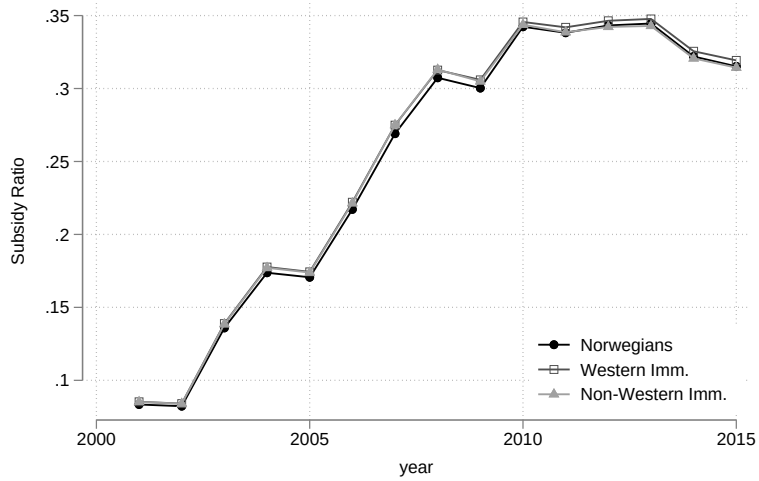
Panel B: Base vs Net Dues, Illustrative Example



Panel C: Net Dues Relative to Average Income



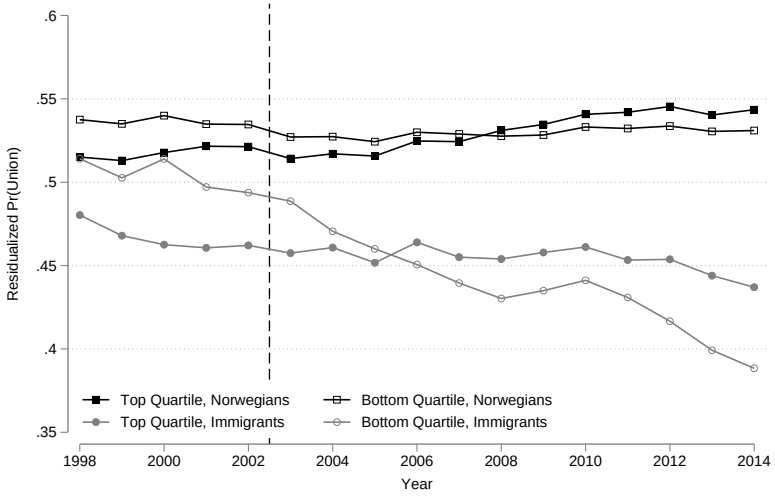
Panel D: Subsidy as a Share of Net Dues by Immigrant Status



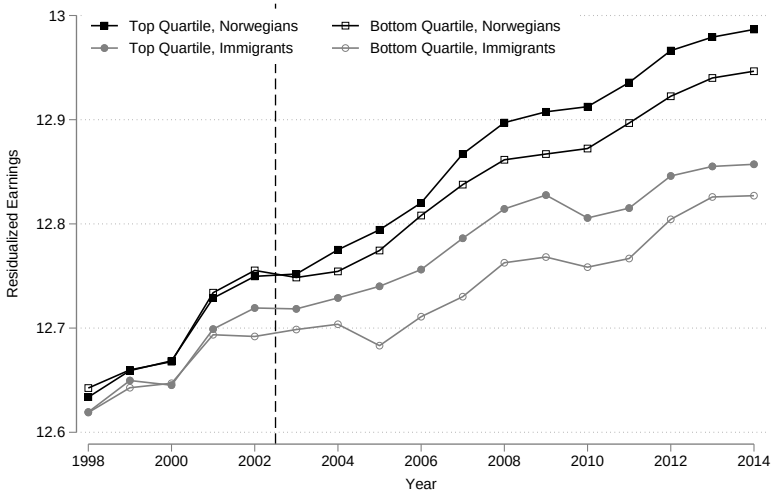
Source: Authors' calculations of Norwegian registry data and illustrative examples.
Notes: Panel A presents the legislated maximum deductions over time. Panel B is an illustrative example of the difference between base dues and dues net of the subsidy assuming a 42% tax rate for illustrative purposes. In Panel C, net dues relative to income is calculated as the net dues after the subsidy in Section 5.2 divided by average annual earnings by base firm over time. In Panel D, the subsidy as a share of net dues is a measure of relative generosity and is defined by the value of the imputed subsidy divided by the net dues following the description each variable in Section 5.2.

Figure 4: Trends in Union Membership Rates and Earnings by Instrument Intensity and Immigrant Status

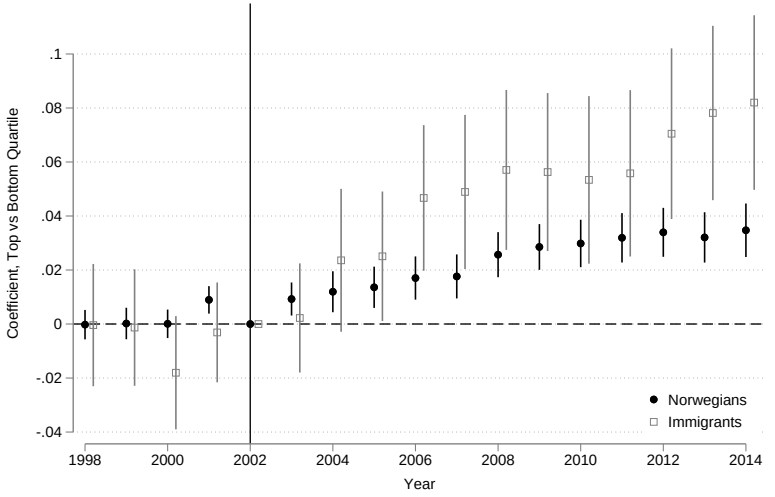
Panel A: Union Membership | Base Firm



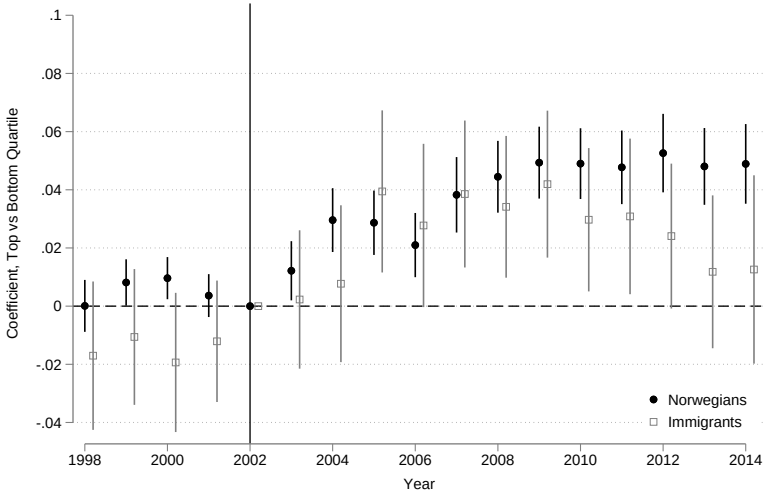
Panel B: Log Earnings | Base Firm



Panel C: Union Membership | Event Study

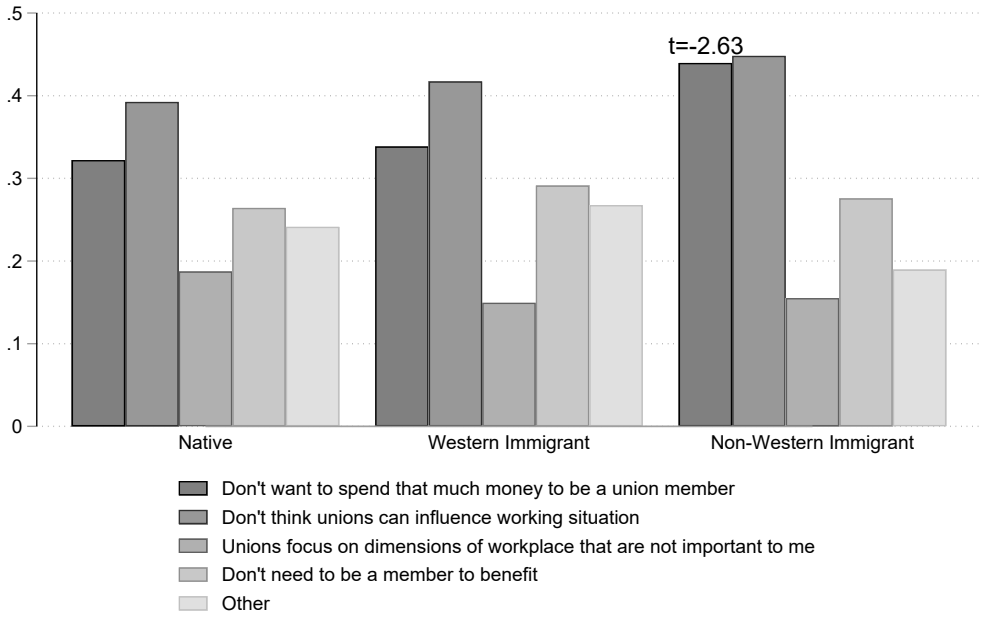


Panel D: Log Earnings | Event Study



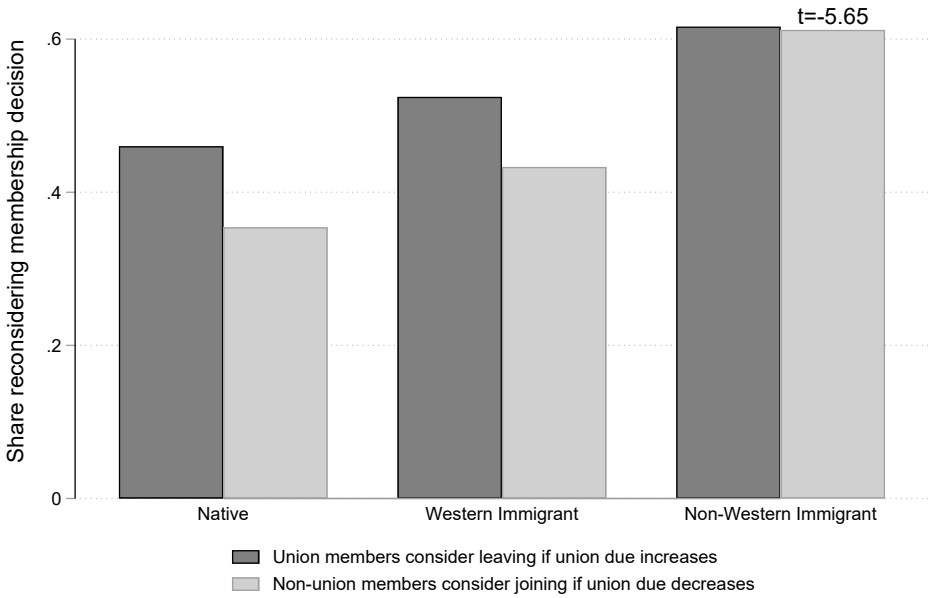
Source: Authors' calculations of Norwegian registry data from 1997 to 2014.
Notes: The "Top" group denotes workers whose base firm was in the top quartile for reductions in net union dues from 2002-2010, while "Bottom" denotes workers whose base firm was in the bottom quartile of net dues reductions over the same period. Panels A and B account for fixed effects for a worker's base firm. Panels C and D formalize the comparisons in an event study model by plotting the coefficients of separate models for immigrants and natives with top quartile indicators interacted with year dummies while also controlling for year and firm fixed effects.
The p-values for joint significance of the pre-2002 coefficients for immigrants in panels C and D are 0.46 and 0.42, respectively.

Figure 5: Nonunionized Workers Reason For Not Unionizing by Immigration Status (Survey One)



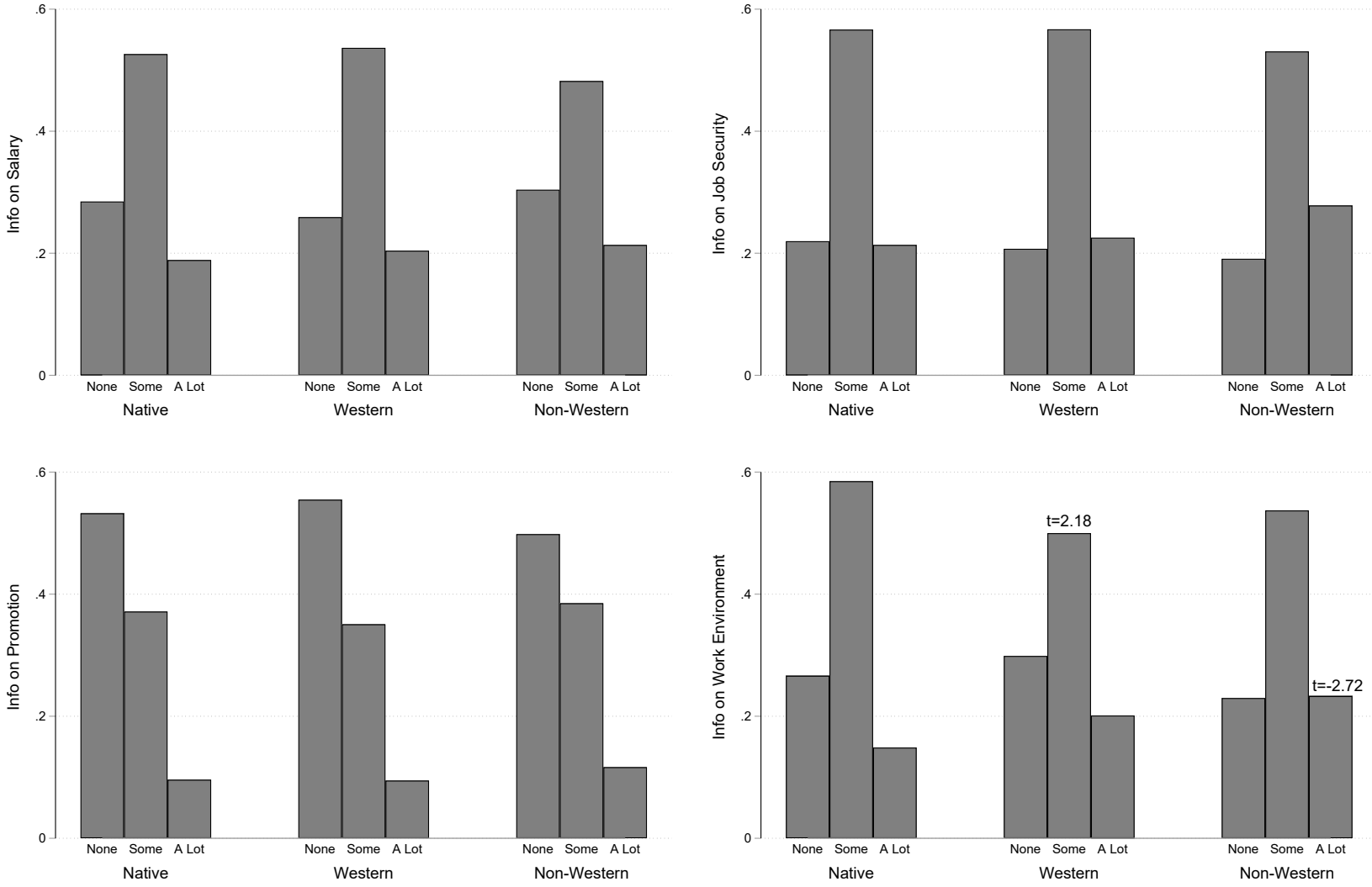
Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "The purpose of this question is to understand the reason why you do not join a union. Check all the boxes that apply." T tests are performed separately for each outcome between natives and western immigrants, and natives and non-western immigrants. The t value is labelled if the difference is significant at the 95% level.

Figure 6: Price Sensitivity to Union Membership by Immigration Status (Survey One)



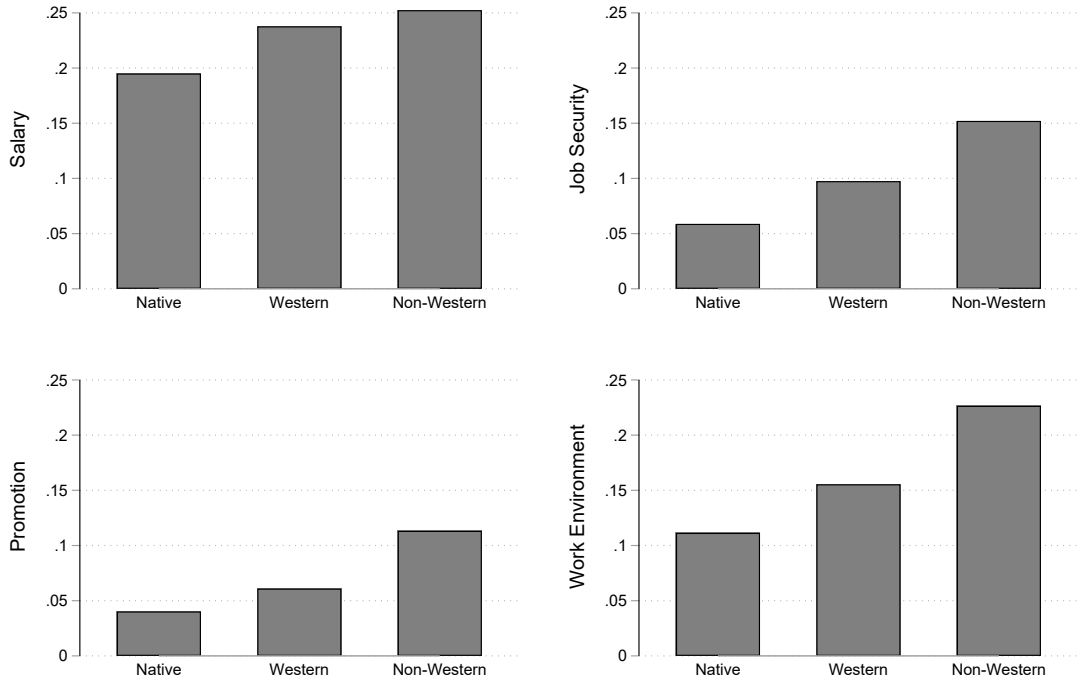
Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "If your after-tax dues for union membership were reduced [increased] by [XYZ] NOK, would you reconsider your decision to join a union?" T tests are performed separately for each outcome between natives and western immigrants, and natives and non-western immigrants. The t value is labelled if the difference is significant at the 95% level.

Figure 7: Information Provided to Union Members (Survey Two)

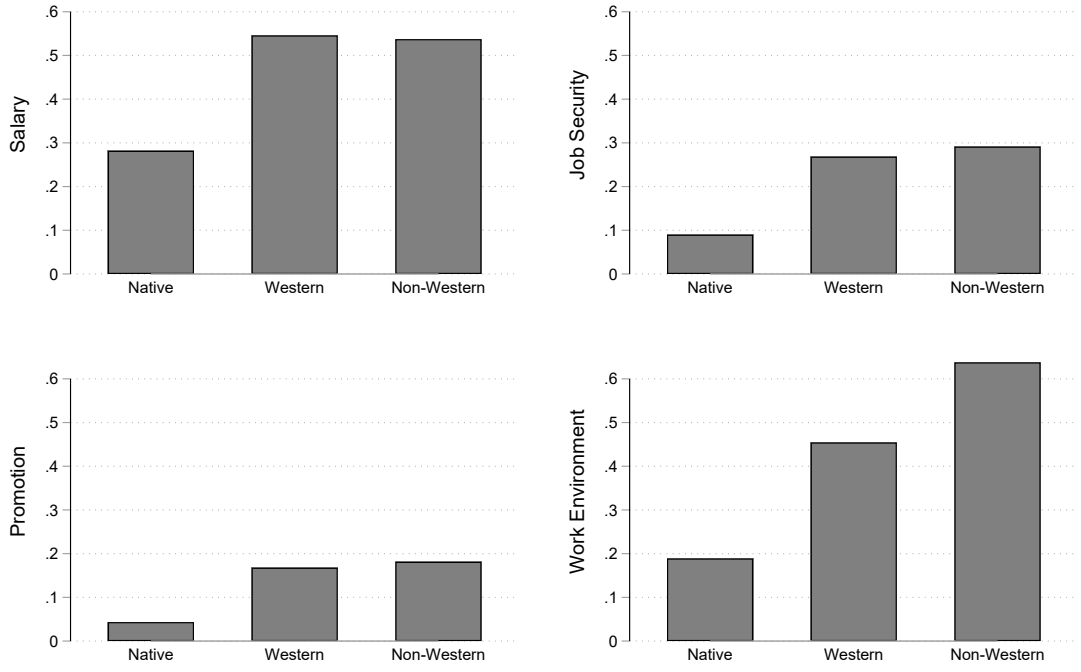


Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The question on the survey asked, "After you joined the union, how much information did your local union provide on the following aspects in which the union might be able to help: Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality." T tests are performed separately for each outcome between natives and western immigrants, and natives and non-western immigrants. The t value is labelled if the difference is significant at the 95% level.

Figure 8: Rates of Contacting Union Representatives (Survey Two)
Panel A: Share of Members Making Any Contact

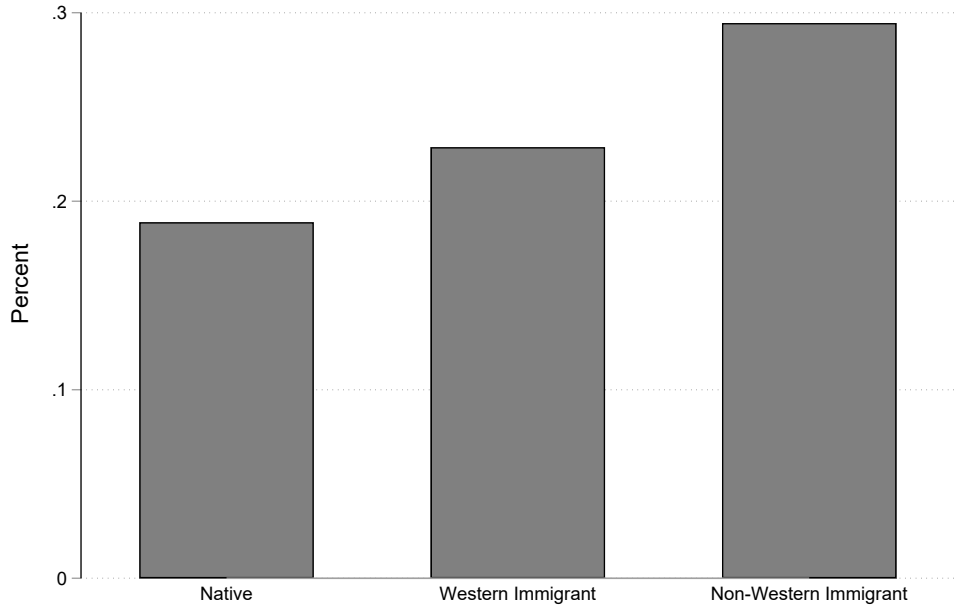


Panel B: Mean Number of Contacts

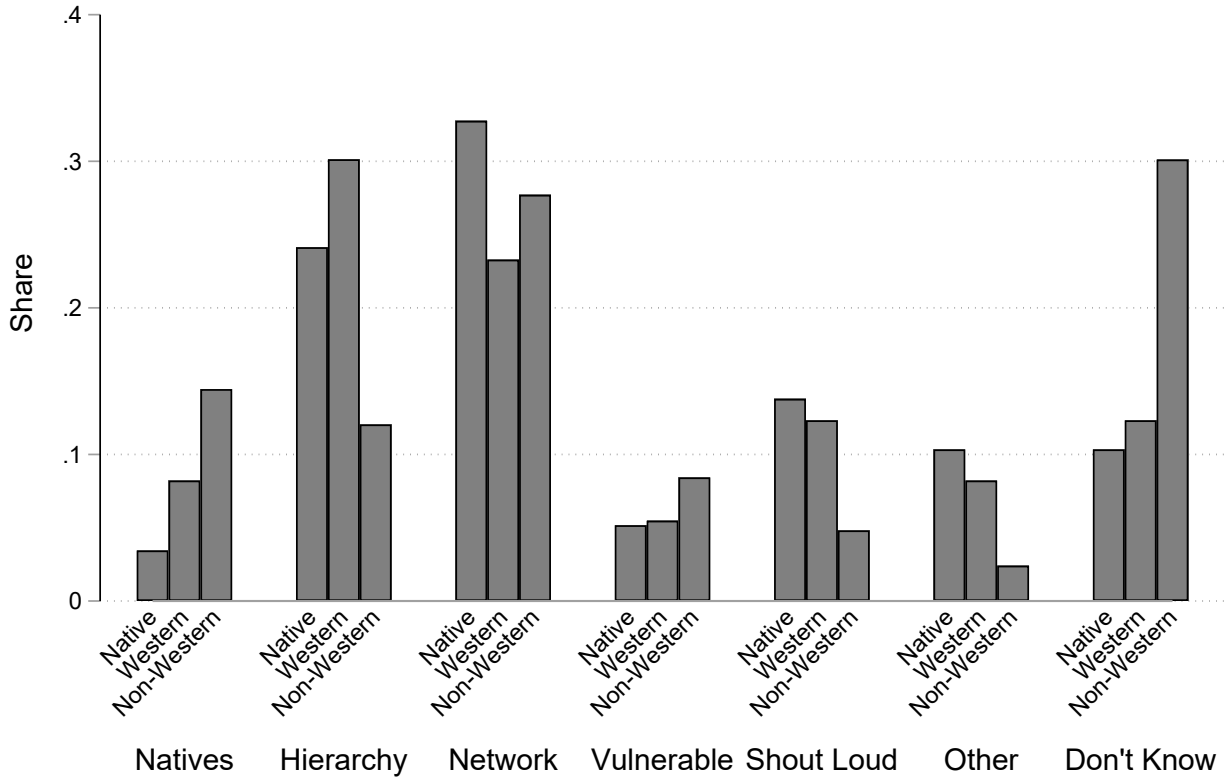


Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The question on the survey asked, "In the past 12 months, how many times have you contacted your local union about the following aspects of your job? Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality."

Figure 9: Perceptions that Unions Do Not prioritise All Members Equally (Survey Two)
Panel A: Share Responding That Unions prioritise Certain Members Over Others



Panel B: Categories of Write-In Responses: Who is prioritised?



Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The question on the survey asked, "How do you think union members are treated at your workplace by the local union?" Response options were, "I think all union members are treated equally by the local union" and "I think some members are prioritised over other members by the local union." If they responded that some members are prioritised over others, the open response question was, "Who do you think is being prioritised by your local union?"

A Tables Appendix

Table A1: Descriptive Statistics of Survey One Respondents

	Natives	Western	Non-Western
Age	44.47	46.33	39.19
Female	0.50	0.52	0.58
College	0.59	0.68	0.72
Public sector	0.39	0.33	0.42
Years of work experience	21.10	21.88	14.44
Work fulltime	0.87	0.83	0.84
Union member	0.55	0.44	0.50
Years of union membership	16.54	13.87	10.76
Non-work union benefits (lower mortgage rate, etc.) 0-100	63.58	53.38	56.32
Observations	4742	226	232

Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.

Table A2: Effect of Union Membership on Career Outcomes
Baseline Specification with Alternative SE Clustering

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.115	4.444**	-15,119***	-6,619	0.122	-0.125
	(0.0993)	(2.214)	(3,461)	(6,651)	(0.0744)	(0.131)
Union×Western Immigrant	-0.0641***	-0.444*	2,188***	6,485***	0.00267	0.0224***
	(0.0188)	(0.253)	(627.5)	(1,455)	(0.00729)	(0.00834)
Union×Non-Western Immigrant	-0.101**	-0.290	7,163***	-14,416***	-0.0159	-0.0120
	(0.0455)	(0.661)	(1,531)	(3,672)	(0.0188)	(0.0519)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	29.54	22.95	29.83	29.83	29.49	29.49

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4. Standard errors are clustered at the firm level. Outcomes are measured in year *t+1*. The model includes fixed effects for immigrant status by year, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A3: Instrument Intensity and Baseline Characteristics, Correlations

Panel A: Correlations and Gaps by Exposure

	(1) Raw Correlation with Reduction in Net Dues	(2) Conditional Correlation with Reduction in Net Dues	(3) Conditional Gap Above-Median Exposure Minus Below-Median
Log Real Earnings	-0.110	-0.003	-0.011
Native Norwegian	-0.034	-0.015	0.009
Non-Western Immigrant	0.038	0.015	-0.007
Female	0.189	0.010	0.013
Age	0.005	0.000	0.150
Less than High School	0.023	0.011	-0.010
High School Diploma	-0.074	-0.016	0.008
Bachelors Degree +	0.058	0.008	0.003
Observations	3,241,467	3,241,467	3,198,410

Panel B: Baseline Union Membership (2001-2002) by Immigration Status and Major Industry

	Norwegians	Western Imm.	Non-Western Imm.
Overall	0.559	0.401	0.398
Agriculture and Fishing	0.271	0.179	0.167
Mining	0.557	0.319	0.483
Manufacturing	0.600	0.432	0.506
Construction	0.434	0.271	0.342
Wholesale, Retail Trade	0.251	0.173	0.223
Transportation	0.585	0.441	0.475
Fin., R. Estate, Busin., Hotels	0.387	0.230	0.256
Public Sector	0.758	0.574	0.500

Panel C: Placebo Tests on Pre-Reform Outcomes

	Log(Earnings)
Union Membership	31.35
	(49.79)
Union Mem. x W. Imm.	-0.845
	(1.319)
Union Mem. x NWI	1.516
	(2.454)
Observations	8,721,498
1st Stage K-P rk Wald	3.793

Source: Authors' calculations of Norwegian registry data from 2002 to 2010.
Notes: Panel A correlations are between the reduction in net dues within a worker's base firm in the data between 2002 and 2010 and a set of baseline characteristics for each worker in the base firm. Conditional correlations are for the reduction in net dues after residualizing on controls for occupation by industry cell, age group, and "always union" status. Panel B shows the union membership rates in 2001-2002 by each major industry group broken down by immigration status. In Panel C, we estimate a placebo regression in which we regress log annual earnings in the period prior to 2002 on the value of the instrument shifted backward in time 6 years, or in other words, to tests if later exposure to the instrument predicts any union membership or earnings responses prior to 2002. We include fixed effects for firm, year, age group, and baseline occupation-industry cell all at their pre-2002 levels in order to match the structure of Equations 4 and 3 with the available data in the pre-reform period. The first stage F statistic is only 3.79, far below any conventional rules of thumb for identifying weak instruments. This strengthens our confidence that our model is not reflecting pre-existing differences or trends.

Table A4: Compliers Analysis

Panel A: Demographic Characteristics, Pr(Complier | X)

	(1) Native Norwegians	(2) Western Immigrants	(3) Non-Western Immigrants
Overall	0.071	0.060	0.065
Female	0.108	0.085	0.112
Less than High School	0.067	0.057	0.065
High School Diploma	0.063	0.056	0.049
Bachelors Degree +	0.081	0.069	0.087
White Collar	0.087	0.069	0.090
Manufacturing	0.052	0.038	0.053
Public Sector	0.136	0.141	0.180
Collective Agreement	0.050	0.043	0.053

Panel B: Income Quartile, Pr(Complier | Income Quartile)

	(1) Native Norwegians	(2) Western Immigrants	(3) Non-Western Immigrants
Earnings Q1	0.092	0.075	0.064
Earnings Q2	0.078	0.061	0.064
Earnings Q3	0.071	0.058	0.066
Earnings Q4	0.047	0.048	0.068

Source: Authors' calculations of Norwegian registry data from 2001 to 2014.
Notes: Compliance is calculated in three steps. First, we residualize the one-year change in net dues on each of our fixed effects (except always union status). Second, we estimate our first-stage regression of union membership status on this measure of instrument exposure for each subgroup. Third, we use the parameters of the model to estimate predicted union membership at the 1st and 99th percentile of the instrument and characterise the share of workers changing status as compliers for that subgroup.

Table A5: Effect of Union Membership on Log Annual Earnings, Reweighting Complier NWIs to Match Characteristics of Complier Norgweians

	(1) Reweighting All to Norwegian Compliers' Individual Characteristics
Union Membership	0.162***
	(0.0456)
Union Membership x Western Imm.	-0.0632***
	(0.0171)
Union Membership x NWI	-0.150***
	(0.0299)
Western Imm.	0.0536***
	(0.00856)
NWI	0.122***
	(0.0149)
Total Union Membership Effect for NWI	0.0123
	(0.0584)
Observations	11,580,967

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4 when applying model weights. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for immigrant status, age group, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues.
To calculate the weights, we estimate our first stage regression separately for Norwegians and NWIs and include the individual binarized characteristics as covariates (see below) in the model and predict compliance. We then use the average propensity scores within each subgroup in the model (e.g. each binary cut) by immigration status to calculate inverse probability weights in which we reweigh the distribution of subgroup characteristics of the complier NWIs to match the characteristics of the complier Norwegians.
Individual characteristics include education bins (less than high school, high school, Bachelor's or more), white collar status, manufacturing and construction industry indicators, public sector status, collective agreement coverage, above- vs below-median firm-specific tenure, and each worker's quartile of the earnings distribution.

Table A6: Effect of Union Membership on Career Outcomes
Baseline Specification with Education Control

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.0993**	4.629***	-14,957***	-6,363	0.111***	-0.120***
	(0.0440)	(0.855)	(1,884)	(5,305)	(0.0248)	(0.0256)
Union×Western Immigrant	-0.0623***	-0.377	2,042***	6,319***	-0.00719	0.0119*
	(0.0174)	(0.237)	(573.2)	(1,482)	(0.00638)	(0.00683)
Union×Non-Western Immigrant	-0.0918***	-0.183	7,159***	-10,284***	-0.0346***	-0.0220**
	(0.0235)	(0.370)	(987.4)	(2,332)	(0.00991)	(0.0109)
Observations	11,552,820	10,675,763	12,433,594	12,448,584	12,480,263	12,480,263
Kleibergen-Paap F stat	156.29	120.23	163.39	163.81	163.2	163.2

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4 with additional education control. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for immigrant status by year, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A7: OLS Correlations between Union Membership and Career Outcomes

VARIABLES	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.0158***	0.652***	-290.8***	3,737***	-0.00370***	0.00171***
	(0.000641)	(0.0113)	(17.49)	(60.40)	(0.000259)	(0.000288)
Union×Western Immigrant	-0.0102***	-0.333***	-67.82	613.1***	0.0103***	0.00807***
	(0.00233)	(0.0406)	(57.22)	(210.2)	(0.000803)	(0.000900)
Union×Non-Western Immigrant	0.0602***	0.502***	-789.0***	4,078***	0.00668***	0.00374***
	(0.00174)	(0.0339)	(64.04)	(175.2)	(0.000669)	(0.000768)
Observations	11,642,166	10,749,095	12,537,473	12,554,065	12,595,270	12,595,270

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the OLS specification with all controls and fixed effects from Equations 3 and 4. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for immigrant status by year, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A8: Characteristics of Union Members in Survey, by Immigrant Status

VARIABLES	(1) Norwegians	(2) Western Immigrants	(3) Non-Western Immigrants
Age (% within Imm. Group)			
25-29	13.3	3	10.3
30-34	12.8	10.1	15.5
35-39	12.4	9.1	25.9
40-44	12.2	17.2	22.4
45-49	12.8	17.2	8.6
50-54	13	13.1	6
55-59	12.3	19.2	8.6
60-64	11.2	11.1	2.6
Region of Norway (% within Imm. Group)			
Mid-Norway	15.2	10.1	10.3
Northern Norway	10	8.1	6.9
Oslo	14.4	22.2	25
Southern Norway	11.6	15.2	12.1
Western Norway	19.4	16.2	14.7
Eastern Norway	29.5	28.3	31
Industry (% within Imm. Group)			
Other Services	3.5	5.1	6.9
Other	16.7	9.1	12.9
Construction	3.2	3	0.9
Professional, scientific and technical services	4.1	6.1	4.3
Financial and insurance activities	2.3	2	0
Health and Social Services	20.9	19.2	29.3
Manufacturing	5.8	3	4.3
Information and communication	3.5	2	5.2
Cultural, entertainment and leisure activities	1.5	5.1	0
Public administration and defence	11.6	11.1	12.1
Transport and storage	3.5	5.1	3.4
Education	14.2	17.2	15.5
Trade, repair of motor vehicles	3.5	6.1	1.7
Full-Time Share	88.4	82.8	88.8
Mean Work Experience in Norway	20.5	21.8	16.5
Share Female	55.7	57.6	59.5

Source: Authors' calculations of the characteristics of union members in the first survey wave.

Table A9: Effect of Union Membership on Career Outcomes
Interacting with labour Market Concentration

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.0688	3.602***	-11,018***	-6,069	0.169***	-0.114***
	(0.0420)	(0.807)	(1,915)	(4,995)	(0.0235)	(0.0244)
Union×HighHHI	0.255***	3.838***	-20,257***	-1,298	-0.160***	-0.0216
	(0.0294)	(0.548)	(1,395)	(3,158)	(0.0155)	(0.0166)
Union×Western Imm	-0.104***	-0.660*	4,200***	10,476***	0.00204	0.0350***
	(0.0279)	(0.338)	(970.9)	(2,487)	(0.0115)	(0.0115)
Union×Non-Western Imm	-0.117***	0.706	14,124***	-12,732***	-0.0338**	0.0402**
	(0.0325)	(0.459)	(1,578)	(3,328)	(0.0146)	(0.0159)
Union×Western Imm*HighHHI	0.0222	0.431	-4,388	517.9	-0.0759***	-0.00617
	(0.0686)	(1.011)	(2,965)	(6,003)	(0.0278)	(0.0285)
Union×Non-Western Imm*HighHHI	-0.424***	-4.742***	-13,679**	-25,257*	-0.228***	-0.425***
	(0.140)	(1.611)	(6,390)	(13,665)	(0.0654)	(0.0866)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	61.47	49.27	63.98	64.51	64.54	64.54

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A10: Effect of Union Membership on Career Outcomes
Baseline Specification with Immigrant Group by Year FE

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.140***	4.296***	-15,470***	-4,293	0.127***	-0.114***
	(0.0443)	(0.857)	(1,904)	(5,332)	(0.0251)	(0.0257)
Union × Western Immigrant	-0.0630***	-0.435*	1,776***	5,970***	0.0102*	0.0285***
	(0.0170)	(0.229)	(558.5)	(1,431)	(0.00621)	(0.00646)
Union × Non-Western Immigrant	-0.0695***	-0.207	4,149***	-10,748***	0.0232**	0.0161
	(0.0240)	(0.382)	(998.8)	(2,359)	(0.0103)	(0.0107)
Total Effect for Immigrant Groups After Linear Combination of Coefficients						
Western Immigrant	0.0767	3.861***	-13,694***	1,677	0.137***	-0.0852***
	(0.0473)	(0.888)	(1,986)	(5,532)	(0.0261)	(0.0267)
Non-Western Immigrant	0.0702	4.089***	-11,321***	-15,041**	0.150***	-0.0976***
	(0.0555)	(1.036)	(2,355)	(6,518)	(0.0304)	(0.0311)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	156.22	118.63	161.87	162.18	161.49	161.49

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4 with additional interactions for immigrant group by year dummies. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for immigrant status by year, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A11: Marginal Effect of Union Membership on Career Outcomes
By labour Market Concentration with Immigrant by Year FE

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Natives in Low HHI Firms	0.0925**	3.469***	-11,327***	-3,957	0.172***	-0.105***
	(0.0423)	(0.813)	(1,945)	(5,023)	(0.0237)	(0.0244)
Western Imm in Low HHI Firms	-0.0128	2.824***	-7,409***	5,667	0.183***	-0.0646**
	(0.0497)	(0.874)	(2,134)	(5,481)	(0.0259)	(0.0264)
Non-Western Imm in Low HHI Firms	0.00431	4.284***	-251.1	-13,565**	0.176***	-0.0384
	(0.0549)	(0.981)	(2,566)	(6,239)	(0.0291)	(0.0302)
Natives in High HHI Firms	0.353***	7.236***	-31,365***	-4,717	0.00965	-0.126***
	(0.0596)	(1.173)	(2,749)	(6,990)	(0.0334)	(0.0344)
Western Imm in High HHI Firms	0.271***	7.018***	-32,462***	4,534	-0.0376	-0.0802*
	(0.0854)	(1.483)	(3,880)	(8,794)	(0.0413)	(0.0427)
Non-Western Imm in High HHI Firms	-0.0641	3.377	-40,427***	-30,912**	-0.132**	-0.423***
	(0.142)	(2.066)	(7,336)	(14,525)	(0.0658)	(0.0817)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	62.24	48.28	63.72	64.15	64.06	64.06

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates represent the marginal effects from the two-stage least squares specification in Equations 3 and 4, and are interpreted independently. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for immigrant status by year, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues.

Table A12: Effect of Union Membership on Career Outcomes
Alternative Immigration Categorization

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.115***	4.443***	-15,115***	-6,641	0.122***	-0.124***
	(0.0440)	(0.852)	(1,884)	(5,306)	(0.0249)	(0.0256)
Union × Scandinavian Imm	-0.0523**	-0.516*	1,965***	6,862***	0.0162*	0.0206**
	(0.0225)	(0.293)	(760.6)	(2,048)	(0.00860)	(0.00903)
Union×Western Imm	-0.0753***	-0.373	2,401***	6,130***	-0.0106	0.0240**
	(0.0253)	(0.342)	(827.4)	(2,044)	(0.00906)	(0.00973)
Union×Non-Western Imm	-0.102***	-0.290	7,166***	-14,424***	-0.0160	-0.0120
	(0.0250)	(0.385)	(1,043)	(2,522)	(0.0106)	(0.0115)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	117.67	90.46	122.69	122.95	122.45	122.45

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. The coefficients on the interaction terms (Union × Scandinavian Imm, Union × Western Imm, Union × Non-Western Imm) represent relative differences in the union effect compared to natives.

Table A13: Effect of Union Membership on Career Outcomes
Interacting with Immigrant Worker Share

	(1) Log total earnings	(2) Hours	(3) Unemployment benefits	(4) Sick leave benefits	(5) Promotion	(6) Firm upgrade
Union	0.117***	4.864***	-17,104***	-5,674	0.133***	-0.114***
	(0.0444)	(0.869)	(1,923)	(5,338)	(0.0251)	(0.0259)
Union×High ImmWorker	0.0102	-0.744***	4,147***	-311.1	-0.0140***	-0.0438***
	(0.00876)	(0.151)	(386.3)	(1,016)	(0.00503)	(0.00539)
Union×Western Imm	-0.0555**	-0.219	500.7	4,771**	-0.0167*	-0.00184
	(0.0232)	(0.313)	(739.3)	(2,077)	(0.00914)	(0.00954)
Union×Non-Western Imm	0.0141	-0.752	-1,539	-6,470*	-0.0143	-0.0804***
	(0.0349)	(0.555)	(1,567)	(3,678)	(0.0166)	(0.0177)
Union×Western Imm*High ImmWorker	-0.0317	-0.123	3,207***	3,864	0.0286**	0.0479***
	(0.0316)	(0.455)	(1,179)	(3,070)	(0.0140)	(0.0154)
Union×Non-Western Imm*High ImmWorker	-0.195***	1.411**	15,242***	-16,145***	-0.0247	0.0827***
	(0.0450)	(0.709)	(2,217)	(4,849)	(0.0215)	(0.0237)
Observations	11,640,111	10,747,267	12,535,083	12,551,670	12,592,849	12,592,849
Kleibergen-Paap F stat	78.92	59.77	82.38	82.57	82.3	82.3

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the two-stage least squares specification in Equations 3 and 4. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A14: Marginal Earnings Effect of Union Membership on Log Earnings by Firms' Baseline Union Density and NWI Share

	(1) X = Pre-2004 Union Density	(2) X = NWI Shares
Natives in Low X Firms	0.0894**	0.117***
	(0.0435)	(0.0444)
Western Imm in Low X Firms	-0.00422	0.0620
	(0.0472)	(0.0517)
Non-Western Imm in Low X Firms	0.00819	0.132**
	(0.0559)	(0.0628)
Natives in High X Firms	0.101**	0.128***
	(0.0436)	(0.0433)
Western Imm in High X Firms	0.0454	0.0405
	(0.0461)	(0.0498)
Non-Western Imm in X Firms	0.0491	-0.0530
	(0.0511)	(0.0608)
Observations	10,436,593	11,640,111

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates represent the marginal effects from the two-stage least squares specification in Equations 3 and 4, and are interpreted independently. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. Current union status is instrumented by the base firm's net union dues. Column (1) definitions of high versus low are above- and below-median union density in the firm before the expansion of the European Union (pre-2004) in the industry. In column (2), it is defined as above- vs below-median share of non-Western immigrants in the firm.

Table A15: Firm-Level Descriptive Statistics by Immigrant Worker Share

	Below median	Above median
Firm labour market power (HHI)	0.06	0.05
Public sector	0.08	0.09
Firm size	15.33	43.62
Firm union density	0.19	0.16
Share with BA	0.25	0.28
Average Earnings	417,295	400,083
Average Hours	31.95	32.16
Average Unemployment benefit	3,601	4,394
Average Sick leave benefit	14,452	15,255
Average Promotion	0.07	0.09
Average Firm upgrade	0.13	0.15
Observations (firm-year)	1,372,990	462,229

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Median immigrant worker share at a firm is 0.11. Education statistics of immigrants are self-reported. Earnings are measured as pre-tax income from labour and self-employment. Firm labour market power (HHI) is the sum of squared employment shares across firms in each occupation and labour market, and ranges from 0 to 1. A value of 1 implies a perfectly monopolistic market, and a value of 0 implies a perfectly competitive market. We fix each firm's HHI at the first year in which the firm appears in the data. Unemployment benefit and sick leave benefit are calculated based on the cumulative amount of benefits received in a given year.

Table A16: OLS Correlations between Union Membership and Career Outcomes, Including Controls Present in the IV Reweighting Exercise

VARIABLES	(1) Log Total Earnings	(2) Hours	(3) Unemployment Benefits	(4) Sick Leave Benefits	(5) Promotion	(6) Firm Upgrade
Union Membership	-0.00502***	0.192***	-114.5***	3,243***	-0.00290***	0.00399***
	(0.000437)	(0.00955)	(17.35)	(60.47)	(0.000259)	(0.000286)
Union Membership x Western Imm.	-0.0161***	-0.317***	-115.4**	568.9***	0.00771***	0.00589***
	(0.00156)	(0.0340)	(58.10)	(213.4)	(0.000818)	(0.000909)
Union Membership x NWI	0.0137***	-0.0329	-765.7***	3,587***	0.00565***	0.00422***
	(0.00113)	(0.0308)	(65.29)	(178.6)	(0.000684)	(0.000780)
Total Union Membership Effect, NWI	0.0087***	0.1595***	-880.19***	6830.10***	0.0027***	0.0082***
	(0.0011)	-0.0301	(65.30)	(174.46)	(0.0007)	-0.0008
Observations	11,570,476	10,690,731	12,453,996	12,469,115	12,501,591	12,501,591

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from the OLS specification with all controls and fixed effects from Equations 3 and 4. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for immigrant status by year, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. We also add controls for the characteristics in our reweighting exercise: binarized characteristics of education (less than high school, high school, Bachelors+), white collar status, work in the manufacturing, construction, and public sectors, coverage by collective agreements, firm-specific tenure, and income quartile. The coefficients on the interaction terms (Union × Western Immigrant, Union × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A17: Public Goods Effect of Union Density on Non-Switchers

Outcome: Log Annual Earnings	(1) Pooled	(2) Total Effect	(3) x Always Union	(4) Total Effect
Predicted Union Density (Norwegians)	0.0045***	0.0045***	0.00352***	0.00352***
	(0.00117)	(0.00117)	(0.00121)	(0.00121)
Predicted Union Density x Western Imm.	-0.00113***	0.0025**	-0.001***	0.0025**
	(0.000232)	(0.0012)	(0.000227)	(0.00122)
Predicted Union Density x NWI	-0.00203***	0.0015	-0.00199***	0.0015
	(0.000472)	(0.0012)	(0.000462)	(0.0012)
Predicted Union Density (Norwegians) x Always Union			0.000905***	0.0044***
			(0.000222)	(0.00117)
Predicted Union Density x Western Imm. X Always Union			-0.000140***	0.0033***
			(0.0000521)	(0.0012)
Predicted Union Density x NWI x Always Union			0.000990***	0.0034***
			(0.0000538)	(0.0012)
	10,470,765	10,470,765	10,470,765	10,470,765

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.
Notes: Estimates come from calculating the predicted union density from Equation 4 and estimating a variation of Equation 3 in which we replace individual union membership with union density and limit the estimation sample only to those who did not switch their union status over the prior three years (a simple proxy for likely compliers). Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status. The coefficients on the interaction terms (Union Density × Western Immigrant, Union Density × Non-Western Immigrant) represent relative differences in the union effect compared to natives.

Table A18: Comparison of Baseline to Inclusion of Firm-Level Union Instrumented Controls

	(1)	(2)
First Stage Coefficient, Net Dues on Own Membership	-0.1147***	-0.0961***
	(0.0053)	(0.00756)
Second Stage Coefficient, Own Membership	0.115**	0.155**
	(0.0447)	(0.0635)
First Stage F Stat	466.05	96.0
Observations	11,640,138	11,429,145

Source: Authors' calculations of Norwegian registry data from 2001 to 2015.

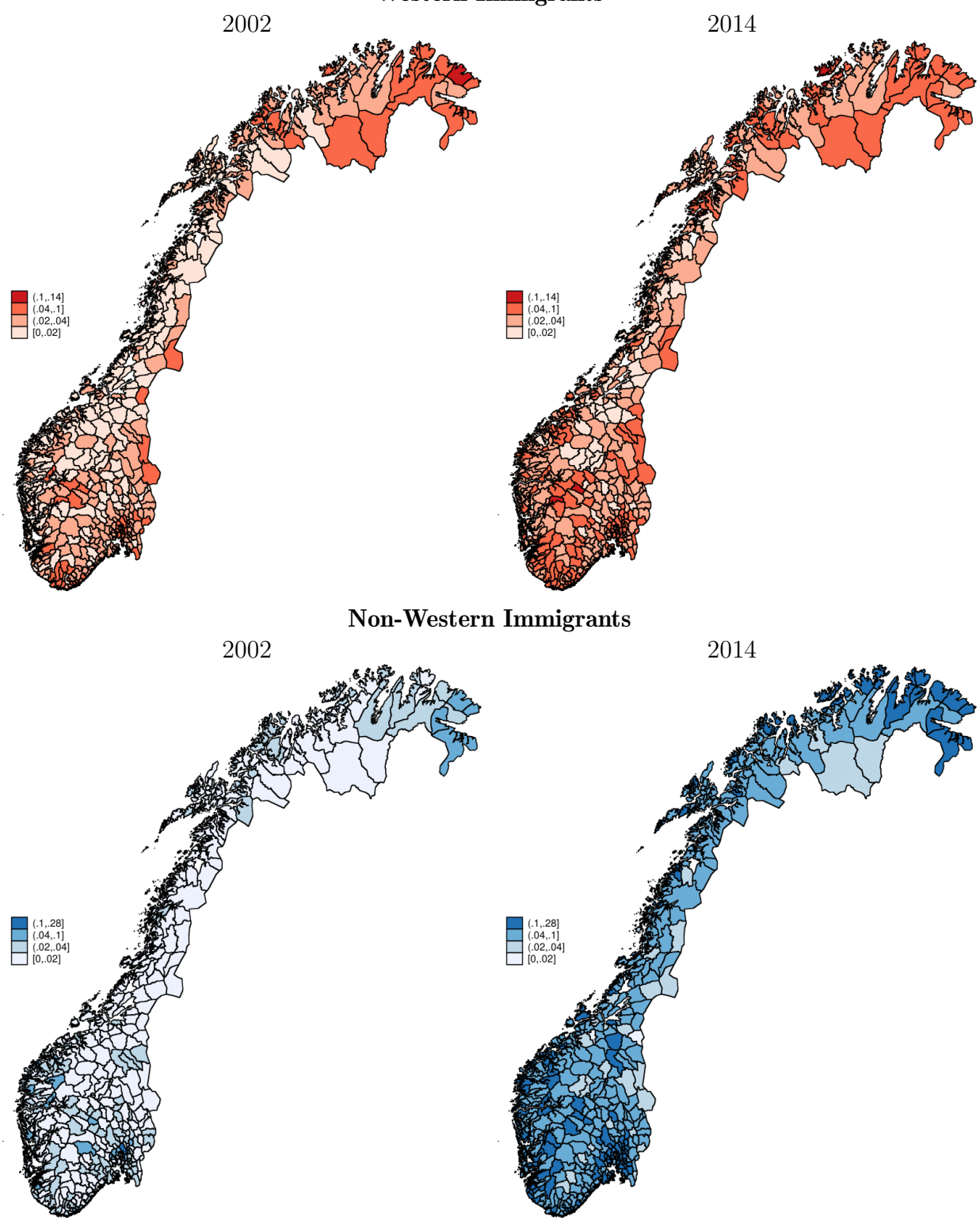
Notes: Estimates come from the two-stage least squares specification in Equations 4 and 3 for column (1). In column (2), we instrument for leave-out firm-level union density by constructing a firm-level analog of our individual instrument. We define this by fixing the firm-level base imputed dues as the average imputed dues within the firm in the *firm's* first year in the data (rather than the worker's first year), which we then adjust to nominal NOK and calculate the net dues based on these imputed dues using the same adjustments as the individual instruments. Standard errors are clustered at the individual level. Outcomes are measured in year *t+1*. The model includes fixed effects for year, immigrant status, base and current occupation-by-industry cell, base and current firm, base and current age group, and always union status.

Table A19: Characteristics of Workers and Managers in Union and Membership organisations

	(1) All Industries	(2) Unions and Membership organisations
2002 % Norwegian	91.7	92.7
2002 % NWI	5.2	2.9
2010 % Norwegian	86.4	90.6
2010 % NWI	9.5	4.7
Average Share NWI of Prior Years' Coworkers, Norwegians	0.055	0.037
Management Occupations		
2002 % Norwegian	95.2	96.2
2002 % NWI	1.3	0.9
2010 % Norwegian	93.7	95.3
2010 % NWI	2.2	1.7
Average Share NWI of Prior Years' Coworkers, Norwegian Managers	0.041	0.031

Figures Appendix

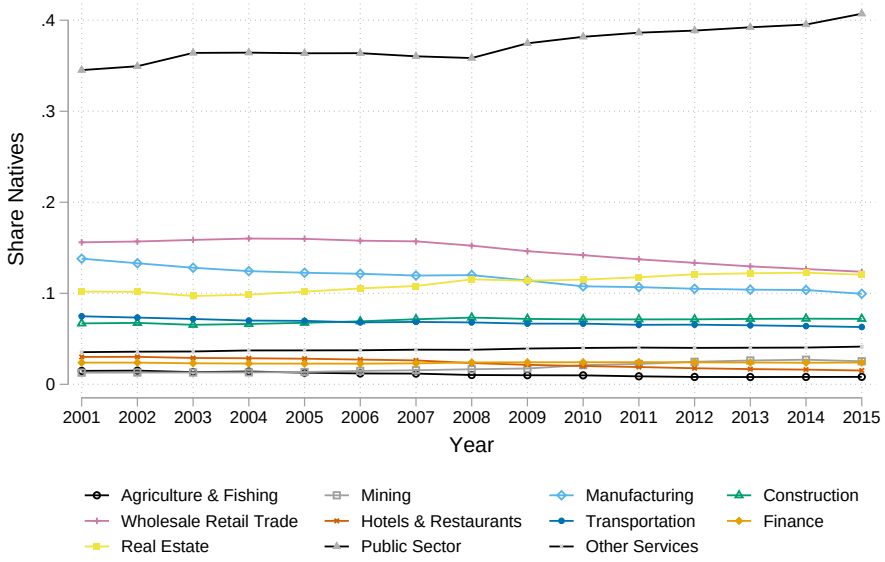
Figure A1: Share of Immigrants in Municipalities by Immigration Status



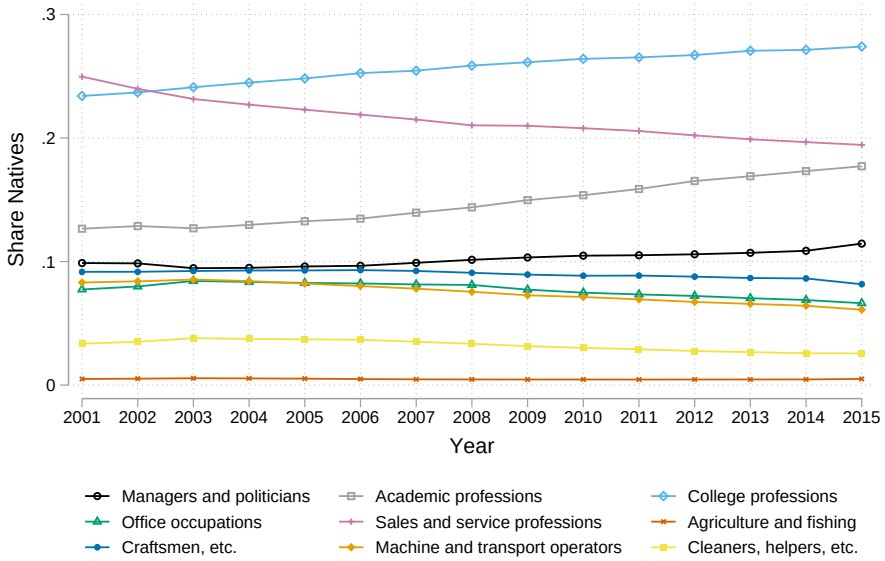
Source: Authors' calculations of Norwegian registry data.

Figure A2: Native Worker Industry and Occupation Share Over Time

Panel A: Industry

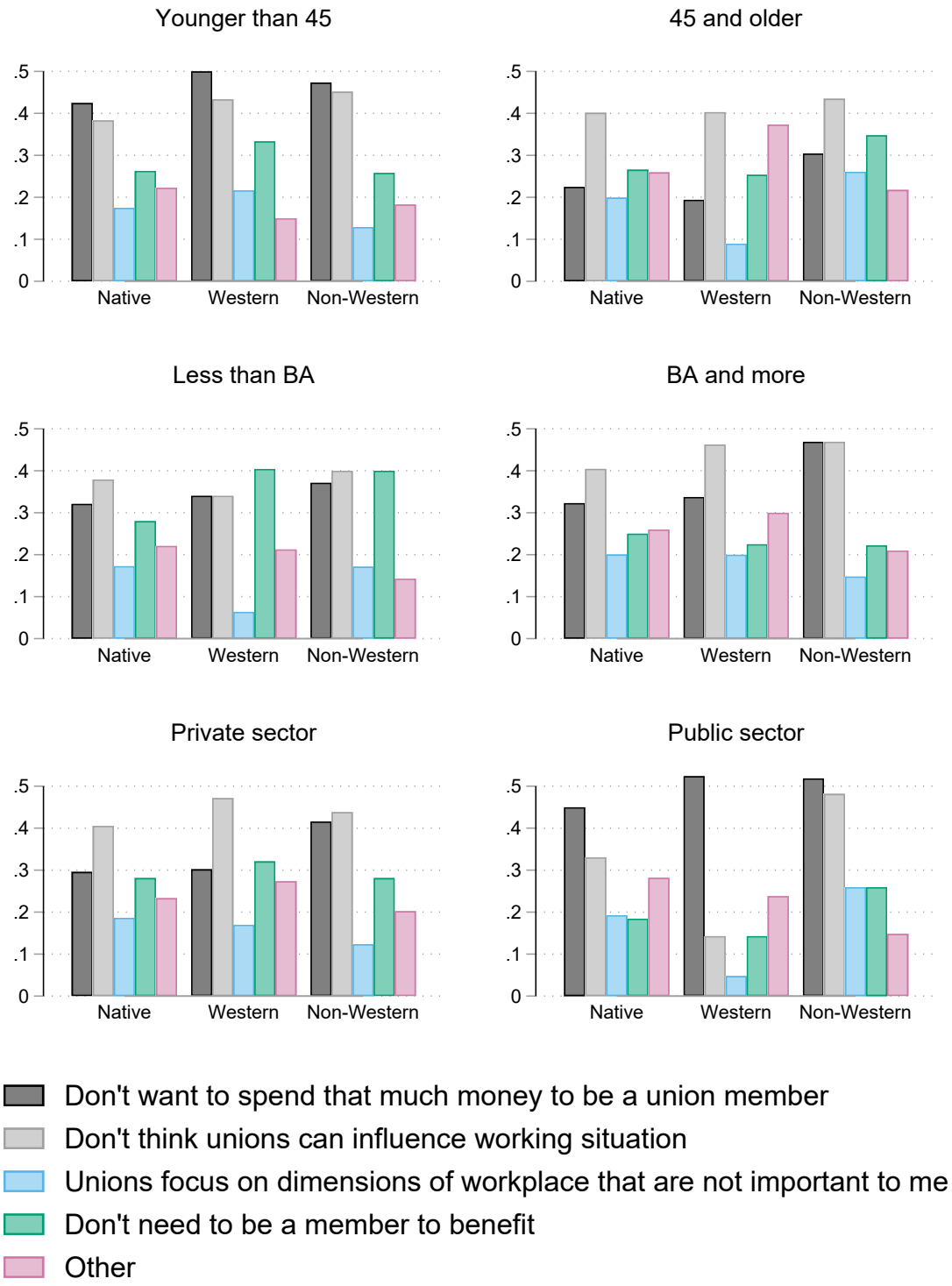


Panel B: Occupation



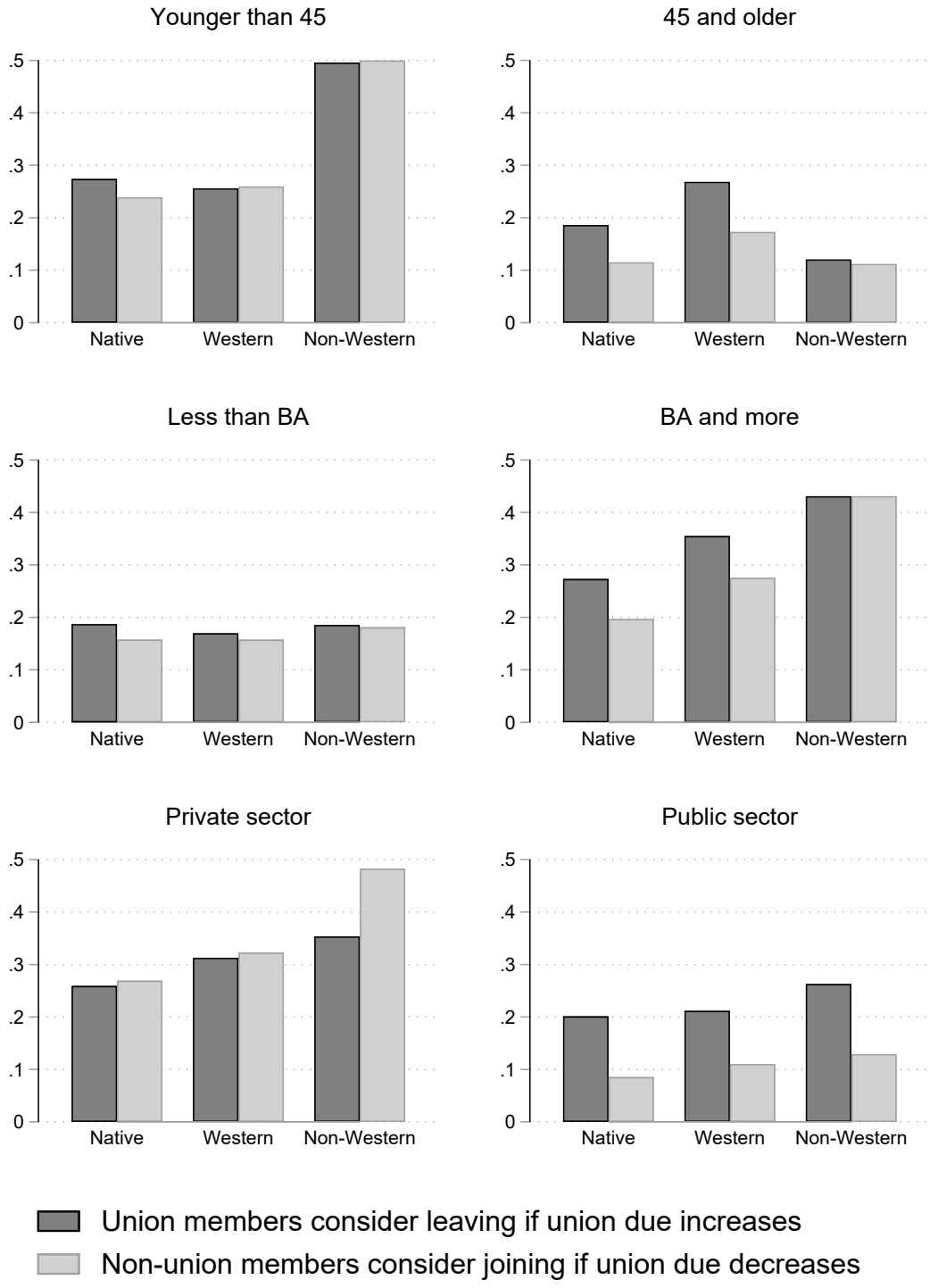
Source: Authors' calculations of Norwegian registry data from 2001 to 2015.

Figure A3: Nonunionized Workers Reason For Not Unionizing by Immigration Status, Age, Education, and Sector (Survey One)



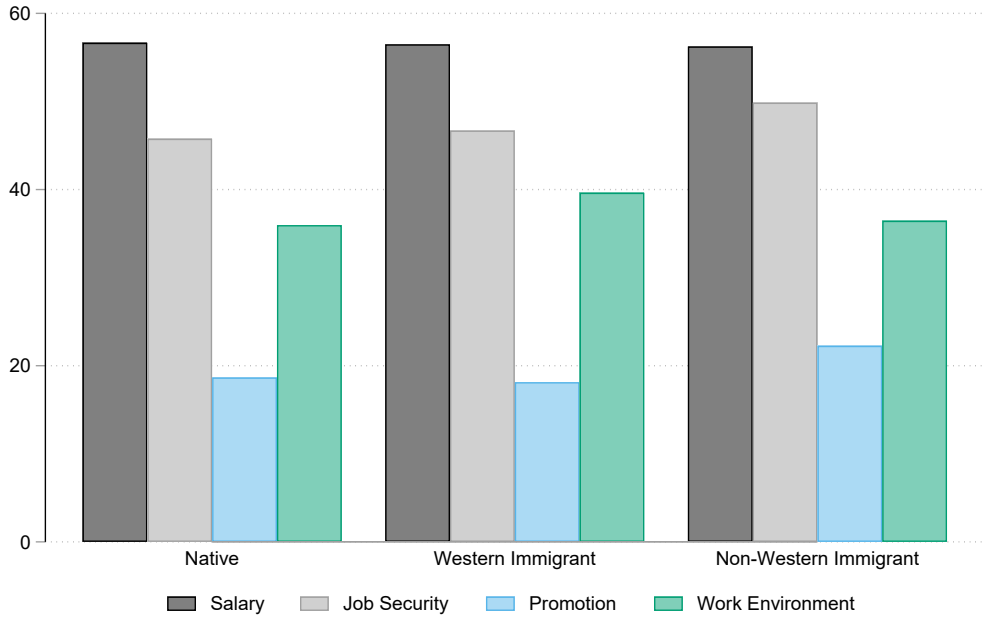
Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "The purpose of this question is to understand the reason why you do not join a union. Check all the boxes that apply."

Figure A4: Price Sensitivity to Union Membership by Immigration Status, Age, Education, and Sector (Survey One)



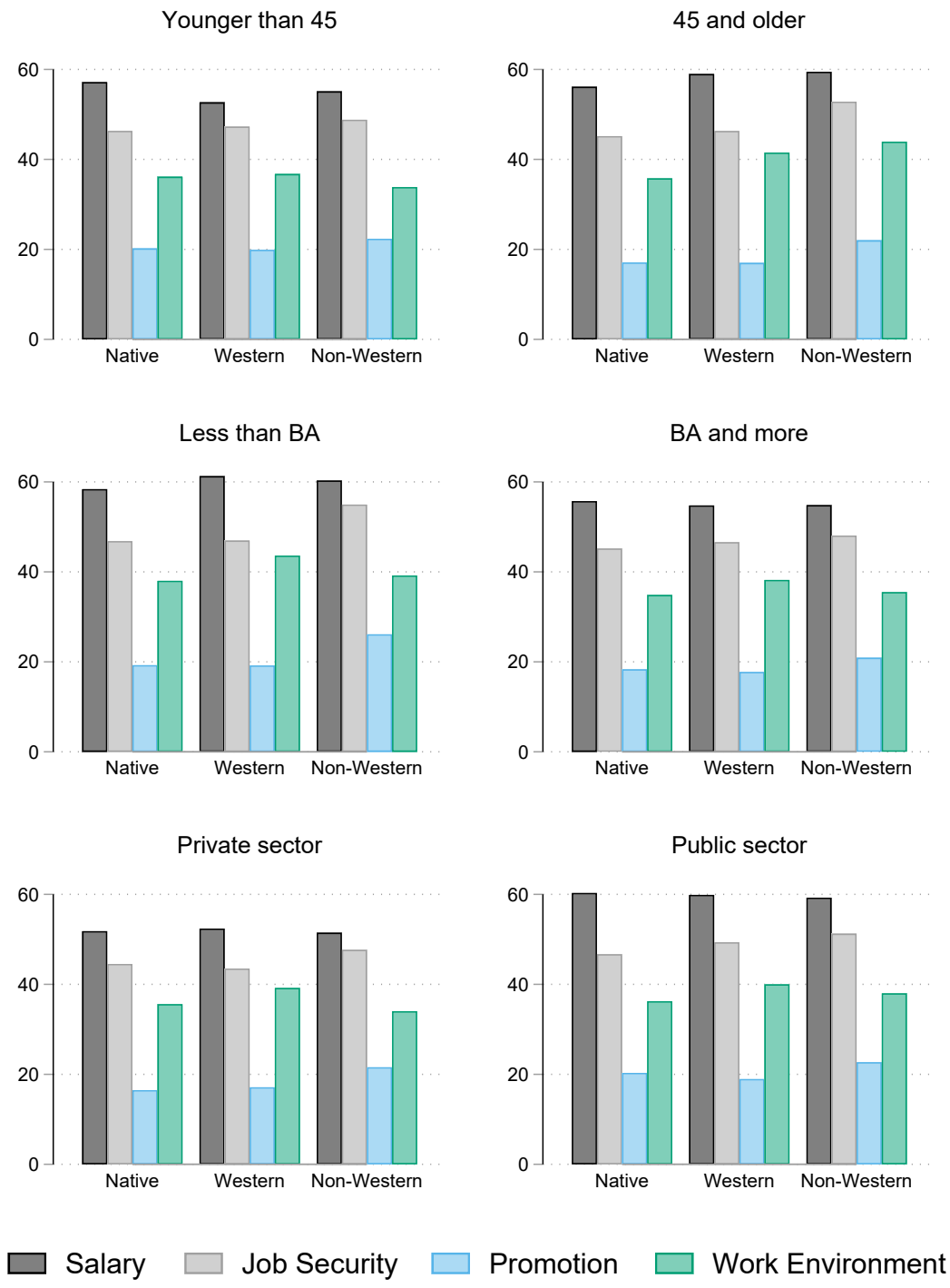
Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "If your after-tax dues for union membership were reduced [increased] by [XYZ] NOK, would you reconsider your decision to join a union?" When we control for each of these categories (age group, public sector, and education), the average NWI is still 9.9 percentage points (s.e. = 3.3) more likely than Norwegians to reconsider their union membership if the price of union membership increased/decreased by at least 500 NOK. The average Western immigrant is 3.65 percentage points (s.e. = 3.3) more sensitive than Norwegians.

Figure A5: Union Member Perception of Union Influence Over Career Outcomes by Immigration Status (Survey One)



Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "How important do you think the union is to improving your pay, job security, promotion potential and work environment quality? 0 means 'not at all' and 100 means 'completely.' The total for all four need NOT be 100." T tests are performed separately for each outcome between natives and western immigrants, and natives and non-western immigrants. The t value is labelled if the difference is significant at the 95% level.

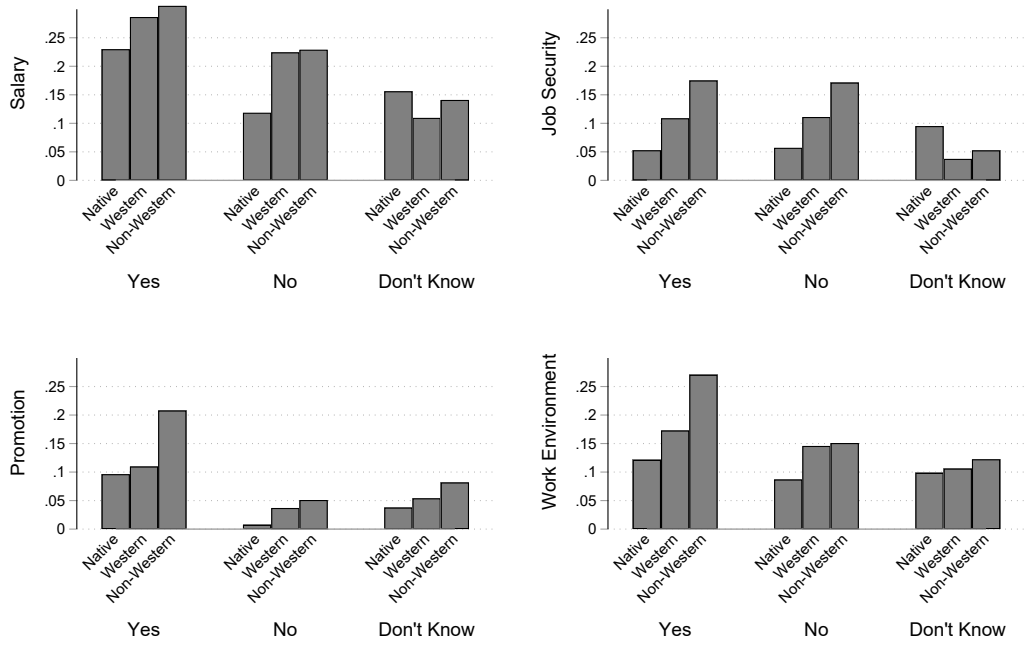
Figure A6: Union Member Perception of Union Influence Over Career Outcomes by Immigration Status, Age, Education, and Sector (Survey One)



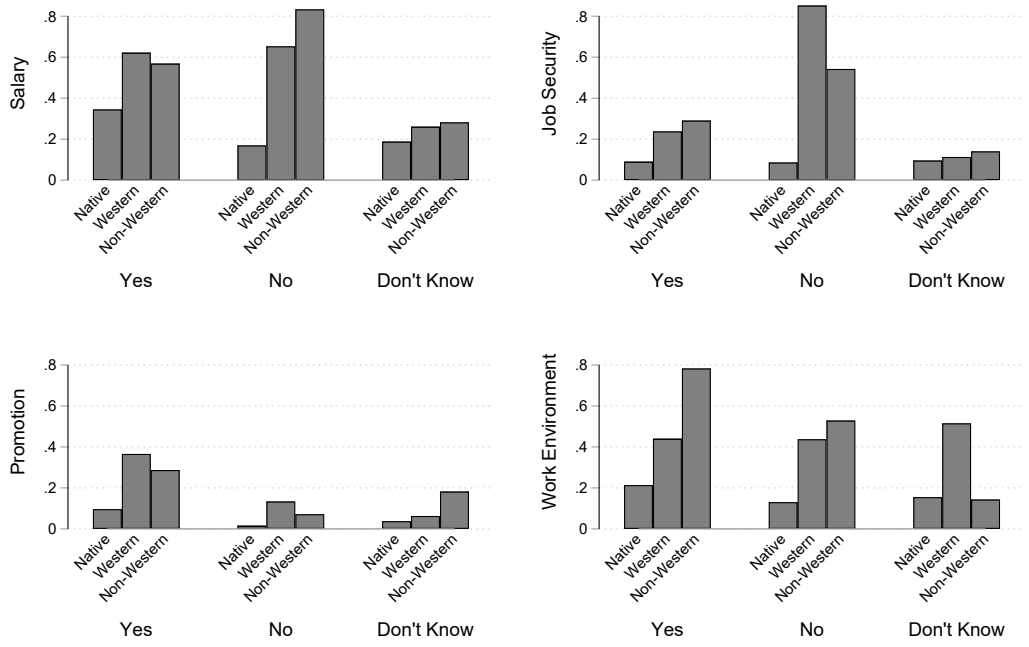
Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "How important do you think the union is to improving your pay, job security, promotion potential and work environment quality? 0 means 'not at all' and 100 means 'completely.' The total for all four need NOT be 100."

Figure A7: Rates of Contacting Union Representatives by Immigrant Status Conditional on Belief of Union Effectiveness (Survey Two)

Panel A: Share of Members Making Any Contact



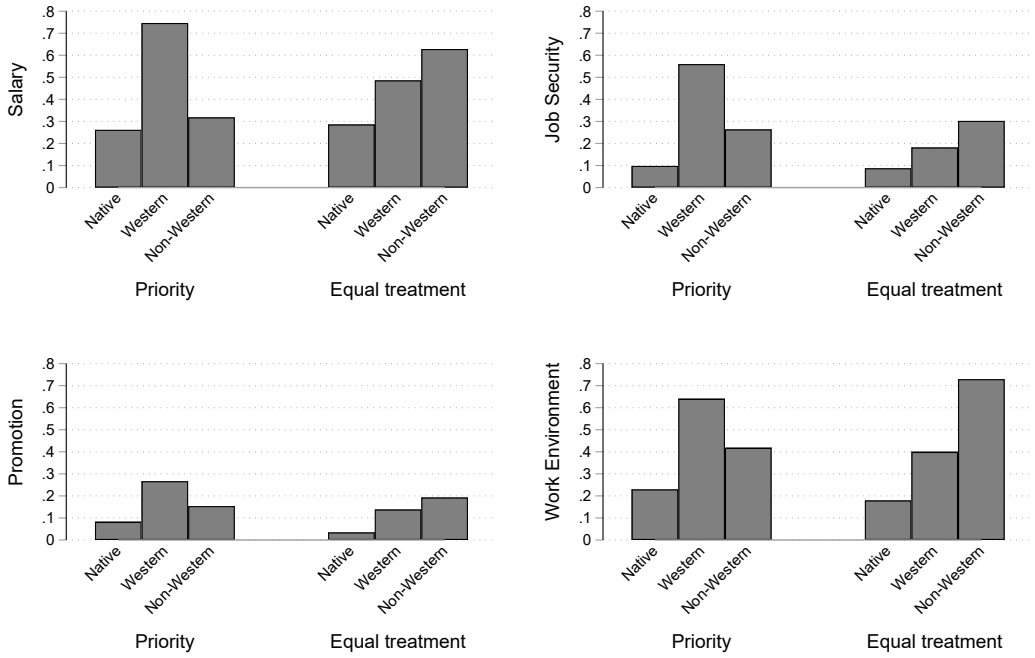
Panel B: Mean Number of Contacts



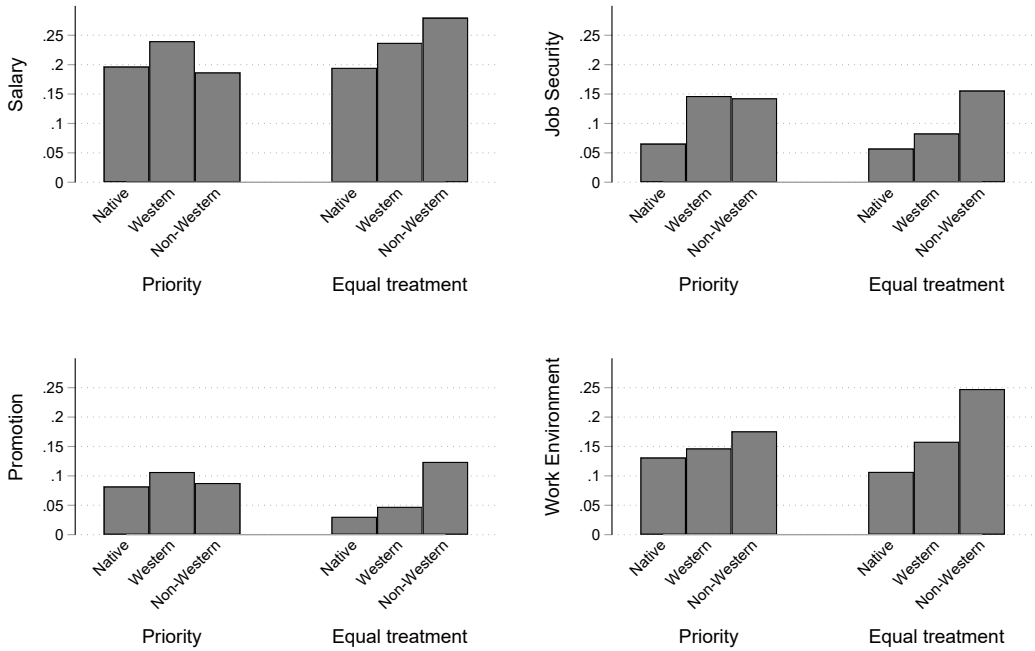
Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The effectiveness question on the survey asked, "If you were to contact your local union regarding any of the following matters, do you think they would be effective at helping you individually? Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality." The contact question on the survey asked, "In the past 12 months, how many times have you contacted your local union about the following aspects of your job? Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality." In all figures, the x-axis (Yes, No, Don't know) shows the answers to the effectiveness question.

Figure A8: Rates of Contacting Union Representatives by Immigrant Status Conditional on Belief of Union Treatment (Survey Two)

Panel A: Mean Number of Contacts

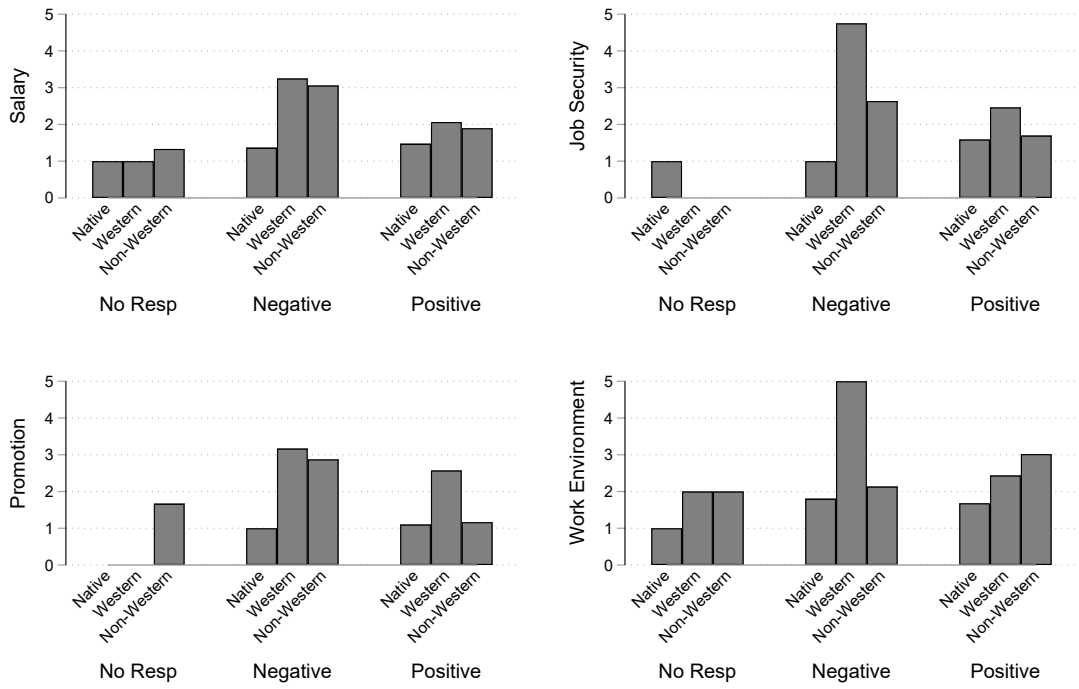


Panel B: Share of Members Making Any Contact



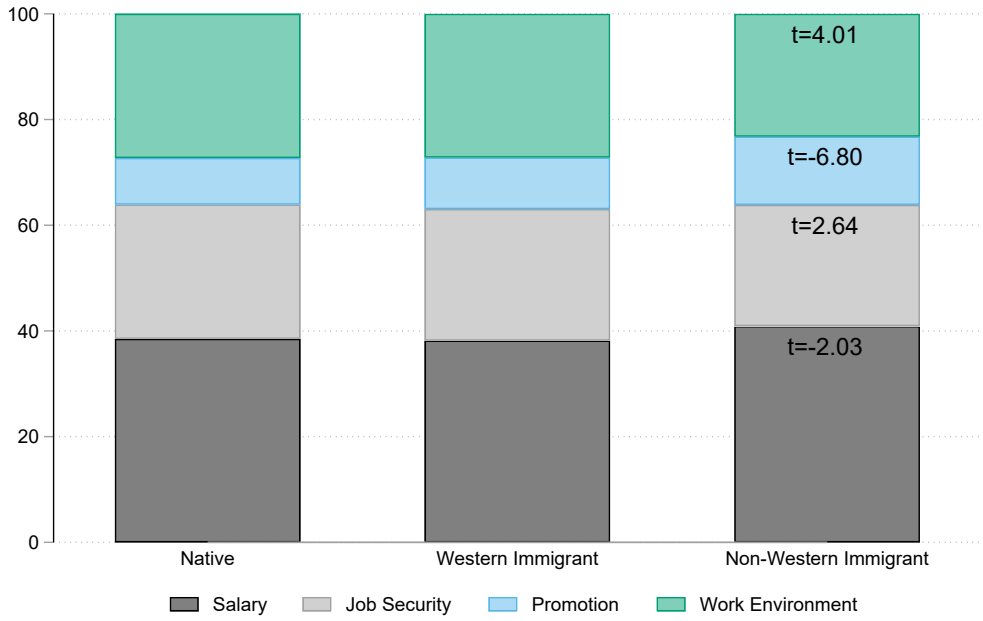
Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The union treatment question on the survey asked, "How do you think union members are treated at your workplace by the local union? I think all union members are treated equally by the local union. Or, I think some members are prioritised over other members by the local union." The contact question on the survey asked, "In the past 12 months, how many times have you contacted your local union about the following aspects of your job? Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality." In all figures, the x-axis (Priority, Equal treatment) shows the answers to the union treatment question.

Figure A9: Mean Number of Contacts by Immigrant Status Conditional on Experience (Survey Two)



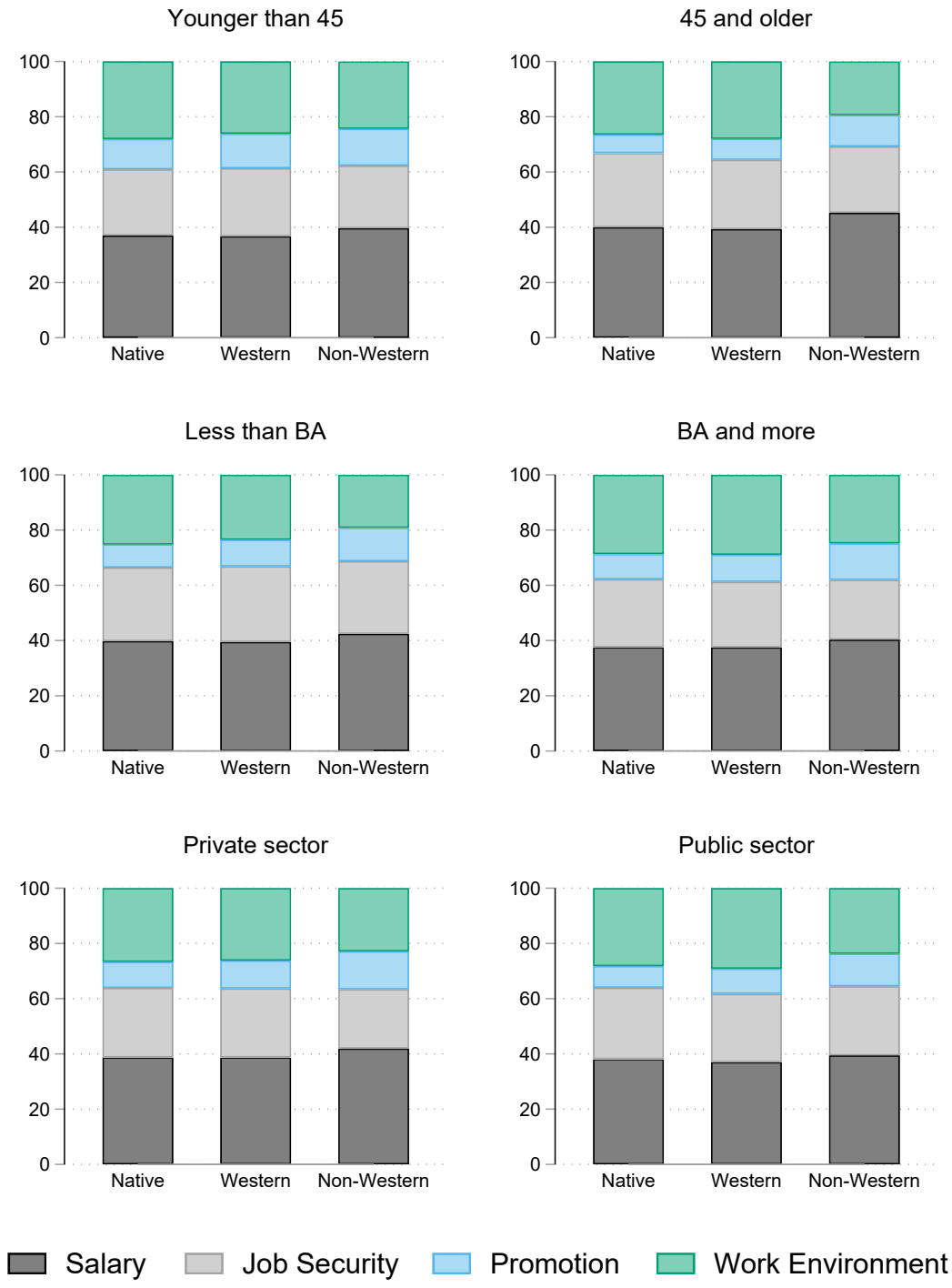
Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The contact question on the survey asked, "In the past 12 months, how many times have you contacted your local union about the following aspects of your job? Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality." The experience question on the survey asked, "Please rate your experiences with your local union." The x-axis (No resp, Negative, Positive) shows the answers to the experience question. Negative consists of very negative and somewhat negative; positive consists of somewhat positive and very positive.

Figure A10: Worker Valuation of Career Amenities by Immigration Status (Survey One)



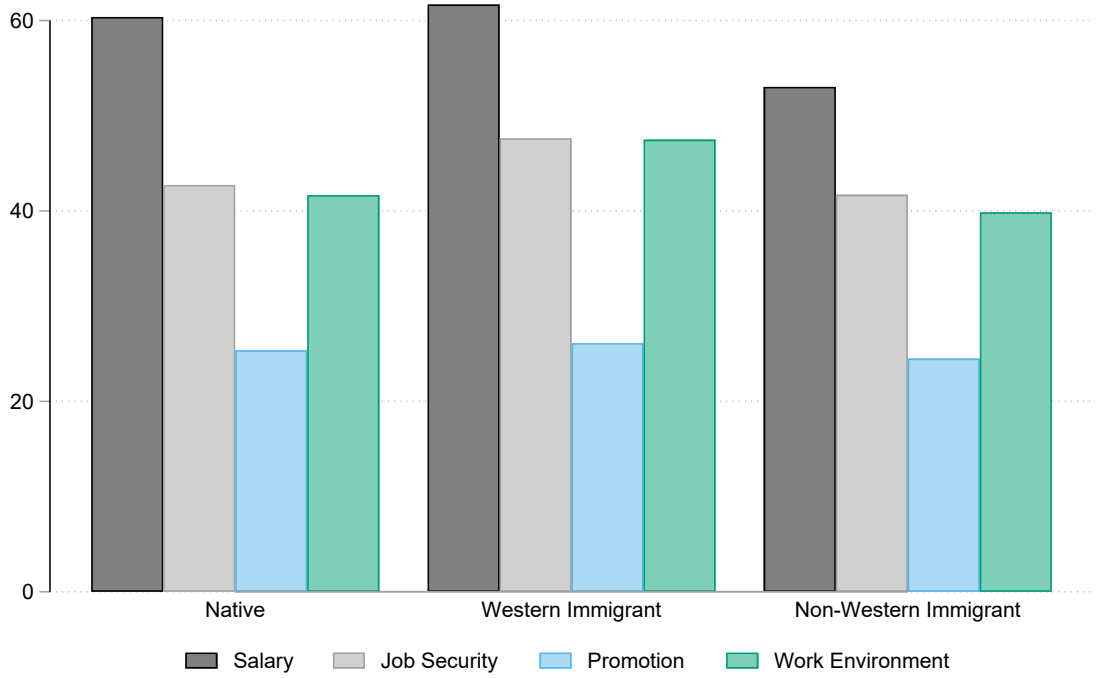
Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "Rank the following job characteristics based on importance to your future career and well-being: Salary, Job Safety, Promotion Potential and Work Environment Quality. Here we ask you to award 100 points across the four categories. You can assign anything between 0 and 100 to any of the categories, as long as the total amount of points for all four categories is 100." T tests are performed separately for each outcome between natives and western immigrants, and natives and non-western immigrants. The t value is labelled if the difference is significant at the 95% level.

Figure A11: Worker Valuation of Career Amenities by Immigration Status, Age, Education, and Sector (Survey One)



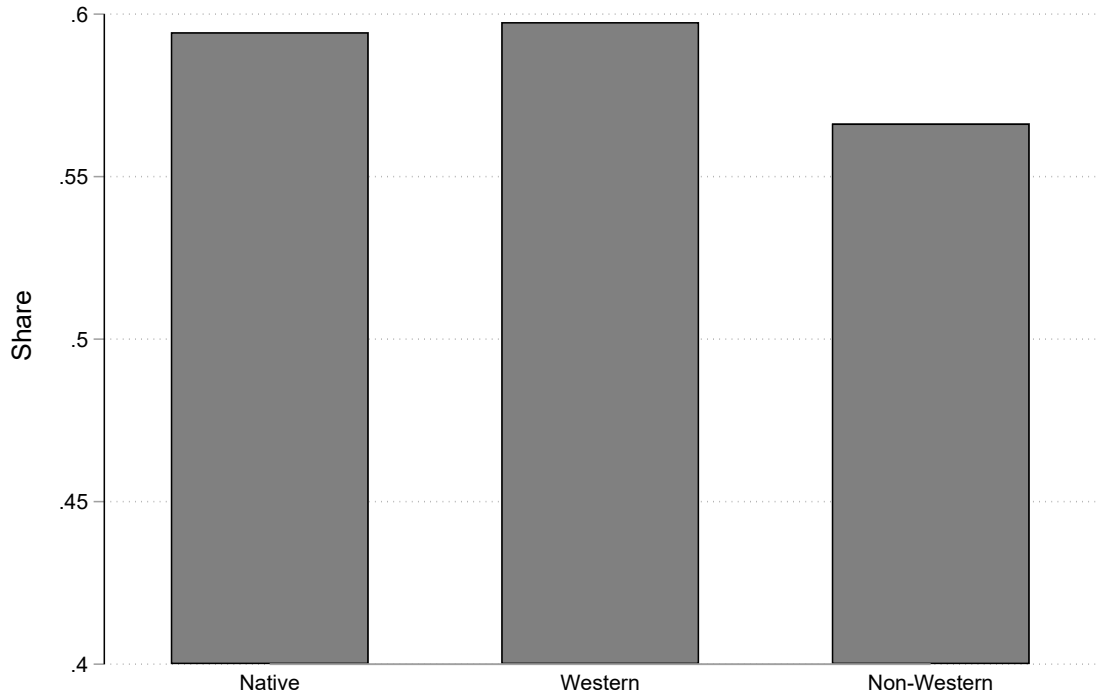
Source: Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "Rank the following job characteristics based on importance to your future career and well-being: Salary, Job Safety, Promotion Potential and Work Environment Quality. Here we ask you to award 100 points across the four categories. You can assign anything between 0 and 100 to any of the categories, as long as the total amount of points for all four categories is 100."

Figure A12: Perception of Union as a Public Good by Immigrant Status (Survey One)



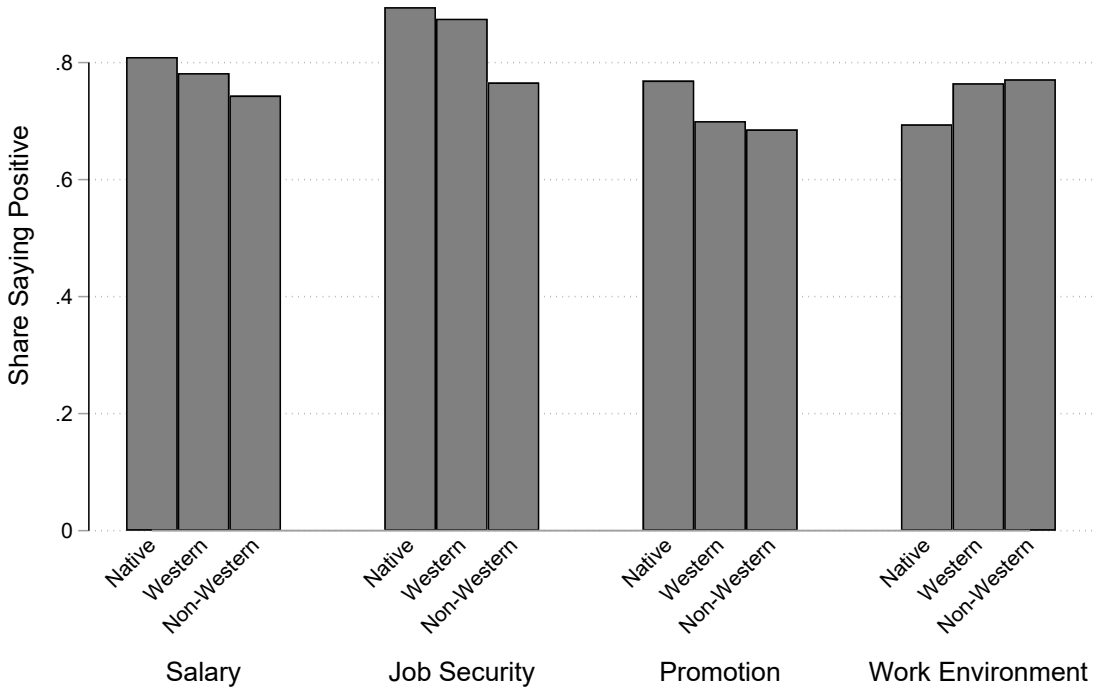
Authors' calculations based on survey data collected by NORSTAT on behalf of the authors.
Notes: The question on the survey asked, "Compared to members, the extent to which do you think non-members in your workplace can benefit from the presence of unions along these four dimensions: Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality. [0-100]"

Figure A13: Share of Members Agreeing that Union Membership Has Made Them Happier (Survey Two)



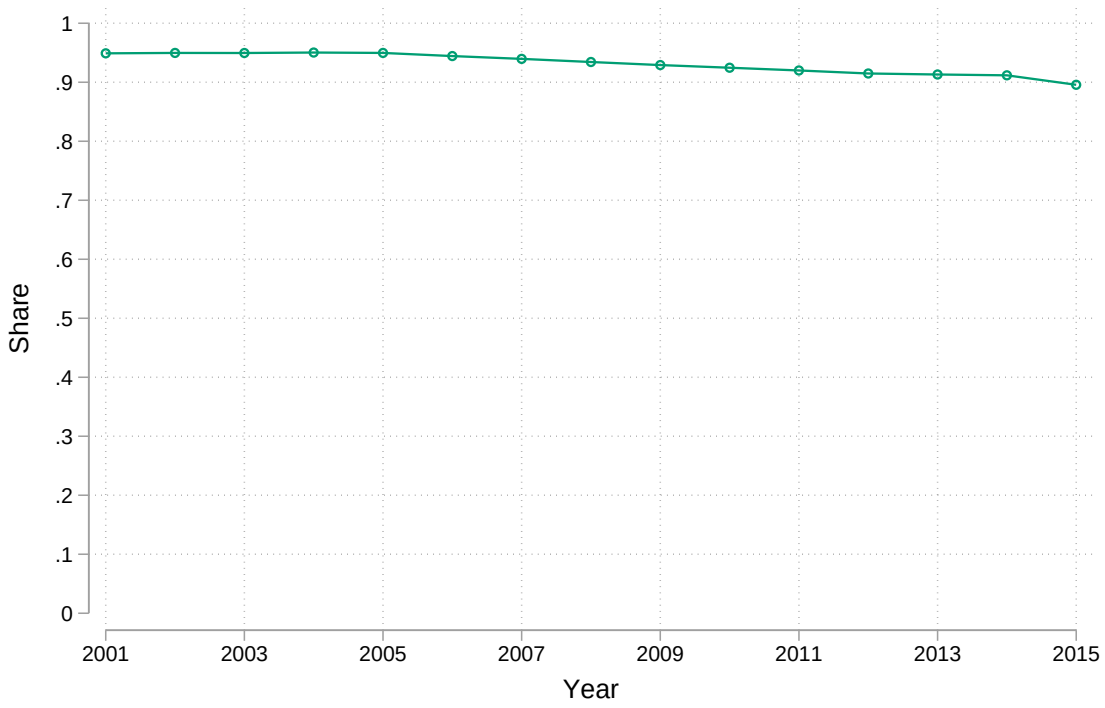
Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The question on the survey asked, "To what extent do you agree with the following statement? Being a union member has led to me feeling happier/more satisfied at work."

Figure A14: Share of Members Rating Their Experience After Contacting Their Union as Positive (Survey Two)



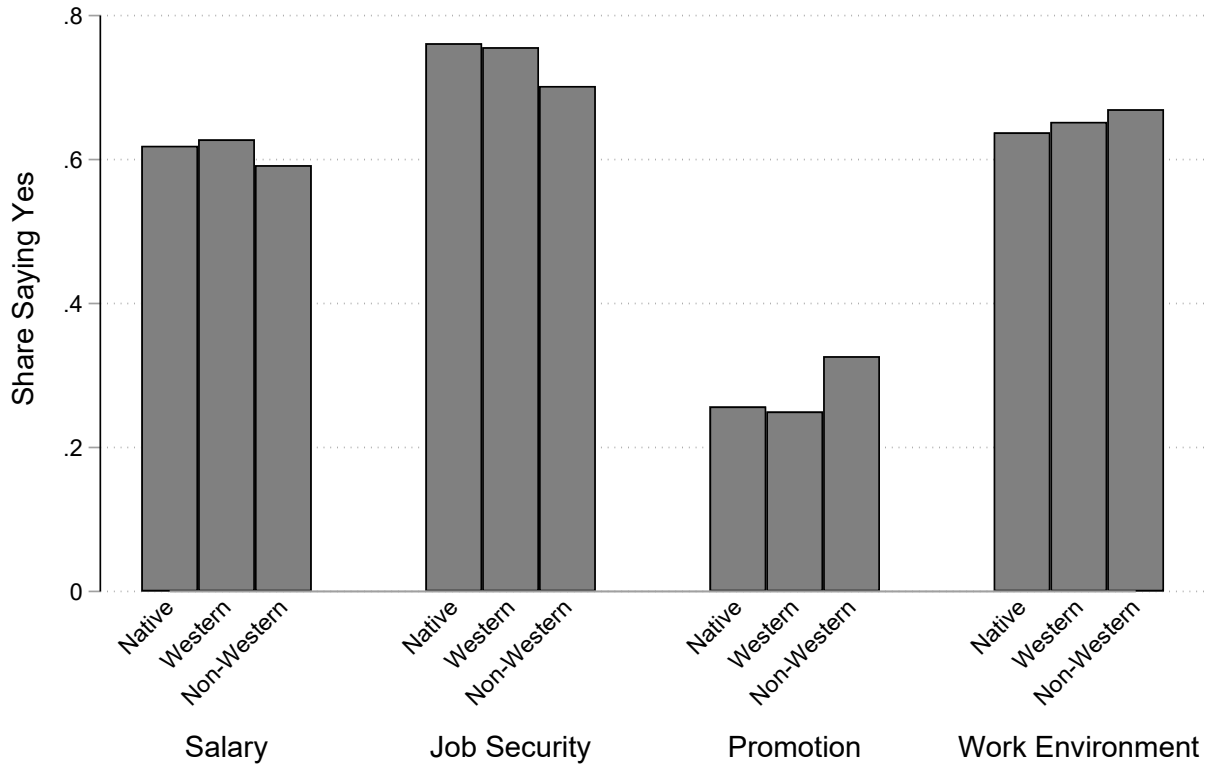
Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The question on the survey asked, "Please rate your experiences with your local union."

Figure A15: Share of Union Dues Paid by Natives, 2001-2015



Source: Authors' calculations of the average share of union dues paid by natives at the firm level in Norway over time using Norwegian register data.

Figure A16: Share of Members Stating the Union Would be Effective in Helping (Survey Two)



Source: Authors' calculations based on survey data collected by IPSOS on behalf of the authors.
Notes: The question on the survey asked, "If you were to contact your local union regarding any of the following matters, do you think they would be effective at helping you individually?"

First Survey Instrument

[INTRO1] This is a survey that Norstat conducts on behalf of the Norwegian School of Economics and Business Administration. The results will be used in a research project.

All information collected through the survey is anonymized and will not be disclosed to any third party. As part of scientific publishing, anonymised data may be shared in open scientific repositories.

If you want more information about the project, you can choose the option below. If you want to start the survey, you choose it.

[R1] I want more information
[R2] I want to start the survey

[R1] Information and declaration of consent

Purpose of the project

We want to understand how individuals in Norway value their work environment and how they view unions. The results of the study will increase our understanding of workplace preferences and their relative importance.

Who is responsible for the project?

The Norwegian School of Economics (NHH) is the responsible institution for the project. Alexander Willen, professor at NHH, is the project manager. The other project members are Kjell G. Salvanes, professor at NHH, Samuel Dodini, postdoctoral fellow vid NHH, and Julia Zhu, postdoctoral fellow at NHH. If you have any questions about the project, you can contact NHH via Alexander Willen (alexander.willen@nhh.no).

What does participation mean for you?

If you choose to participate in the project, you will be asked to answer a survey by completing an online questionnaire. It takes about 7 minutes. The survey includes questions about your work situation, union status, and your job preferences. In addition, we will ask some basic demographic questions about, for example, age and gender. Participation in the survey is voluntary and you can withdraw your consent at any time without giving any reason. All information collected through the survey is anonymized and will not be disclosed to any third party. As part of scientific publishing, anonymised data may be shared in open scientific repositories. There will be no negative consequences if you choose not to participate or decide to withdraw at a later date.

Declaration of consent

I have received and understood information about the survey and hereby consent:

• to participate in the online survey.
• to enable researchers to process my anonymised data and use them for publications in scientific journals and other scientific dissemination.

[R2] Survey

[Age] What is your age?
[Gender] Are you male or female?
[Zip code] What is your zip code?
[Fylke] Which county do you live in?

What is your highest completed education?
[R1] Primary school/primary school
[R2] Upper secondary school (incl. former vocational school)
[R3] Vocational school, trade certificate/journeyman's certificate and other 1-2 year education after upper secondary school
[R4] University/college up to 3 years (Bachelor's degree)
[R5] University/college 4 years or more (Master's degree and higher)
[R98] Other

Where were you born?
[R1] Norway
[R2] Outside Norway
[R3] Don't want to answer

Can you state which country you were born in?

At what age did you move to Norway?

How many years of full-time work experience do you have?

Are you currently in part-time or full-time work?
[R1] Part-time (less than 30 hours per week)
[R2] Full-time (at least 30 hours per week)
[R3] Not working

What industry is your main job in?

Do you work in the public or private sector?
[R1] Public sector
[R2] Private sector

How many people work at your workplace?
Row:
[R1] 1-5
[R2] 6-10
[R3] 11-50
[R4] 51-100
[R5] More than 100
[R6] Don't want to answer

Rank the following job characteristics based on importance to your future career and well-being: Salary, Job Safety, Promotion Potential and Work Environment Quality.
Here we ask you to award 100 points across the four categories. You can assign anything between 0 and 100 to any of the categories, as long as the total amount of points for all four categories is 100.
Row:
[R1] Salary: Everything associated with the financial payment of your work (base salary, bonuses, overtime pay, generosity with retirement plans, etc.)
[R2] Job security: Protection and support (legal and otherwise) against being laid off and fired, both in the event of mass closures and individual layoffs (wrongful or not)
[R3] Promotion potential: Potential to move up the career ladder in the company
[R4] Work environment quality: The day-to-day quality of your work environment, including physical environment (e.g. equipment and facilities), company culture (e.g. support, feedback, collaboration, potential to influence) and working conditions (e.g. workplace safety, conditions employment, work-life balance)

Are you a member of a trade union?
[R1] Yes
[R2] No
[R3] Don't want to answer

For how many years have you been a member?

Have you been a member continuously during that time, or have you changed in and out of membership over the years?
[R1] Continuous
[R2] Not continuously

How important do you think the union is to improving your pay, job security, promotion potential and work environment quality?
0 means "not at all" and 100 means "entirely". The total for all four need NOT be 100.
[R1] Monetary compensation
[R2] Job security
[R3] Promotion potential
[R4] Working environment quality

Compared to members, the extent to which do you think nonmembers in your workplace can benefit from the presence of unions along these four dimensions
0 means "not at all" and 100 means "complete". The total for all four need NOT be 100.
[R1] Monetary compensation
[R2] Job security
[R3] Promotion potential
[R4] Working environment quality

Have you found a union membership useful for receiving non-work benefits such as lower mortgage rates, access to cheaper/better insurance, etc.?

How important has this been for your decision to join a union?

If your after-tax dues for union membership increased by [XYZ] dollars, would you reconsider the decision to join a union?
Row:
[R1] Yes
[R2] No

The purpose of this question is to understand the reason why you do not join a union. Check all the boxes that apply.
Row:
[R1] I don't want to spend so much money being a union member
[R2] I don't think unions can affect my work situation
[R3] I find that unions focus on dimensions of the workplace that are not important to me.
[R4] I don't think I need to be a member of a union to take advantage of the influence unions have on my work situation and well-being
[R5] Other reason, note:

If your after-tax dues for union membership were reduced by [XYZ] NOK, would you reconsider your decision to join a union?
Row:
[R1] Yes
[R2] No

Second Survey Instrument

[INTRO1] This is a survey that IPSOS conducts on behalf of the Norwegian School of Economics and Business Administration. The results will be used in a research project.

All information collected through the survey is anonymized and will not be disclosed to any third party. As part of scientific publishing, anonymized data may be shared in open scientific repositories.

If you want more information about the project, you can choose the option below. If you want to start the survey, you choose it.

[R1] I want more information
[R2] I want to start the survey

[R1] Information and declaration of consent

Purpose of the project

We want to understand how individuals in Norway value their work environment and how they view unions. The results of the study will increase our understanding of workplace preferences and their relative importance.

Who is responsible for the project?

The Norwegian School of Economics (NHH) is the responsible institution for the project. Alexander Willen, professor at NHH, is the project manager. The other project members are Samuel Dodini, postdoctoral fellow at NHH, and Julia Zhu, postdoctoral fellow at NHH. If you have any questions about the project, you can contact NHH via Alexander Willen (alexander.willen@nhh.no).

What does participation mean for you?

If you choose to participate in the project, you will be asked to answer a survey by completing an online questionnaire. It takes about 4 minutes. The survey includes questions about your experiences with your local union. In addition, we will ask some basic demographic questions about, for example, age and gender. Participation in the survey is voluntary and you can withdraw your consent at any time without giving any reason. All information collected through the survey is anonymized. As part of scientific publishing, anonymized data may be shared in open scientific repositories. There will be no negative consequences if you choose not to participate or decide to withdraw at a later date.

Declaration of consent

I have received and understood information about the survey and hereby consent:

- to participate in the online survey.
- to allow researchers to process my anonymized data and use them for publications in scientific journals and other scientific dissemination.

[R2] Survey

[Proceed only for union members]

[Age] What is your age?
[Gender] Are you male or female?
[Zip code] What is your zip code?
[Fylke] Which county do you live in?

What is your highest completed education?
[R1] Primary school/primary school
[R2] Upper secondary school (incl. former vocational school)
[R3] Vocational school, trade certificate/journeyman's certificate and other 1-2 year education after upper secondary school
[R4] University/college up to 3 years (Bachelor's degree)
[R5] University/college 4 years or more (Master's degree and higher)
[R98] Other

Where were you born?
[R1] Norway
[R2] Outside Norway
[R3] Don't want to answer

Can you state which country you were born in?

At what age did you move to Norway?

How many years of full-time work experience do you have?

Are you currently in part-time or full-time work?
[R1] Part-time (less than 30 hours per week)
[R2] Full-time (at least 30 hours per week)
[R3] Not working

What industry is your main job in?

How many people work at your workplace?
Row:
[R1] 1-5
[R2] 6-10
[R3] 11-50
[R4] 51-100
[R5] More than 100
[R6] Don't want to answer

After you joined the union, how much information did your local union provide on the following aspects in which the union might be able to help: Monetary Compensation, Job Security, Promotion Potential and Work Environment Quality.
For each row, we give respondents the choices of "None," "Some information," "A lot of information."
Row:
[R1] Monetary compensation: Everything associated with the financial payment of your work (base salary, bonuses, overtime pay, generosity with retirement plans, etc.)
[R2] Job security: Protection and support (legal and otherwise) against being laid off and fired, both in the event of mass closures and individual layoffs (wrongful or not)
[R3] Promotion potential: Potential to move up the career ladder in the company
[R4] Work environment quality: The day-to-day quality of your work environment, including physical environment (e.g. equipment and facilities), company culture (e.g. support, feedback, collaboration, potential to influence) and working conditions (e.g. workplace safety, conditions employment, work-life balance)

How familiar are you with the ways you can contact your local union representative?
Row:
[R1] Very unfamiliar
[R2] Somewhat unfamiliar
[R3] Somewhat familiar
[R4] Very familiar

If you were to contact your local union regarding any of the following matters, do you think they would be effective at helping you individually?
For each row, we give respondents three options: No, Yes, I don't know
Row:
[R1] Monetary compensation
[R2] Job security
[R3] Promotion potential
[R4] Working environment quality

In the past 12 months, how many times have you contacted your local union about the following aspects of your job?
For each row, we give respondents an empty text box to type in a number.

Row:
[R1] Monetary compensation
[R2] Job security
[R3] Promotion potential
[R4] Working environment quality

[Triggered by nonzero response to previous question]
Please rate your experiences with your local union
Row:
[R1] Very negative
[R2] Somewhat negative
[R3] Somewhat positive
[R4] Very positive
[R5] They never responded

How do you think union members are treated at your workplace by the local union?
Row:
[R1] I think all union members are treated equally by the local union
[R2] I think some members are prioritized over other members by the local union

---

[If answer R2 in last question] Who do you think are being prioritized by your local union?

To what extent do you agree with the following statement?
Unions have improved the work environment at my workplace ***in general****.*

Row:

[R1] Strongly disagree
[R2] Somewhat disagree
[R3] Somewhat agree
[R4] Strongly agree

[SAME PAGE]

To what extent do you agree with the following statement?
Unions have improved the work environment at my workplace ***for union members****.*

Row:

[R1] Strongly disagree
[R2] Somewhat disagree
[R3] Somewhat agree
[R4] Strongly agree

[SAME PAGE]

To what extent do you agree with the following statement?
Unions have improved the work environment at my workplace ***for me individually****.*

Row:

[R1] Strongly disagree
[R2] Somewhat disagree
[R3] Somewhat agree
[R4] Strongly agree

To what extent do you agree with the following statement?
Being a union member has led to me feeling happier/more satisfied at work.
Row:
[R1] Strongly disagree
[R2] Somewhat disagree
[R3] Somewhat agree
[R4] Strongly agree
